SOUNDNESS, FREE MODELS AND DEPTH-ONE GRADED MONADS FOR GRADED NOMINAL THEORIES

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Editorial scope. Only two principal groups counted for this source. Game correctness and its complete proof are retained as context: authors transfer FordEA22 Theorem6.6 and prove the required kernel property; it is not an extra counted H2 target.

Allowed standard prerequisites (editorial).

- Nominal sets and abstraction, graded monads, coalgebraic trace semantics and standard categorical limits. Original definitions and author-cited prerequisites retained. Abstract prior game theorem is cited as context, not separately counted.

1. Introduction

In processes and formal languages, infinite alphabets serve as an abstraction of data types that are either infinite or prohibitively large. Typical examples include data values in XML documents [31], object identities [20], channel names [22], process identifiers [5], parameters of method calls [23], or nonces in cryptographic protocols [26]. Transition systems and automata models processing infinite alphabets are conveniently described as systems based on nominal sets [4, 32], in which, roughly speaking, the underlying set of names plays the role of the infinite alphabet, and system states store a bounded number of names in analogy to the standard register paradigm. For instance, non-deterministic orbit-finite automata (NOFA) [4] are equivalent to register automata [24] with non-deterministic update [25].

The nominal setting naturally supports explicit name-allocating transitions, subject to α-conversion of bound names; these are featured, for instance, in *nominal transition systems* [32] and in *regular non-deterministic nominal automata (RNNA)* [37], a nominal automata model that is equivalent to a subclass of register automata characterized by a lossiness condition called *name dropping*. RNNA trade some of the expressiveness of register automata for computational tractability; in particular, they allow inclusion checking in elementary complexity (a problem that is undecidable for the full class of register automata). Indeed, name-allocating transitions support two distinct notions of freshness, determined by the discipline imposed on renaming bound names: Under *global freshness*, as featured, for instance, in the register-based model of *session automata* [5], newly read names are required to be fresh w.r.t. all previous names. In the nominal setting, this is enforced by a discipline of clean renaming (avoiding shadowing of bound names). On

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the other hand, *local freshness*, i.e. freshness w.r.t. (some of) the names currently stored in the registers, is induced by unrestricted renaming according to the usual rules of α-equivalence familiar from λ-calculus. These concepts have been extended to nominal automata models processing trees [35] and infinite words [40], as well as to alternating nominal automata [14].

Our main concern in the present work is to capture the various semantics of nominal systems in the general categorical framework of *graded semantics* [30], which provides a unified treatment of spectra of semantics of automata models and concurrent systems, such as the well-known linear-time / branching-time spectrum of semantics of labelled transition systems [8]. This is achieved by combining the setting of universal coalgebra [36] with additional algebraic laws introduced via the choice of a *graded monad* [38], a categorical structure that abstractly represents a *graded algebraic theory*. Here, the notion of *grade* intuitively corresponds to depth of look-ahead, measured in (exact) numbers of transition steps. A typical example of a graded algebraic equation is the equation $a(x + y) = a(x) + a(y)$, which occurs in the axiomatization of the trace semantics of labelled transition systems [8] and says that taking an a-labelled step distributes over nondeterministic choice $+$. Here, one assigns depth 1 to the unary operation $a(-)$ representing a-steps and depth 0 to the choice operator $+$, so that the overall depth of the equation is 1. The notion of graded theory has originally been introduced for set-based graded monads [38]; it has been extended to graded monads on the category of posets [10], the category of metric spaces [12], and more generally to graded monads on categories of relational structures [9, 13], while coverage of nominal systems is currently missing. Benefits of casting a given semantics as a graded semantics include instantiation of generic results obtained for graded semantics, such as logical characterizations [30, 8, 10, 12] and game characterizations [11, 13].

In this context, our main contributions in the present work are as follows.

- We introduce a notion of graded nominal algebra, based on a simplification of Gabbay and Mathijsen's framework of nominal algebra [19]. We show that graded nominal theories induce graded monads on the category of nominal sets and, crucially, that graded nominal theories of depth 1 induce graded monads that are depth-1 in an abstract categorical sense [30, 8], a key prerequisite for most technical results in graded semantics (e.g. [30, 8, 11, 13]).
- Besides more basic examples, we then present graded nominal theories capturing the global and local freshness semantics of name allocation in nominal transition systems and RNNA; in particular the case of local freshness turns out to be quite non-trivial even though it is characterized by a single simple additional axiom.
- We transfer the game characterization of graded semantics [11] to nominal sets, obtaining as instances of this result Spoiler-Duplicator games for trace equivalence of nominal transition systems under both global and local freshness. Notably, establishing the technical conditions for correctness of the game in the case of local freshness requires a maybe unexpected twist to the setup of the corresponding graded nominal-algebraic theory, which as it turns out needs to include name restriction [34] as a depth-0 operation in order to unblock certain α-renamings, in analogy to name-dropping constructions for RNNA [37].

Related work. We have already mentioned previous work on graded semantics [30, 8, 10, 11, 12, 13] and nominal algebra [19]; details on the relationship with the latter are discussed in Remark 5.11. We note that alternative systems of varying expressiveness for universal algebra over nominal sets have been proposed [7, 28]. Some of these systems [28, 29] indeed use standard multisorted universal algebra, which however needs to be restricted to *uniform* theories and terms, and then corresponds to a subclass of (finitary) monads on nominal sets, the so-called fb-monads [29]. While many nominal-algebraic systems derive freshness judgements [18, 19, 7], sometimes in synchrony with equalities [7], our system is entirely concerned with equalities, with freshness implicit in the context of equalities. The global freshness semantics of RNNA has been modelled coalgebraically using categorical distributive laws (in what has been termed Eilenberg-Moore-type [1] and Kleisli-type [21] coalgebraic trace semantics), which by general results [30] entails a modelling as a graded semantics, without however providing a syntactic algebraic presentation [15]. Categorical distributive laws have also been used to obtain up-to techniques for proving language equivalence of NOFAs [6].

2. Preliminaries

We assume basic familiarity with category theory (e.g. [2]). We proceed to recall requisite background on nominal sets and universal coalgebra.
Nominal Sets. serve the principled treatment of *names* and *renaming* in syntax and semantics [34]; a key insight behind the framework is that many issues around renaming are best handled by permuting names. Fix a countably infinite set $\mathbb{A}$ of *names*, and let $\mathsf{Perm}(\mathbb{A})$ be the group of finite permutations on $\mathbb{A}$, which is generated by the permutations $(a\ b)$ that swap $a, b \in \mathbb{A}$. A $\mathsf{Perm}(\mathbb{A})$-*set* $(X, \cdot)$, or just X, is a set X equipped with a left action $(\pi, x) \mapsto \pi \cdot x$ of $\mathsf{Perm}(\mathbb{A})$. Then, X decomposes disjointly into the *orbits* of its elements $x \in X$, i.e. the sets $\{\pi \cdot x \mid \pi \in \mathsf{Perm}(\mathbb{A})\}$. If X has only finitely many orbits, then X is *orbit-finite*.

For $x \in X$, $A \subseteq X$, we write

$$\mathsf{fix}(x) = \{\pi \in \mathsf{Perm}(\mathbb{A}) \mid \pi \cdot x = x\} \quad \text{and} \quad \mathsf{Fix}(A) = \textstyle\bigcap_{x \in A} \mathsf{fix}(x);$$

that is, $\mathsf{fix}(x)$ is the set of permutations fixing the element x, and $\mathsf{Fix}(A)$ is the set of permutations fixing the set A pointwise. Notice that $\mathbb{A}$ itself is a $\mathsf{Perm}(\mathbb{A})$-set with the action given by $\pi \cdot a = \pi(a)$. A set $S \subseteq \mathbb{A}$ is a *support* of $x \in X$ if $\mathsf{Fix}(S) \subseteq \mathsf{fix}(x)$, i.e. if every permutation that fixes all names in S fixes also x; in this case, we think of S as giving an upper bound on the names occurring in x. We say that x is *finitely supported* if x has some finite support. A $\mathsf{Perm}(\mathbb{A})$-set X is a *nominal set* if every $x \in X$ is finitely supported. One can then show that every $x \in X$ has a *least* finite support, denoted $\mathsf{supp}(x)$; we think of $\mathsf{supp}(x)$ as the set of names occurring in x. A name $a \in \mathbb{A}$ is *fresh* for $x \in X$ (in symbols, $a \# x$) if $a \notin \mathsf{supp}(x)$. Every set X can be made into a *discrete* nominal set by equipping X with the trivial $\mathsf{Perm}(\mathbb{A})$-action $\pi \cdot x = x$. Then, $\mathsf{supp}(x) = \emptyset$ for all $x \in X$. For every $n \in \mathbb{N}$, $\mathbb{A}^n$ is a nominal set under the componentwise action; then, $\mathsf{supp}(a_1, \ldots, a_n) = \{a_1, \ldots, a_n\}$.

A map $f\colon X \to Y$ between $\mathsf{Perm}(\mathbb{A})$-sets X, Y is *equivariant* if $f(\pi \cdot x) = \pi \cdot f(x)$ for all $x \in X$, $\pi \in \mathsf{Perm}(\mathbb{A})$. (This term extends to subsets $Y \subseteq X$ of a nominal set X by viewing them as predicates; explicitly, Y is equivariant if $y \in Y$ implies $\pi \cdot y \in Y$ for all $\pi \in \mathsf{Perm}(\mathbb{A})$.) Nominal sets and equivariant maps form a category **Nom**.

For every $\mathsf{Perm}(\mathbb{A})$-set X, the finitely supported elements of X form a nominal set. For instance, the powerset $\mathcal{P}(X)$ of a nominal set X is a $\mathsf{Perm}(\mathbb{A})$-set under the action $\pi \cdot A = \{\pi \cdot x \mid x \in A\}$. We obtain a nominal set $\mathcal{P}_{\mathsf{fs}}(X)$, the *finitely supported powerset*, consisting of the finitely supported elements of $\mathcal{P}(X)$. Similarly, for nominal sets X, Y, the set $X \to Y$ of maps from X to Y is a $\mathsf{Perm}(\mathbb{A})$-set under the action $(\pi \cdot f)(x) = \pi \cdot f(\pi^{-1} \cdot x))$. The set $X \to_{\mathsf{fs}} Y$ of finitely supported elements of $X \to Y$ forms a nominal set, the set of *finitely supported maps* $X \to Y$. A set $A \in \mathcal{P}_{\mathsf{fs}}(X)$ is *uniformly finitely supported* if $\mathsf{supp}(x) \subseteq \mathsf{supp}(A)$ for all $x \in A$; the *uniformly finitely supported powerset* $\mathcal{P}_{\mathsf{ufs}}(X)$ consists of the uniformly finitely supported subsets of X. If X is orbit-finite, then the uniformly finitely supported subsets of X are precisely the finite subsets. (Contrastingly, for instance, a subset of $\mathbb{A}$ is finitely supported iff it is either finite or cofinite.)

Given a nominal set X, the *abstraction set* $[\mathbb{A}]X$ is the quotient of $\mathbb{A} \times X$ modulo the equivalence relation $\sim$ defined by

$$(a, x) \sim (b, y) \iff (c\ a) \cdot x = (c\ b) \cdot y \text{ for some fresh } c \in \mathbb{A}.$$

We write $\langle a \rangle x$ for the equivalence class of (a, x) under $\sim$. We think of $\langle a \rangle x$ as arising from x by abstracting the name a in analogy to λ-abstraction in λ-calculus or name abstraction in the π-calculus, i.e. by allowing a to be α-renamed. Indeed we have $\mathsf{supp}(\langle a \rangle x) = \mathsf{supp}(x) \setminus \{a\}$.

Universal Coalgebra. Recall that a *functor* $F\colon \mathbf{C} \to \mathbf{C}$ on a category $\mathbf{C}$ assigns to each $\mathbf{C}$-object C a $\mathbf{C}$-object FC and additionally acts on $\mathbf{C}$-morphisms, compatibly with domains and codomains of morphisms as well as identities and composition. It is useful to think of FC as consisting of structured objects made up of inhabitants of C. We will primarily be interested in the case $\mathbf{C} = \mathbf{Nom}$. Indeed, all constructions of nominal sets recalled above extend to functors. For instance, the construction of the abstraction set $[\mathbb{A}]X$ extends to an *abstraction functor* $[\mathbb{A}](-)$ taking an equivariant map $f\colon X \to Y$ to the equivariant map $[\mathbb{A}]f\colon [\mathbb{A}]X \to [\mathbb{A}]Y$ given by $[\mathbb{A}]f(\langle a \rangle x) = \langle a \rangle f(x)$.

Universal coalgebra [36] provides a unified framework for automata models and concurrent systems by abstracting the type of a system as a functor $F\colon \mathbf{C} \to \mathbf{C}$ on a base category $\mathbf{C}$. An *F-coalgebra* (C, γ) then consists of a $\mathbf{C}$-object C, thought of as an object of *states*, and a $\mathbf{C}$-morphism $\gamma\colon C \to FC$ embodying the *transition* structure of the system. As a first example, we have a functor G on **Nom** defined on objects by $GX = 2 \times \mathcal{P}_{\mathsf{fs}}(\mathbb{A} \times X)$ where $2 = \{\top, \bot\}$ is a two-element discrete nominal set. A G-coalgebra $\gamma\colon C \to GC$ then assigns to each state $c \in C$ a binary value, read as determining finality, and a set of label/successor pairs; it is thus equivalently described as a triple $(C, F, \to)$ where $F \subseteq C$ is an equivariant set of final states and $\to \subseteq C \times \mathbb{A} \times C$ is an equivariant $\mathbb{A}$-labelled transition relation. Thus, orbit-finite G-coalgebras are precisely non-deterministic orbit-finite automata (NOFAs) over the alphabet $\mathbb{A}$ [4].

A $\mathbf{C}$-morphism $f\colon C \to D$ between F-coalgebras (C,γ) and (D,δ) is an *F-coalgebra morphism* if $Ff \cdot \gamma = \delta \cdot f$. Such F-coalgebra morphisms are thought of as behaviour-preserving maps. Correspondingly, universal coalgebra yields a canonical notion of behavioural equivalence: States $c \in C$, $d \in D$ in F-coalgebras (C,γ), (D,δ) are *behaviourally equivalent* if there exists an F-coalgebra (E,ϵ) and F-coalgebra morphisms $f\colon C \to E$, $g\colon D \to E$ such that $f(c) = g(d)$. This notion typically captures branching-time equivalences, such as bisimilarity on labelled transition systems. However, one is often interested in more coarse-grained notions of equivalence, such as trace equivalence on labelled transition systems or language equivalence of automata. There have been quite a number of approaches to modelling such equivalences in universal coalgebra (e.g. [21, 1, 30]); the specific framework of graded semantics is recalled in detail in Section 4.

3. Nominal Systems with Name Allocation

We proceed to recall regular non-deterministic nominal automata (RNNA) [37] and the associated subclass of nominal transition systems [32], discussing their global and local freshness semantics. The main driving principle behind RNNA is to let automata natively accept words with bound names indicating that a fresh name is read from the input string, so-called bar strings, and then generate data words from bar strings by taking all instantiations of the bound letters. Here, instantiation of bound names proceeds in two steps: First, form all α-equivalent variants of the bar string, and then forget that bound names are bound. Bound names are indicated by a preceding '|'; that is, $|a$ indicates that a fresh letter a is read from the input. Thus, a *bar string* is just a word over the extended alphabet $\overline{\mathbb{A}} = \mathbb{A} \cup \{|a \mid a \in \mathbb{A}\}$. Formally, an occurrence of $|a$ in a bar string binds the name a, with the scope of the binding extending to the end of the string. Bound names in this sense can be renamed according to the usual discipline of α-equivalence known from λ-calculus or π-calculus; explicitly:

Definition 3.1 (α-Equivalence of bar strings)**.** The relation of *α-equivalence* on bar strings is the equivalence $\equiv_\alpha$ generated by

$$w|av \equiv_\alpha w|bu \quad \text{if } \langle a\rangle v = \langle b\rangle u \text{ in } [\mathbb{A}]\,\overline{\mathbb{A}}^*. \tag{3.1}$$

The α-equivalence class of a bar string $w \in \overline{\mathbb{A}}^*$ is denoted $[w]_\alpha$. The set $\mathsf{FN}(w)$ of *free names* in w is given recursively by $\mathsf{FN}(\epsilon) = \emptyset$ (where as usual ϵ denotes the empty string), $\mathsf{FN}(aw) = \{a\} \cup \mathsf{FN}(w)$, and $\mathsf{FN}(|aw) = \mathsf{FN}(w) \setminus \{a\}$. A bar string w is *closed* if $\mathsf{FN}(w) = \emptyset$, and *clean* if all bound names $|a$ in w are mutually distinct and distinct from all free names in w.

For instance, $|a|b \equiv_\alpha |a|c \equiv_\alpha |a|a$, while $|a|ba \equiv_\alpha |a|ca \not\equiv_\alpha |a|aa$. Crucially, the renaming of $|c$ into $|a$ in $|a|ca$ is *blocked* by the free occurrence of a in $|ca$; formally, this is due to the (straightforward) observation that $\mathsf{supp}([w]_\alpha) = \mathsf{FN}(w)$.

We will be interested in three types of languages: *Data languages* are subsets of $\mathbb{A}^*$, whose elements we refer to as *data words*; *literal languages* are subsets of $\overline{\mathbb{A}}^*$; and *bar languages* are subsets of $\overline{\mathbb{A}}^*/\equiv_\alpha$. We convert bar languages into data languages in two alternative ways respectively corresponding to *global* and *local freshness* semantics of name binding. We define the

auxiliary operation $\mathsf{ub}\colon \overline{\mathbb{A}}^* \to \mathbb{A}^*$ recursively by $\mathsf{ub}(\epsilon) = \epsilon$, $\mathsf{ub}(aw) = a\mathsf{ub}(w)$, and $\mathsf{ub}(|aw) = a\mathsf{ub}(w)$; that is, ub removes all bars '|' from a bar string. Then given a bar language L, we define data languages $N(L)$, $D(L)$ by

$$N(L) = \{\mathsf{ub}(w) \mid [w]_\alpha \in L, w \text{ clean}\} \quad \text{and} \quad D(L) = \{\mathsf{ub}(w) \mid [w]_\alpha \in L\}.$$

We briefly write $D(w) = D(\{w\})$, $N(w) = N(\{w\})$. For instance, $N(|a|b) = \{ab \mid a, b \in \mathbb{A}, a \neq b\}$, $D(|a|b) = \{ab \mid a, b \in \mathbb{A}\}$, and $D(|a|ba) = \{aba \mid a, b \in \mathbb{A}, a \neq b\}$. Roughly speaking, global freshness semantics, embodied by the N operator, enforces instantiations of bound letters to be distinct from all letters seen previously in the input word, while local freshness, embodied by the D operator, is more permissive, enforcing distinctness only from previously seen letters that are expected to be seen again (in an automaton model, this implies in particular that the previous letter must still be in memory). In fact the N operator is injective on languages of closed bar strings, so global freshness semantics is essentially equivalent to a semantics given directly in terms of bar languages.

The automata model of RNNA is designed to accept bar languages, subsequently equipped with global or local freshness semantics as indicated above.

Definition 3.2. A *regular nondeterministic nominal automaton (RNNA)* is a tuple $A = (Q, \to, s, F)$ consisting of an orbit-finite set Q of *states*; an equivariant subset $\to \subseteq Q \times \overline{\mathbb{A}} \times Q$, the *transition relation*; an *initial state* $s \in Q$; and an equivariant subset $F \subseteq Q$ of *final states* such that the following conditions are satisfied:

- The relation $\to$ is *α-invariant*: If $s \xrightarrow{|a} q$ and $\langle a\rangle q = \langle a'\rangle q'$, then $s \xrightarrow{|a'} q'$.
- The relation $\to$ is *finitely branching up to α-equivalence*: For every state $q \in Q$, the sets $\{(a, q') \mid q \xrightarrow{a} q'\}$ and $\{\langle a\rangle q' \mid q \xrightarrow{|a} q'\}$ are finite.

The relation $\to$ is extended to bar strings in the usual manner; a bar string w is *accepted* by a state $q \in Q$ if there exists a state $q' \in F$ such that $q \xrightarrow{w} q'$. The *literal language* $L_0(A, q)$ accepted by q is the set of bar strings accepted by q. Correspondingly, the *bar language* $L(A, q)$ accepted by q is the set $L_0(A, q)/{\equiv_\alpha}$ of equivalence classes of bar strings accepted by q. The *data languages* accepted by q under *global* and *local freshness semantics* are the sets $N(L(A, q))$ and $D(L(A, q))$, respectively. When mention of q is omitted, we assume $q = s$; e.g. $L_0(A) = L_0(A, s)$ is the literal language accepted by A, etc. A *regular nominal transition system (RNTS)* is an RNNA A as above with $F = Q$, i.e. with all states final; we then refer to elements of $L_0(A, q)$, $L(A, q)$, $N(L(A, q))$, or $D(L(A, q))$ as *literal traces*, *bar traces*, or *data traces* (under global/local freshness) of q, respectively, and similarly to languages as *trace sets*. States q, q' in regular nominal transitions A, A', respectively, are *trace equivalent under global (local) freshness semantics* if $N(L(A, q)) = N(L(A', q'))$ (resp. $D(L(A, q)) = D(L(A', q'))$).

For readability, we will restrict the technical development to RNTS (see however Remark 4.4), which can be seen as a slight simplification of nominal transition systems in the sense of Parrow et al. [32] given by omitting propositional atoms (*state predicates*), restricting the number of bound

names per transition to at most one, and imposing finite branching. RNTS can be viewed as orbit-finite coalgebras (cf. Section 2) for the functor $H\colon \mathbf{Nom} \to \mathbf{Nom}$ given on objects by

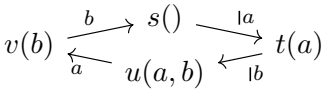
$$HX = \mathcal{P}_{\mathsf{ufs}}(\mathbb{A} \times X) \times \mathcal{P}_{\mathsf{ufs}}([\mathbb{A}]X). \tag{3.2}$$

The induced coalgebraic notion of behavioural equivalence can be described concretely as a form of bisimilarity in a fairly straightforward manner, and as such in particular does not coincide with trace equivalence under either local or global freshness semantics. Instead, we discuss in Sections 4 and 6 how these semantics are captured using graded semantics.

An important property of RNTS is that names are obtained only by reading them from the input word; formally, this is reflected in the properties that

$$q \xrightarrow{a} q' \implies \{a\} \cup \mathsf{supp}(q') \subseteq \mathsf{supp}(q) \quad \text{and} \quad q \xrightarrow{|a} q' \implies \mathsf{supp}(q') \subseteq \mathsf{supp}(q) \cup \{a\}.$$

The following diagram depicts an example RNTS A by picking representatives of orbits:



The bar trace set $L(A, s())$ of $s()$ consists of the α-equivalence classes of all bar strings of the form $(|a|bab)^n w$ where $w \in \{\epsilon, |a, |a|b, |a|ba\}$. Correspondingly, the data trace set $D(L(A, s()))$ under local freshness consists of the data traces of the form $a_1 b_1 a_1 b_1 \dots a_n b_n a_n b_n w$ where $a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{A}$, $a_i \neq b_i$ for $i = 1, \dots, n$, and $w \in \mathbb{A}^*$ is a data trace of length at most 3 in which the third letter, if any, equals the first and differs from the second.

4. Graded Monads and Graded Theories

We next briefly summarize the framework of *graded semantics* [30, 8], which serves to unify system semantics of varying granularity, such as found, for example, on the well-known linear-time / branching-time spectrum [41]. The central notion of the framework is the following.

Definition 4.1. A *graded monad* [38] on a category $\mathbf{C}$ is a tuple $\mathbb{M} = ((M_n)_{n\in\mathbb{N}}, \eta, (\mu^{nk})_{n,k\in\mathbb{N}})$ consisting of

- a family of endofunctors $M_n\colon \mathbf{C} \to \mathbf{C}$, indexed over $n \in \mathbb{N}$;
- a natural transformation $\eta : \mathsf{Id} \to M_0$, the *unit*; and
- a family of natural transformations $\mu^{nk} : M_n M_k \to M_{n+k}$, indexed over $n, k \in \mathbb{N}$, the *multiplication*

satisfying graded variants of the usual *unit* and *associative* laws ($\mu^{0,n} \cdot \eta M_n = id_{M_n} = \mu^{n,0} \cdot M_n \eta$, $\mu^{n,k+m} \cdot M_n \mu^{k,m} = \mu^{n+k,m} \cdot \mu^{n,k} M_m$). The *graded Kleisli star* operation $(-)_k^*$ of $\mathbb{M}$, indexed over $k \geq 0$, lifts a morphism $f\colon X \to M_n Y$ to the morphism $f_k^* = \mu^{kn} \cdot M_k f\colon M_k X \to M_{k+n} Y$.

We recall how finitary graded monads on the category **Set** of sets and maps can be generated from algebraic presentations [38, 30]; one of our present contributions is to generalize this principle to graded monads on the category **Nom** of nominal sets (Section 5). A *graded signature* Σ is a collection of function symbols f with assigned *arity* $\mathsf{ar}(f)$ and *depth* $d(f)$.

This induces a notion of term of uniform depth: Variables are terms of uniform depth 0, and if $d(f) = k$ and $\mathsf{ar}(f) = n$, then $f(t_1, \dots, t_n)$ is a term of uniform depth $m + k$ if all t_i are terms of uniform depth m. (Hence, if $\mathsf{ar}(c) = 0$ and $d(c) = k$, then c is a term of uniform depth m for every $m \geq k$.) A *graded theory* (Σ, E) then consists of a graded signature Σ and a set E of equational axioms of *uniform depth*, i.e. for every equation $s = t$ in E, there is some k such that s and t have uniform depth k. Such a graded theory induces a graded monad $((M_n), \eta, (\mu^{n,k}))$ on **Set** where M_nX consists of terms over X of uniform depth n modulo derivable equality, η converts variables into terms, and $\mu^{n,k}$ collapses layered terms [30].

Generally, we think of M_nX as a set of n-step behaviours ending in post-states in X, and of M_n1 (where 1 is a terminal object) as a set of n-step behaviours. Correspondingly, one has the following notion of *graded semantics*:

Definition 4.2. A *graded semantics* $(\mathbb{M}, \beta)$ for a functor G on a category $\mathbf{C}$ with a terminal object 1 consists of a graded monad $\mathbb{M} = ((M_n), \eta, (\mu^{nk}))$ and a natural transformation $\beta : G \to M_1$. Given a G-coalgebra $\gamma\colon X \to GX$, the *behaviour maps* $(\gamma^{(n)}\colon X \to M_n1)_{n<\omega}$ are given recursively as the composites

$$\gamma^{(0)} = (X \xrightarrow{\eta_X} M_0X \xrightarrow{M_0!} M_01),$$
$$\gamma^{(n+1)} = (X \xrightarrow{\beta_X \cdot \gamma} M_1X \xrightarrow{M_1\gamma^{(n)}} M_1M_n1 \xrightarrow{\mu_X^{1n}} M_{n+1}1).$$

States $x \in X$, $y \in Y$ in G-coalgebras $\gamma\colon X \to GX$ and $\delta\colon Y \to GY$ are *(β)-behaviourally equivalent* if $\gamma^{(n)}(x) = \delta^{(n)}(y)$ for all $n \in \mathbb{N}$.

Example 4.3 (Trace semantics of LTS [8])**.** *We can view finitely branching $\mathcal{A}$-labelled transition systems as coalgebras for the functor $G\colon$ **Set** $\to$ **Set** given by $GX = \mathcal{P}_\omega(\mathcal{A} \times X)$ where $\mathcal{P}_\omega$ denotes finite powerset. We have a graded monad $\mathbb{M}$ with functors M_n given on objects by $M_nX = \mathcal{P}_\omega(\mathcal{A}^n \times X)$ and with evident unit and multiplications, and obtain a graded semantics $(\mathbb{M}, \beta)$ for G where β is just the identity on $G = M_1$. This graded semantics captures exactly the trace semantics of LTS; in particular, two states are β-behaviourally equivalent iff they are trace equivalent in the usual sense. The graded monad $\mathbb{M}$ is induced by the graded algebraic theory that has a constant 0 and a binary function $+$ of depth 0, and for every $a \in \mathcal{A}$ a unary function $a(-)$ of depth 1; the equations are the usual join-semilattice axioms for 0 and $+$, and depth-1 equations $a(0) = 0$ and $a(x + y) = a(x) + a(y)$.*

Remark 4.4. *The coalgebraic modelling of automata models differs from that of transition systems in having an additional output or acceptance component. Graded monads for the language semantics of automata retain a trace component capturing words that have a run; the language semantics is then obtained by applying a canonical projection $M_n1 \to M_n0$ to the graded semantics [27, 30]. In the algebraic view, accepted words are captured by a depth-1 constant for acceptance, and the canonical projection substitutes the single variable in the set 1 by the bottom element. In the present discussion, we elide this complication purely in the interest of readability.*

One has the following graded analogues of monad algebras:

Definition 4.5 (Graded algebras)**.** Given $n \in \mathbb{N}$ and a graded monad $\mathbb{M} = ((M_n), \eta, (\mu^{n,k}))$ on $\mathbf{C}$, an *M_n-algebra* A consists of

- a family $(A_k)_{0\leq k\leq n}$ of $\mathbf{C}$-objects A_k called *carriers*, and
- a family $(a^{m,k})_{0\leq m+k\leq n}$ of morphisms $a^{m,k} : M_m A_k \to A_{m+k}$

satisfying evident graded variants of the usual laws ($a^{0,m} \cdot \eta_{A_m} = id_{A_m}$, $a^{m,r+k} \cdot M_m a^{r,k} = a^{m+r,k} \cdot \mu^{m,r}_{A_k}$). A *homomorphism* $A \to B$ of M_n-algebras A, B is a family $(f_k)_{0\leq k\leq n}$ of morphisms $f_k \colon A_k \to B_k$ such that, for $m+k \leq n$, $b^{m,k} \cdot M_m f_k = f_{m+k} \cdot a^{m,k}$.

Most results on graded semantics, such as logical and game-based characterizations of behavioural equivalence [30, 8, 11, 12, 13], hinge on the monad being *depth*-1 in the following sense.

Definition 4.6. A graded monad $\mathbb{M} = ((M_n), \eta, (\mu^{nk}))$ is *depth-1* if the following diagram (which commutes by the associative law of $\mathbb{M}$) is a coequalizer in the category of M_0-algebras for all X and $n \in \mathbb{N}$:

$$M_1 M_0 M_n X \underset{\mu^{1,0}_{M_n X}}{\overset{M_1 \mu^{0,n}_X}{\rightrightarrows}} M_1 M_n X \xrightarrow{\mu^{1,n}_X} M_{1+n} X. \tag{4.1}$$

On **Set**, a graded monad is depth-1 iff it is generated by a *depth*-1 *graded theory*, i.e. one whose operations and axioms all have depth at most 1 [30, Proposition 7.3] (where the 'only if' direction depends on size considerations, i.e. on either allowing infinitary operations or restricting to finitary graded monads). For instance, the presentation of the graded monad for trace semantics (Example 4.3) implies that the graded monad is depth-1. Similar results (in particular for the 'if' direction) have been obtained for graded monads on posets [10], relational structures [9], and metric spaces [12], and indeed we prove the 'if' direction for our notion of graded nominal theory, introduced next (Theorem 5.13). In the universal-algebraic view, the coequalizer condition in Definition 4.6 says roughly that all identifications caused by collapsing depth-1 terms over depth-n terms into depth-$(n+1)$ terms come from shifting depth-0 operations between the depth-1 layer and the depth-n layer, in combination with the depth-0 axioms.

5. Graded Nominal Algebra

We now present our framework of *graded nominal algebra*, a universal-algebraic system in which theories define graded monads on **Nom**. The system builds on *nominal algebra* [18, 19], from which it differs most prominently by adding the grading. Further differences are discussed in Remark 5.11.

Definition 5.1. A *nominal graded signature* Σ is a disjoint union of sets Σ_0, Σ_{f}, and Σ_{b} of *pure*, *free*, and *bound* operations, respectively. Each operation f is equipped with an *arity* $\mathsf{ar}(f) \in \mathbb{N}$ and a *depth* $d(f) \in \mathbb{N}$. We write $f/n \in \Sigma_0$ if $f \in \Sigma_0$ and $\mathsf{ar}(f) = n$, similarly for Σ_{f}, Σ_{b}, and Σ.

Definition 5.2 (Nominal Terms)**.** Given a nominal set X of *variables* and a graded signature Σ, we define *nominal Σ-terms $t, \dots$ (over X)* by the grammar

$$t ::= x \mid f(t_1, \dots, t_p) \mid a.g(t_1, \dots, t_p) \mid \nu a.\, h(t_1, \dots, t_p)$$

where $x \in X$, $a \in \mathbb{A}$, $f/p \in \Sigma_0$, $g/p \in \Sigma_{\mathsf{f}}$, and $h/p \in \Sigma_{\mathsf{b}}$. For brevity, we occasionally unify the three last clauses notationally as $\eta.f(t_1, \dots, t_p)$ for $f \in \Sigma$, where η is either nothing or of the form a or νa, with use of the notation implying the assumption of well-formedness.

Note in particular that we work with given nominal structures on sets of variables; cf. Remark 5.11. In terms $a.g(t_1, \dots, t_p)$, a is a *free* name, while in terms $\nu a.\, h(t_1, \dots, t_p)$, the name a is *bound*. This reading gives rise to an evident notion of *free names* in a term, where we count both explicit occurrences of names and occurrences of names in the support of variables; we write $\mathsf{FN}(t)$ for the set of free names in t. Explicitly, $\mathsf{FN}(x) = \mathsf{supp}(x)$, $\mathsf{FN}(f(t_1, \dots, t_p)) = \mathsf{FN}(t_1) \cup \dots \cup \mathsf{FN}(t_p)$, $\mathsf{FN}(a.g(t_1, \dots, t_p)) = \mathsf{FN}(t_1) \cup \dots \cup \mathsf{FN}(t_p) \cup \{a\}$, and $\mathsf{FN}(\nu a.\, h(t_1, \dots, t_p)) = (\mathsf{FN}(t_1) \cup \dots \cup \mathsf{FN}(t_p)) \setminus \{a\}$.

We have a notion of *uniform depth* of terms defined completely analogously to the set-based case (Section 4). Then, a substitution σ (a map assigning nominal terms $\sigma(x)$ to variables $x \in X$) has *uniform depth* if all $\sigma(x)$ are of the same uniform depth. As usual, we write $t\sigma$ for the application of σ to a nominal term t, which just replaces all occurrences of variables x in t with $\sigma(x)$. Like in nominal algebra [17, 18, 19], application of substitutions thus explicitly does *not* avoid capture of names; these issues are instead dealt with at the level of the derivation system. We write $\mathsf{Term}_{\Sigma,m}(X)$ for the set of nominal terms of uniform depth m over X; equipped with the evident action of $\mathsf{Perm}(\mathbb{A})$ (in particular, $\pi \cdot \nu a.\, f(t_1, \dots, t_n) = \nu \pi(a).\, f(\pi \cdot t_1, \dots, \pi \cdot t_n)$), $\mathsf{Term}_{\Sigma,m}(X)$ is a nominal set.

Definition 5.3 (Equations and Theories)**.** A *depth-n Σ-equation*, written $X \vdash_n t = u$, consists of an orbit-finite nominal set X of *variables* and terms $t, u \in \mathsf{Term}_{\Sigma,n}(X)$. We often omit mention of n. A *graded (nominal) theory* $T = (\Sigma, E)$ consists of a graded signature Σ and a set E of Σ-equations, referred to as *axioms*.

Given a graded theory $T = (\Sigma, E)$, we define *derivable Σ-equations* inductively by the derivation rules in Figure 1. Note that the (ax) rule allows instantiation of axioms both under a permutation τ of the names and under a substitution σ of the variables. We refer to the condition imposed on σ in the premise of the rule as *derivable equivariance*. This condition ensures *equivariance of derivability* (i.e., if $X \vdash_m t = u$ is derivable, then so is $X \vdash_m \pi \cdot t = \pi \cdot u$ for every $\pi \in \mathsf{Perm}(\mathbb{A})$).

Remark 5.4. *Rule (ax) as phrased in Figure 1 has infinitely many premisses. However, it can be equivalently replaced with a finitary version where σ is defined only on the finitely many variables occurring in r and s (and on one representative of every orbit of Y, in the maybe pathological case that Y has orbits not mentioned in r, s) and then extended to all of Y ensuring derivable equivariance. This is made possible by imposing premisses ensuring well-definedness up-to-derivability of such an extension. The set*

$$\text{(refl)}\ \frac{}{X \vdash_0 x = x} \qquad \text{(symm)}\ \frac{X \vdash_m u = t}{X \vdash_m t = u}$$

$$\text{(trans)}\ \frac{X \vdash_m t = v \qquad X \vdash_m v = u}{X \vdash_m t = u}$$

$$\text{(cong)}\ \frac{X \vdash_m t_i = u_i \qquad (i = 1, \dots, p)}{X \vdash_{m+d(f)} \eta.f(t_1, \dots, t_p) = \eta.f(u_1, \dots, u_p)}\ (f/p \in \Sigma)$$

$$\text{(ax)}\ \frac{X \vdash_l \pi \cdot \sigma(y) = \sigma(\pi \cdot y) \qquad \forall \pi \in \mathsf{Perm}(\mathbb{A}), y \in Y}{X \vdash_{m+l} (\tau \cdot r)\sigma = (\tau \cdot s)\sigma}\ (Y \vdash_m r = s \in E, \tau \in \mathsf{Perm}(\mathbb{A}))$$

$$\text{(perm)}\ \frac{X \vdash_m t_i = (a\ b) \cdot u_i \qquad (i = 1, \dots, p)}{X \vdash_{m+d(f)} \nu a.f(t_1, \dots, t_p) = \nu b.f(u_1, \dots, u_p)}\ (f/p \in \Sigma_{\mathsf{b}}, a \neq b \in \mathbb{A}, a \# u_i)$$

FIGURE 1. Equational inference rules over a graded theory $T = (\Sigma, E)$.

of these premisses is orbit-finite, so that in a simultaneous inductive proof with derivable equivariance, we may restrict to finitely many representative premisses.

Like in standard universal algebra, the notion of graded nominal theory comes equipped with a notion of model for which we will establish soundness and completeness:

Definition 5.5 (Graded nominal algebras)**.** Given a graded signature Σ and a depth $n < \omega$, a *nominal* (Σ, n)*-algebra* A consists of a family $(A_i)_{0 \leq i \leq n}$ of nominal sets, called the *carriers*, and for each $f/p \in \Sigma$ and $m \leq n - d(f)$, an equivariant function $f_{A,m}$ of type $A_m^p \to A_{m+d(f)}$ if $f \in \Sigma_0$, of type $\mathbb{A} \times A_m^p \to A_{m+d(f)}$ if $f \in \Sigma_{\mathsf{f}}$, and of type $[\mathbb{A}]A_m^p \to A_{m+d(f)}$ if $f \in \Sigma_{\mathsf{b}}$. A *homomorphism* $A \to B$ of (Σ, n)-algebras A, B is a family $(h_i)_{0 \leq i \leq n}$ of equivariant functions $h_i \colon A_i \to B_i$ satisfying the following equalities for $f/p \in \Sigma$, $a \in \mathbb{A}$, and $x = (x_1, \dots, x_p) \in A_m^p$:

$$\begin{aligned} h_{m+d(f)}(f_{A,m}(x)) &= f_{B,m}(h_m(x_1), \dots, h_m(x_p)) && \text{if } f \in \Sigma_0 \\ h_{m+d(f)}(f_{A,m}(a, x)) &= f_{B,m}(a, h_m(x_1), \dots, h_m(x_p)) && \text{if } f \in \Sigma_{\mathsf{f}} \\ h_{m+d(f)}(f_{A,m}(\langle a \rangle x)) &= f_{B,m}(\langle a \rangle (h_m(x_1), \dots, h_m(x_p))) && \text{if } f \in \Sigma_{\mathsf{b}}. \end{aligned}$$

We write $\mathsf{Alg}(\Sigma, n)$ for the category of nominal (Σ, n)-algebras and their homomorphisms. We also define (Σ, ω)-algebras and their homomorphisms; definitions are the same except that the range of indices is $i, m < \omega$ (rather than $\leq n$)

Algebras actually satisfying the axioms of the theory are singled out as follows:

Definition 5.6 (Term evaluation, satisfaction, models)**.** Given a nominal (Σ, n)-algebra A (where $n \leq \omega$) and a base depth $k \in \mathbb{N}_0, k \leq n$, the *value* $[\![t]\!]^\iota_m \in A_{k+m}$ of a depth-m term t under an equivariant *valuation* $\iota\colon X \to A_k$ is defined by

$$[\![x]\!]^\iota_0 = \iota(x) \qquad [\![f(t_1,\dots,t_p)]\!]^\iota_{m+d(f)} = f_{A,k+m}([\![t_1]\!]^\iota_m,\dots,[\![t_p]\!]^\iota_m)$$
$$[\![a.f(t_1,\dots,t_p)]\!]^\iota_{m+d(f)} = f_{A,k+m}(a,[\![t_1]\!]^\iota_m,\dots,[\![t_p]\!]^\iota_m)$$
$$[\![\nu a.f(t_1,\dots,t_p)]\!]^\iota_{m+d(f)} = f_{A,k+m}(\langle a\rangle([\![t_1]\!]^\iota_m,\dots,[\![t_p]\!]^\iota_m))$$

We say that A *satisfies* a Σ-equation $X \vdash_m t = u$ if $[\![t]\!]^\iota_m = [\![u]\!]^\iota_m$ for every valuation $\iota\colon X \to A_k$ with $k+m \leq n$ (a vacuous condition if $m > n$). Finally, given a graded theory $T = (\Sigma, E)$, a (T, n)-*model* is a nominal (Σ, n)-algebra satisfying every axiom in E. We write $\mathsf{Alg}(T, n)$ for the full subcategory of (T, n)-models in $\mathsf{Alg}(\Sigma, n)$.

We obtain soundness w.r.t. this notion of model:

Theorem 5.7 (Soundness)**.** *The derivation system in Figure 1 is sound: If an equation is derivable in a graded theory T, then it is satisfied by every (T, n)-model.*

We prove completeness of the system by constructing free graded nominal algebras, which at the same time yields our construction of a graded monad on **Nom** from a graded nominal theory. We write $\sim$ for the equivalence relation on terms defined by derivable equality, which by equivariance of derivability is equivariant, and denote the equivalence class of a depth-n term t by $[t]_n$. Given a nominal set X and $n \leq \omega$, we define a (Σ, n)-algebra FX by carriers $(FX)_i = \mathsf{Term}_{\Sigma,i}(X)/{\sim}$ (for $i \leq n$ if $n < \omega$, or for $i < \omega$ if $n = \omega$), and for $f/p \in \Sigma$:

$$f_{FX,m}([t_1]_m,\dots,[t_p]_m) = [f(t_1,\dots,t_p)]_{m+d(f)} \qquad \text{if } f \in \Sigma_0,$$
$$f_{FX,m}(a,[t_1]_m,\dots,[t_p]_m) = [a.f(t_1,\dots,t_p)]_{m+d(f)} \qquad \text{if } f \in \Sigma_{\mathsf{f}},$$
$$f_{FX,m}(\langle a\rangle([t_1]_m,\dots,[t_p]_m)) = [\nu a.f(t_1,\dots,t_p)]_{m+d(f)} \qquad \text{if } f \in \Sigma_{\mathsf{b}}.$$

This algebra is equipped with a *canonical valuation* $\eta\colon X \to (FX)_0$ given by $\eta(x) = [x]_0$. The rules of the system, specifically (cong) and (perm), guarantee that this structure is well-defined.

Theorem 5.8 (Free (T, n)-models)**.** *For a graded nominal theory T and $n \leq \omega$, the algebra FX is a free (T, n)-model over X w.r.t. the forgetful functor that sends a (T, n)-model A to the nominal set A_0, with η being the universal arrow.*

Since FX identifies terms precisely when they are derivably equal, we obtain

Corollary 5.9 (Completeness)**.** *The derivation system in Figure 1 is complete: If a depth-m equation is satisfied by every (T, n)-model of a graded nominal theory T where $m \leq n \leq \omega$, then it is derivable in T.*

From the adjunction of Theorem 5.8 for the case $n = \omega$, we obtain a graded monad by standard results [16, Section 3]. Specifically, we have a (strict)

action $\star$ of the discrete monoidal category $(\mathbb{N}, +, 0)$ on $\mathsf{Alg}(T, \omega)$ given on objects by taking $n \star A$, for a (T, ω)-model A, to be the (T, ω)-model whose i-th carrier is A_{i+n} and whose interpretation $f_{n\star A,i}$ of $f \in \Sigma$ is $f_{A,i+n}$. We then obtain a graded monad $\mathbb{M}^T$ by sandwiching this action between the two adjoint functors as per Theorem 5.8. In particular, the functor parts M_n^T of $\mathbb{M}^T$ are given as $M_n^T X = (n \star FX)_0 = (FX)_n$; that is, we have $M_n^T X = \mathsf{Term}_{\Sigma,n}(X)/\sim$ as expected. The unit of $\mathbb{M}$ is just the canonical valuation η, and the multiplication is obtained from the counit of the adjunction; intuitively, η thus converts variables into terms, and the multiplication collapses layered terms.

Example 5.10 (Traces in nominal systems)**.** *As a first example of a graded nominal theory, take* Σ_0 *to consist of depth-*0 *operations* 0, $+$, *with* 0 *a constant and* $+$ *being binary and written in infix notation, and* Σ_{f} *to consist of a single unary operation* pre *of depth* 1*. We impose depth-*0 *equations ensuring that* 0, $+$ *form a* nominal join-semilattice *(i.e. a nominal set carrying an equivariant join-semilattice structure), explicitly* $X \vdash_0 x + 0 = x$, $X \vdash_0 x + x = x$, $X \vdash_0 x + y = y + x$, *and* $X \vdash_0 (x + y) + z = x + (y + z)$ *for all orbit-finite* X *(more precisely, representatives of all such* X *modulo isomorphism to ensure that the axioms form a set, a point we will elide henceforth) and* $x, y, z \in X$*. Additionally, we have depth-*1 *axioms*

$$X \vdash_1 a.\mathsf{pre}(x + y) = a.\mathsf{pre}(x) + a.\mathsf{pre}(y) \quad \text{and} \quad \emptyset \vdash_1 a.\mathsf{pre}(0) = 0 \tag{5.1}$$

for all $a \in \mathbb{A}$*, all orbit-finite* X*, and all* $x, y \in X$*. Notice here that postulating the equalities for* all X, x, y *ensures that they apply to all elements of all nominal sets (see also Remark 5.11); on the other hand, postulating the equalities for all rather than just some* a *is in fact inessential, since rule (ax) allows applying permutations to axioms. Similarly as in Example 4.3, the induced graded monad* $\mathbb{M}$ *has functor parts* M_n *given on objects by* $M_n X = \mathcal{P}_\omega(\mathbb{A}^n \times X)$*. Coalgebras for the functor* $G = M_1 = \mathcal{P}_\omega(\mathbb{A} \times (-))$ *are finitely branching* non-deterministic nominal transition systems*, that is, NOFAs (Section 2) in which all states are accepting. The graded semantics* $(\mathbb{M}, id)$ *for* G *captures precisely the trace semantics of such systems (i.e. the language semantics of the associated NOFA). We see an example with name-binding operations in Section 6.*

Remark 5.11. *The necessity of combining operations with free or bound names instead of having names and name binding as separate syntactic constructs is owed to the grading. For instance, in Example 5.10, it is crucial that* $a.\mathsf{pre}(-)$ *has depth* 1 *as a whole, rather than being composed of two components, one of which would necessarily have depth* 0*.*

In some previous systems of nominal algebra [18, 19]*, contexts of equations are presented in terms of variables and freshness conditions, which in our terms essentially amounts to restricting to contexts* X *that are* strong *nominal sets* [39]*, i.e. for every* $\pi \in \mathsf{Perm}(\mathbb{A})$ *and* $x \in X$*, one has* $\pi \cdot x = x$ *if and only if* $\pi(a) = a$ *for all* $a \in \mathsf{supp}(x)$*. Roughly speaking, strong nominal sets thus do not have symmetries; more formally, every orbit in a strong nominal set is isomorphic to some nominal set of the form* $\mathbb{A}^{\#n}$ *for* $n \in \mathbb{N}$*, i.e. the set of* n*-tuples over* $\mathbb{A}$ *with pairwise distinct entries* [33, Lemma 2.4.2].

For instance, the set $\{\{a,b\} \subseteq \mathbb{A} \mid a \neq b\}$ *fails to be strong, due to the symmetry embodied in the equality* $(a\ b) \cdot \{a,b\} = \{a,b\}$. *Our system is thus more expressive in that it allows equations to be conditioned on symmetries in the context; e.g. we could impose an equation* $f(x) = x$ *conditioned on* $(a\ b) \cdot x = x$. *We do not currently claim to have real-life examples where this is needed; for instance, in Example 5.10 one could equivalently restrict all equations to strong nominal sets of variables.*

Finally, we establish the announced criterion for the graded monad induced by a graded nominal theory to be depth-1 (Definition 4.6).

Definition 5.12. A graded theory T is *depth-1* if all its operations and axioms are of depth at most 1.

Theorem 5.13. *Graded monads induced by depth-*1 *graded nominal theories are depth-*1.

For instance, the graded monad of Example 5.10 is depth-1.

6. Case Study: Graded Monads for Global and Local Freshness

We now present our main examples, in which we cast the global and local freshness semantics of regular nominal transition systems (RNTS; Section 3) as graded semantics. We begin with global freshness, exploiting that the induced notion of state equivalence coincides with bar language semantics. The graded theory T^{bar} for bar language semantics extends the theory for traces of non-deterministic nominal transition systems (Example 5.10) by an additional depth-1 operation $\mathsf{abs} \in \Sigma_{\mathsf{b}}$, defining a signature Σ^{bar}, and additional depth-1 equations

(6.1)
$$X \vdash_1 \nu a.\, \mathsf{abs}(x+y) = (\nu a.\mathsf{abs}(x)) + (\nu a.\mathsf{abs}(y)) \quad \text{and} \quad \emptyset \vdash_1 \nu a.\, \mathsf{abs}(0) = 0$$

for all orbit-finite X and $x, y \in X$. We write $at = a.\mathsf{pre}(t)$ and $\mathsf{l}at = \nu a.\, \mathsf{abs}(t)$ for all terms t. A *pretrace* over a nominal set X is an element of $(w,x) \in \overline{\mathbb{A}}^* \times X$, with x thought of as a poststate reached after executing w. We can literally extend the notion of *α-equivalence* $\equiv_\alpha$ of bar strings to pretraces. We now define a graded monad $\mathbb{M}^{\mathsf{bar}} = ((M_n^{\mathsf{bar}}), \eta, (\mu^{nm}))$ of pretraces modulo α-equivalence by

$$M_n^{\mathsf{bar}} X = \mathcal{P}_\omega((\overline{\mathbb{A}}^n \times X)/\equiv_\alpha),$$

(hence $M_0^{\mathsf{bar}} X = \mathcal{P}_\omega X$) with $\eta(x) = [x]_\alpha$ and $\mu^{nm}(S) = \{[uv]_\alpha \mid [uV]_\alpha \in S, [v]_\alpha \in V\}$.

Theorem 6.1. *The graded monad* $\mathbb{M}^{\mathsf{bar}}$ *is induced by the graded nominal theory* T^{bar}, *and in particular is depth-*1 *by Theorem 5.13.*

Recall that regular nominal transition systems are coalgebras for the functor H as per (3.2). To avoid discussion of infinitary theories, we restrict H to $H_\omega = \mathcal{P}_\omega(\mathbb{A} \times (-)) \times \mathcal{P}_\omega([\mathbb{A}](-))$ (and hence restrict to finitely branching RNTS). We have an obvious isomorphism $\beta\colon H_\omega \to M_1$, giving rise to a graded semantics $(\mathbb{M}^{\mathsf{bar}}, \beta)$ for H. This graded semantics captures global freshness semantics:

Theorem 6.2. *Two states in finitely branching RNTS are trace equivalent under global freshness semantics iff they are behaviourally equivalent w.r.t. the graded semantics* $(\mathbb{M}^{\mathsf{bar}}, \beta)$.

The *local freshness* semantics of an RNTS (Section 3) is computed from its bar language semantics, and imposes additional identifications. We define a graded theory T^{loc} for local freshness semantics by extending T^{bar} with depth-1 axioms

$$X \vdash_1 a.\mathsf{pre}(x) + \nu a.\mathsf{abs}(x) = \nu a.\mathsf{abs}(x) \tag{6.2}$$

for all orbit-finite X and $x \in X$, saying that every bound name can be instantiated by itself. We will see that this is already sufficient to describe local freshness. However, we can simplify some computations by introducing *name restrictions*: Similar to the computationally useful idea of closing RNNA under name dropping and thus closing the literal language under α-equivalence [37], we will introduce an algebraic operation that can arbitrarily restrict the support of a term. This construction will be essential for the description of equivalence games for local freshness semantics in Section 7. We define the graded theory T^{drop} by extending T^{loc} with a bound operation $\mathsf{res}/1$ at depth 0, which hides a name in its argument from the outside. Correspondingly, we add the *name restriction axioms* [34, Chapter 9]:

$$X \vdash_0 \nu a.\mathsf{res}(x) = x \quad \text{for } a \# x \quad \text{and} \quad X \vdash_0 \nu a.\mathsf{res}(\nu b.\mathsf{res}(x)) = \nu b.\mathsf{res}(\nu a.\mathsf{res}(x)). \tag{6.3}$$

We require this operation to be compatible with our join-semilattice structure, in that

$$X \vdash_0 \nu a.\mathsf{res}(x+y) = \nu a.\mathsf{res}(x) + \nu a.\mathsf{res}(y) \quad \text{and} \quad X \vdash_0 x = x + \nu a.\mathsf{res}(x), \tag{6.4}$$

with $X \vdash_0 \nu a.\mathsf{res}(0) = 0$ already implied by (6.3). For the depth-1 operations, we need the axioms

$$X \vdash_1 \nu a.\mathsf{res}(b.\mathsf{pre}(x)) = b.\mathsf{pre}(\nu a.\mathsf{res}(x)) \quad \text{and} \quad X \vdash_1 \nu a.\mathsf{res}(a.\mathsf{pre}(x)) = 0, \tag{6.5}$$

$$X \vdash_1 \nu a.\mathsf{res}(\nu b.\mathsf{abs}(x)) = \nu b.\mathsf{abs}(\nu a.\mathsf{res}(x)) \tag{6.6}$$

for $a \neq b$, where $X \vdash_1 \nu a.\mathsf{res}(\nu a.\mathsf{abs}(x)) = \nu a.\mathsf{abs}(x)$ is again implied by (6.3).

Let $\mathbb{M}^{\mathsf{loc}} = ((M^{\mathsf{loc}}), \eta, (\mu^{nm}))$ denote the graded monad induced by T^{loc} (we refrain from decorating η and μ with more indices) and similarly for $\mathbb{M}^{\mathsf{drop}}$ and T^{drop}. Again, we have by Theorem 5.13 that $\mathbb{M}^{\mathsf{loc}}$ and $\mathbb{M}^{\mathsf{drop}}$ are depth-1. From theory inclusion, we then have canonical natural transformations

$$\theta_n \colon M_n^{\mathsf{bar}} \to M_n^{\mathsf{loc}} \quad \text{and} \quad \kappa_n \colon M_n^{\mathsf{loc}} \to M_n^{\mathsf{drop}}$$

for every n, and obtain graded semantics $(\mathbb{M}^{\mathsf{loc}}, \theta_1 \cdot \beta)$ and $(\mathbb{M}^{\mathsf{drop}}, \kappa_1 \cdot \theta_1 \cdot \beta)$ for H. Both of these capture local freshness:

Theorem 6.3. *Two states in finitely branching RNTS are trace equivalent under local freshness semantics iff they are behaviourally equivalent w.r.t. the graded semantics* $(\mathbb{M}^{\mathsf{loc}}, \theta_1 \cdot \beta)$ *iff they are behaviourally equivalent w.r.t. the graded semantics* $(\mathbb{M}^{\mathsf{drop}}, \kappa_1 \cdot \theta_1 \cdot \beta)$.

Remark 6.4. *Given a nominal set X, $x \in X$, and $N \subseteq \mathsf{supp}(x)$, write $x|_N = \mathsf{Fix}(N) \cdot x = \{\pi \cdot x \mid \pi \in \mathsf{Fix}(N)\}$ for the orbit of x under $\mathsf{Fix}(N)$ (similar to the states in the construction of name-dropping RNNA [37]). We can then describe elements of $M_0^{\mathsf{drop}}X$ as finite sets S of such orbits such that $x|_{N'} \in S$ whenever $x|_N \in S$ and $N' \subseteq N$.*

7. Graded Behavioural Equivalence Games

For illustration of the benefits of having a depth-1 graded semantics, we discuss how graded behavioural equivalence games, previously considered for set-based coalgebras [11], transfer to the nominal setting. We then instantiate the generic game to trace semantics of RNTS (Section 6). The game has a categorical formulation that is analogous to the set-based case. The main novelty in the present work is the formulation of the game in nominal-algebraic terms using our system of graded nominal algebra from Section 5, presented next. To this end, we fix a depth-1 graded semantics $(\mathbb{M}, \beta)$ for a functor F on **Nom**, with $\mathbb{M}$ presented by a graded theory T, and an F-coalgebra (C, γ). Throughout, we exploit that in join semilattices, inequalities $s \leq t$ are expressible as equations $t = s + t$.

Predeterminization. Graded behavioural equivalence games are morally played on a *predetermization* of (C, γ), that is, on a coalgebra $M_0C \to \overline{M}_1M_0C$ for a functor $\overline{M}_1$ on M_0-algebras induced from a depth-1 graded semantics. (The name of the construction is owed to the fact that in case $M_01 = 1$, it yields an actual determinization [11]). For our present purposes, we only need the construction of $\overline{M}_1$. Given an M_0-algebra A, the M_0-algebra $\overline{M}_1(A, a)$ may be described categorically via a coequalizer construction. Algebraically, $\overline{M}_1A$ is obtained by freely extending A to an M_1-algebra and then forgetting A. That is, $\overline{M}_1A$ consists of depth-1 terms over A, quotiented by equalities derivable in the given graded theory from depth-0 equalities that hold over A. It can be shown that $\overline{M}_1(M_nX, \mu^{0n}) \cong (M_{n+1}X, \mu^{0,n+1})$ [11]; we thus briefly write $\overline{M}_1M_n = M_{n+1}$.

Example 7.1. *(1)* Global freshness semantics (graded monad $\mathbb{M}^{\mathsf{bar}}$): *We have $\overline{M}_1A = A^{\mathbb{A}}_{\omega} \times [\mathbb{A}]A$, where the endofunctor $(-)^{\mathbb{A}}_{\omega}$ on M_0-algebras sends A to the set of functions $f\colon \mathbb{A} \to A$ with $f(a) = 0$ for almost all a.*

(2) Local freshness semantics with name-dropping (graded monad $\mathbb{M}^{\mathsf{drop}}$): *An M_0-algebra A is a join semilattice $(A, +, 0)$ with a name restriction operation $[\mathbb{A}]A \to A, \langle a\rangle x \mapsto a \setminus x$. As such, we can describe $\overline{M}_1A$ as the set of finitely supported functions $f\colon \mathbb{A} \to_{\mathsf{fs}} A$ such that $(a\ c) \cdot (a \setminus f(c)) \leq f(a)$ for all $a \in \mathbb{A}$ and $c \# (f, a)$.*

In both cases, the given description implies immediately that $\overline{M}_1$ preserves monomorphisms, as needed for the correctness of the game (Theorem 7.3).

Equivalence games. Graded behavioural equivalence games come in a finite-depth and an infinite-depth variant [11]; we focus on finite depth. Generally, we blur the distinction between terms and elements of M_nC, i.e. equivalence classes of terms. The game is played by *Spoiler (S)* and *Duplicator (D)*, with D trying to prove and S trying to disprove behavioural equivalence. Configurations of the game are pairs $(s, t) \in M_0C \times M_0C$, read as equalities $C \vdash_0 s = t$ claimed by D. Given a set $Z \subseteq M_0C \times M_0C$ of depth-0 equalities,

we write $Z \Rightarrow C \vdash_n s = t$ for terms $s, t \in \mathsf{Term}_{\Sigma,n}(C)$ if the depth-n equation $C \vdash_n s = t$ can be derived from the closure of Z under equivariance in the system of Section 5 (that is, equations in Z may be renamed but not substituted into). We view the map $\beta \cdot \gamma \colon C \to M_1 C$ as a uniform depth-1 substitution σ. The game is parametrized by a number n of rounds:

Definition 7.2. The *n-round $(\mathbb{M}, \beta)$-behavioural equivalence game* $\mathsf{G}_n(\gamma)$ on (C, γ) is played in n rounds. A round starting in configuration (s, t) is played as follows:

(1) Duplicator plays a finite set $Z \subseteq M_0 C \times M_0 C$ such that $Z \Rightarrow C \vdash_1 s\sigma = t\sigma$.
(2) Spoiler picks an element $(s', t') \in Z$, which becomes the next configuration.

Any player who cannot move on their turn, loses. After n rounds have been played, ending in configuration (s, t), D wins if $1 \vdash_0 s\sigma_0 = t\sigma_0$ is derivable, where σ_0 is the depth-0 substitution mapping all variables to $\star \in 1$; we refer to this step as *calling the bluff*. Otherwise, S wins.

The game is intuitively understood as follows. We may regard $\beta \cdot \gamma \colon C \to M_1 C$ as defining the behaviour of states $c \in C$ by algebraic equations $c = \sigma(c)$. The behaviour map $\gamma^{(n)} \colon C \to M_n 1$ as per Definition 4.2 unfolds definitions of states n times and finally applies the substitution σ_0 mapping all variables to $\star$ as above; this defines a depth-n substitution σ_n such that $\gamma^{(n)}(c)$ is the equivalence class of $\sigma_n(c)$ in $M_n 1$. We can thus understand the n-round game from configuration (s, t) as D trying to construct a nominal-algebraic proof of the depth-n equation $1 \vdash_n s\sigma_n = t\sigma_n$, with S challenging selected claims made by D.

Correctness of the game is formulated as follows.

Theorem 7.3 (Game correctness)**.** *Let $(\mathbb{M}, \beta)$ be a graded semantics for a functor F on* **Nom***, with $\mathbb{M}$ presented by a graded nominal theory, and let $\gamma \colon C \to FC$ be a F-coalgebra. Suppose that $\overline{M}_1$ preserves monomorphisms (i.e. injective M_0-homomorphisms). For $n \geq 0$, D wins the configuration (s, t) in $\mathsf{G}_n(\gamma)$ iff $(\gamma^{(n)})_0^*(s) = (\gamma^{(n)})_0^*(t)$. In particular, states $c, d \in C$ are β-behaviourally equivalent iff D wins (c, d) in $\mathsf{G}_n(\gamma)$ for every $n \geq 0$.*

Example 7.4. *For our main examples (Section 6), we give explicit descriptions of the game, which we can rearrange to let S move first. In both cases, we use normal forms of $M_1 C$ implicit in the equation $M_1 = \overline{M}_1 M_0 C$ and the respective description of $\overline{M}_1$ (7.1), and the substitution σ representing the coalgebra structure.*

(1) Global freshness semantics (graded monad $\mathbb{M}^{\mathsf{bar}}$): *Given a term $s \in M_0 C$, i.e. a finite set of states, we can describe $s\sigma \in M_1 C = \overline{M}_1 M_0 C = (M_0 C)^{\mathbb{A}}_{\omega} \times [\mathbb{A}] M_0 C$ as $s\sigma = ((s_a)_{a \in \mathbb{A}}, \langle c \rangle s_{\mathsf{lc}})$ where*

$$s_a = \{y \mid x \in s, x \xrightarrow{a} y\} \quad \text{and} \quad s_{\mathsf{lc}} = \{y \mid x \in s, x \xrightarrow{\mathsf{lc}} y\}) \quad \text{for } c \# s.$$

In a configuration (s, t) with $c \# (s, t)$, S either picks an $a \in \mathbb{A}$ and $x \in s_a$ or an $x \in s_{\mathsf{lc}}$ (or analogously for t). Duplicator then responds with a subset $t' \subseteq t_a$ or $t' \subseteq t_{\mathsf{lc}}$ correspondingly, yielding the new configuration $(t' \cup \{x\}, t')$ corresponding to the claim $x \leq t'$ (and analogously for s). This also means that S cannot switch sides after the initial move.

(2) Local freshness semantics with name-dropping (graded monad $\mathbb{M}^{\mathsf{drop}}$): *Viewing terms $s \in M_0C$ as sets of name restrictions closed under further restriction as detailed in Remark 6.4, we can describe $s\sigma \in M_1C = \overline{M}_1M_0C$ as a function $\mathbb{A} \to M_0C, a \mapsto s_a$ where*

$$s_a = \{y|_{N'} \mid x|_N \in s, x \xrightarrow{a} y, a \in N, N' \subseteq \mathsf{supp}(y) \cap N\}$$
$$\cup\, \{(a\ c) \cdot (y|_{N'\setminus\{a\}}) \mid x|_N \in s, x \xrightarrow{\mathsf{lc}} y, N' \subseteq \mathsf{supp}(y) \cap (N \cup \{c\})\}.$$

In a configuration (s,t), S then picks an $a \in \mathbb{A}$ and $x|_N \in s_a$ (or $y|_N \in t_a$), to which D responds with a subset $t' \subseteq t_a$ closed under further restriction, yielding the new configuration $(t' \cup \{x|_{N'} \mid N' \subseteq N\}, t')$ (or symmetrically), effectively corresponding to the claim $x|_N \leq t'$. Again, this means that S cannot switch sides after the initial move. As a rough summary, S picks a side where they play restricted successors, and D answers with restriction-closed sets of restricted successors under matching labels.

8. Conclusions and Future Work

We have introduced a universal-algebraic system for the presentation of graded monads on the category of nominal sets. In a subsequent case study, we have applied this system to model the global and local freshness semantics of regular non-deterministic transition systems (a slightly simplified form of *regular non-deterministic nominal automata (RNNA)* [37]) in the framework of *graded semantics* [30, 8]. Notably, the axiom distinguishing local from global freshness semantics just states that every bound name subsumes the corresponding free name. We have proved soundness and completeness of our system of *graded nominal algebra*, and we have shown that the graded monad induced by graded nominal theory is *depth*-1 if the graded theory itself is depth-1. In particular, this shows that the graded monads featuring in the local and global freshness semantics of regular non-deterministic nominal automata are depth-1.

In work on graded semantics over sets [30, 8, 11], partial orders [10], metric spaces [12], and relational structures [13], the somewhat technical-looking depth-1 property has played a key role in enabling characterization results for graded semantics in terms of modal logics and Spoiler-Duplicator games. We have illustrated this point by introducing a nominal variant of *graded equivalence games* [11], whose correctness depends on the depth-1 property, and instantiating the generic game to obtain game characterizations of trace equivalence in nominal transition systems under both local and global freshness semantics. Our system of graded nominal algebra plays a key role in graded equivalence games, which like in the set-based case can be viewed as playing out a graded algebraic proof. An important direction for future work is to extend further results in graded semantics to the nominal setting, notably results on characteristic modal logics [30, 8]. We expect that such results will, like the game characterization that we have already materialized in the present work, depend on the involved graded monads being depth-1. An additional point of future research is coverage of additional examples, such as fresh labelled transition systems [3].

Appendix A. Omitted Details and Proofs

In several places in the technical development, we need equivariance of derivability:

Lemma A.1. *Let $\pi \in \mathsf{Perm}(\mathbb{A})$, let $t, s \in \mathsf{Term}_{\Sigma,m}(Y)$, and let $\sigma\colon Y \to \mathsf{Term}_{\Sigma,l}(X)$ be derivably equivariant. Then*

(1) $X \vdash_{m+l} (\pi \cdot t)\sigma = \pi \cdot (t\sigma)$ is derivable.
(2) If $Y \vdash_m t = s$ is derivable, then $Y \vdash_m \pi \cdot t = \pi \cdot s$ is derivable.

Proof. Claim 1 is shown by straightforward induction on t; the base case $t = y \in Y$ is precisely derivable equivariance of σ. Claim 2 is shown by straightforward induction on derivations. In the case for rule (ax), one uses Claim 1. □

Details for Remark 5.4. Explicitly, the orbit-finite version (ax-f) of rule (ax) is the following:

$$\frac{X \vdash_l \pi \cdot \sigma(y) = \pi' \cdot \sigma(y') \quad \forall y, y' \in Y_0, \pi, \pi' \text{ s.t. } \pi \cdot y = \pi' \cdot y'}{X \vdash_{m+l} (\tau \cdot r)\sigma = (\tau \cdot s)\sigma} \ (*)$$

where Y_0 is the set of variables occurring in r or s plus (if necessary) one representative of every orbit of Y not mentioned in r, s, and the side condition $(*)$ requires that $Y \vdash_m r = s \in E, \tau \in \mathsf{Perm}(\mathbb{A})$ and that σ is a map $Y_0 \to \mathsf{Term}_{\Sigma,l}(X)$. Since Y is orbit-finite, Y_0 is finite. To show that (ax-f) is derivable, first extend σ to a substitution $\overline{\sigma}$ defined on all of Y, as follows. Since Y_0 contains at least one representative of every orbit of Y, we can pick, for every $y \in Y \setminus Y_0$, elements $\pi_y \in \mathsf{Perm}(\mathbb{A})$ and $z_y \in Y_0$ such that $y = \pi_y \cdot z_y$ in such a way that for $y \in Y_0$, we have $\pi_y = id$, $z_y = y$. Then, define

$$\bar{\sigma}(y) = \pi_y \cdot \sigma(z_y).$$

We are done once we show that $\bar{\sigma}$ is derivably equivariant, as this will allow deriving the conclusion by (ax). So let $\pi \in \mathsf{Perm}(\mathbb{A})$, $y \in Y$; we have to show that $\pi \cdot \bar{\sigma}(y) = \bar{\sigma}(\pi \cdot y)$ is derivable. The left-hand term is $\pi \cdot \pi_y \cdot \sigma(z_y)$, and the right-hand term is $\pi_{\pi \cdot y} \cdot \sigma(z_{\pi \cdot y})$. So the required equality is actually one of the premisses of (ax-f), since $\pi \cdot \pi_y \cdot z_y = \pi \cdot y = \pi_{\pi \cdot y} \cdot z_{\pi \cdot y}$. For the other direction, given a derivably equivariant substitution $\bar{\sigma}$, the premisses of (ax-f) are derivable for the restriction σ of $\bar{\sigma}$ to Y_0: Given $\pi, \pi' \in \mathsf{Perm}(\mathbb{A})$ and $y, y' \in Y_0$ such that $\pi \cdot y = \pi' \cdot y'$, we have the following chain of derivable or literal equalities over X:

$$\begin{aligned} &\pi \cdot \sigma(y) \\ &= \pi \cdot \bar{\sigma}(y) \\ &= \bar{\sigma}(\pi \cdot y) \\ &= \bar{\sigma}(\pi' \cdot y') \\ &= \pi' \cdot \bar{\sigma}(y') \\ &= \pi' \cdot \sigma(y'). \end{aligned}$$

Thus, we can alternatively apply (ax-f) to σ to obtain the conclusion.

We can thus, altogether, replace (ax) with (ax-f). The set of premisses of (ax-f) is clearly orbit-finite. We prove that we can replace (ax-f) with

an actual finitary version (ax-ff) where we only require one representative premise for each orbit, by simultaneous induction with Lemma A.1.(2) as indicated in Remark 5.4.

Proof of Theorem 5.7. The proof relies on the following property:

Lemma A.2 (Substitution Lemma)**.** *Let A be a nominal (Σ, n)-algebra, $\iota : Y \to A_k$ a valuation, and $\sigma : X \to \mathsf{Term}_{\Sigma,l}(X)$ a substitution. If $\kappa : X \to A_{k+l}$ with $\kappa(x) = [\![\sigma(x)]\!]^\iota_l$ is equivariant, then $[\![t\sigma]\!]^\iota_{m+l} = [\![t]\!]^\kappa_m$ for every term $t \in \mathsf{Term}_{\Sigma,m}(X)$ with $k + m + l \leq n$.*

Proof. By routine induction on t, unfolding the definition of term evaluation at each step. □

The proof of Theorem 5.7 then proceeds as follows:

Let A be a (T, n)-model, and suppose that $X \vdash_m t = u$ is derivable.

We will show that for every equivariant valuation $\iota : X \to A_k$ with $k+m \leq n$, it follows that $[\![t]\!]^\iota_m = [\![u]\!]^\iota_m$, by induction on the derivation of $X \vdash_m t = u$.

- Cases (refl), (symm), (trans), (cong) are straightforward.
- *For (ax):* We know that A satisfies $Y \vdash_{m'} r = s$ with $r, s \in \mathsf{Term}_{\Sigma,m'}(Y)$ (because it is a (T, n)-model), and that $t = (\tau r)\sigma, u = (\tau s)\sigma$ for a derivably equivariant substitution $\sigma : Y \to \mathsf{Term}_{\Sigma,l}(X)$ and a permutation $\tau \in \mathsf{Perm}(\mathbb{A})$.

  Let $\kappa : Y \to A_{k+l}$ be defined as $\kappa(x) = [\![\sigma(x)]\!]^\iota_l$. We will first show that κ is equivariant: For $\pi \in \mathsf{Perm}(\mathbb{A})$ and $x \in Y$, we know that $X \vdash_l \pi\sigma(x) = \sigma(\pi x)$ is derivable. It follows that

  $$\kappa(\pi x) = [\![\sigma(\pi x)]\!]^\iota_l = [\![\pi\sigma(x)]\!]^\iota_l = \pi[\![\sigma(x)]\!]^\iota_l = \pi\kappa(x).$$

  due to the induction hypothesis and equivariance. Thus, we conclude that

  $$\begin{aligned}
  [\![t]\!]^\iota_m &= [\![(\tau r)\sigma)]\!]^\iota_{m'+l} && \\
  &= [\![\tau r]\!]^\kappa_{m'} && \text{Lemma A.2} \\
  &= \tau[\![r]\!]^\kappa_{m'} && \text{equivariance} \\
  &= \tau[\![s]\!]^\kappa_{m'} && \text{model} \\
  &= [\![\tau s]\!]^\kappa_{m'} && \text{equivariance} \\
  &= [\![(\tau s)\sigma]\!]^\iota_{m'+l} && \text{Lemma A.2} \\
  &= [\![u]\!]^\iota_m. &&
  \end{aligned}$$

  where the fourth equation is due to A satisfying $Y \vdash_{m'} r = s$.
- *For (perm):* We know that $t = \nu a.f(t_1, \ldots, t_p)$ and $u = \nu b.f(u_1, \ldots, u_p)$, where $a \# u_i$ and $X \vdash_{m'} t_i = (a\ b)u_i$ is derivable for each $i \in \{1, \ldots, p\}$ with $m = m' + d(f)$. Since $a \neq b$ and, by equivariance of $[\![\cdot]\!]^\iota_{m'}$, $a \# [\![u_i]\!]^\iota_{m'}$ for every i, it follows by standard characterizations of α-equivalence [34] that

$$\langle a\rangle(a\ b)([\![u_1]\!]^\iota_{m'}, \ldots, [\![u_p]\!]^\iota_{m'}) = \langle b\rangle([\![u_1]\!]^\iota_{m'}, \ldots, [\![u_p]\!]^\iota_{m'}). \tag{A.1}$$

It then follows by computation that

$$\begin{aligned}
[\![t]\!]^\iota_m &= [\![\nu a.f(t_1,\dots,t_p)]\!]^\iota_{m'+d(f)} \\
&= f_{A,m'}(\langle a\rangle([\![t_1]\!]^\iota_{m'},\dots,[\![t_p]\!]^\iota_{m'})) && \text{semantics} \\
&= f_{A,m'}(\langle a\rangle([\![(a\ b)u_1]\!]^\iota_{m'},\dots,[\![(a\ b)u_p]\!]^\iota_{m'})) && \text{induction} \\
&= f_{A,m'}(\langle a\rangle((a\ b)[\![u_1]\!]^\iota_{m'},\dots,(a\ b)[\![u_p]\!]^\iota_{m'})) && \text{equivariance} \\
&= f_{A,m'}(\langle b\rangle([\![u_1]\!]^\iota_{m'},\dots,[\![u_p]\!]^\iota_{m'})) && \text{Equation (A.1)} \\
&= [\![\nu b.f(u_1,\dots,u_p)]\!]^\iota_{m'+d(f)} && \text{semantics} \\
&= [\![u]\!]^\iota_m.
\end{aligned}$$

□

Details for Theorem 5.8. We state and prove well-definedness and equivariance of the structure of FX, claimed inline in the main body:

Lemma A.3. *The definition of FX indeed yields a well-defined nominal (Σ,n)-algebra, in that $f_{FX,m}$ is well-defined and equivariant for all $f\in\Sigma$.*

Proof. To show well-definedness for $f\in\Sigma_0$, let $[t_i]_m=[u_i]_m$ for all $i\in\{1,\dots,p\}$. It follows from (cong) that $f(t_1,\dots,t_p)\sim f(u_1,\dots,u_p)$ and thus

$$\begin{aligned}
f_{FX,m}([t_1]_m,\dots,[t_p]_m) &= [f(t_1,\dots,t_p)]_{m+d(f)} \\
&= [f(u_1,\dots u_p)]_{m+d(f)} \\
&= f_{FX,m}([u_1]_m,\dots,[u_p]_m).
\end{aligned}$$

The case $f\in\Sigma_{\mathsf{f}}$ is analogous. For $f\in\Sigma_{\mathsf{b}}$, let $\langle a\rangle([t_1]_m,\dots,[t_p]_m)=\langle b\rangle([u_1]_m,\dots,[u_p]_m)$. By standard properties of α-equivalence [34], we only need to consider two cases:

If $a=b$, then $[t_i]_m=[u_i]_m$ for every $i\in\{1,\dots,p\}$ and the statement follows similar to the above case.

Otherwise, $a\#[u_i]_m$ and $[t_i]_m=(a\ b)[u_i]_m$ for every $i\in\{1,\dots,p\}$. By standard properties of equivariant equivalences [34], we know that there is some $\tilde{u}_i\in\mathsf{Term}_{\Sigma,m}(X)$ with $u_i\sim\tilde{u}_i$ and $a\#\tilde{u}_i$ for every i. Thus, $[t_i]_m=(a\ b)[\tilde{u}_i]_m=[(a\ b)\tilde{u}_i]_m$ for every i. We then have

$$\begin{aligned}
& f_{FX,m}(\langle a\rangle([t_1]_m,\dots,[t_p]_m)) \\
&= [\nu a.f(t_1,\dots,t_p)]_{m+d(f)} && \text{definition} \\
&= [\nu b.f(\tilde{u}_1,\dots,\tilde{u}_p)]_{m+d(f)} && \text{(perm)} \\
&= [\nu b.f(u_1,\dots,u_p)]_{m+d(f)} && \text{(cong)} \\
&= f_{FX,m}(\langle b\rangle([u_1]_m,\dots,[u_p]_m)) && \text{definition.}
\end{aligned}$$

For equivariance of $f\in\Sigma_0$, let $\pi\in\mathsf{Perm}(\mathbb{A})$. Then we have

$$\begin{aligned}
& f_{FX,m}(\pi[t_1]_m,\dots,\pi[t_p]_m) \\
&= f_{FX,m}([\pi t_1]_m,\dots,[\pi t_p]_m) && \text{equivariance} \\
&= [f(\pi t_1,\dots,\pi t_p)]_{m+d(f)} && \text{definition} \\
&= [\pi(f(t_1,\dots,t_p))]_{m+d(f)} && \text{definition} \\
&= \pi[f(t_1,\dots,t_p)]_{m+d(f)} && \text{equivariance} \\
&= \pi(f_{FX,m}([t_1]_m,\dots,[t_p]_m)) && \text{definition.}
\end{aligned}$$

The two remaining cases are again analogous.

□

Lemma A.4. *If $t \in \mathsf{Term}_{\Sigma,m}(X)$, then $[\![t]\!]_m^{\eta X} = [t]_m$.*

Proof. By induction on t.

- *For $t = x$ with $x \in X$:* By definition, we have $[\![t]\!]_m^{\eta X} = [\![x]\!]_0^{\eta X} = \eta_X(x) = [x]_0 = [t]_m$.
- *For $t = f(t_1, \dots, t_p)$ with $f/p \in \Sigma_0$, $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,m'}(X)$, and $m = m' + d(f)$:*

$$\begin{aligned}
[\![t]\!]_m^{\eta X} &= [\![f(t_1, \dots, t_p)]\!]_{m'+d(f)}^{\eta X} \\
&= f_{FX,m'}([\![t_1]\!]_{m'}^{\eta X}, \dots, [\![t_p]\!]_{m'}^{\eta X}) && \text{definition} \\
&= f_{FX,m'}([t_1]_{m'}, \dots, [t_p]_{m'}^{\eta X}) && \text{induction} \\
&= [f(t_1, \dots, t_p)]_{m'+d(f)} && \text{definition} \\
&= [t]_m.
\end{aligned}$$

- *For $t = a.f(t_1, \dots, t_p)$ and $t = \nu a.f(t_1, \dots, t_p)$:* Analogous to the above case.

□

Lemma A.5. *The graded nominal algebra FX is a (T, n)-model.*

Proof. Since we know that FX is a nominal (Σ, n)-algebra (by Lemma A.3), we only need to prove that FX satisfies every axiom in E.

Let $Y \vdash_m t = u \in E$ be an axiom and $\iota : Y \to (FX)_k$ an equivariant valuation. We will show that $[\![t]\!]_m^{\iota} = [\![u]\!]_m^{\iota}$.

Fix a splitting $u_k : (FX)_k \to \mathsf{Term}_{\Sigma,k}(X)$ of the canonical projection (in that $[-]_k \circ u_k = id$) and let $\sigma = u_k \circ \iota$. This definition of σ yields a derivably equivariant substitution: Let $x \in Y$ and $\pi \in \mathsf{Perm}(\mathbb{A})$. Then

$$\begin{aligned}
[\sigma(\pi x)]_k &= [u_k(\iota(\pi x))]_k && \text{by definition of } \sigma \\
&= \iota(\pi x) && \text{because } [-]_k \circ u_k = id \\
&= \pi\iota(x) && \text{because } \iota \text{ is equivariant} \\
&= \pi[u_k(\iota(x))]_k && \text{because } [-]_k \circ u_k = id \\
&= \pi[\sigma(x)]_k && \text{by definition of } \sigma \\
&= [\pi\sigma(x)]_k,
\end{aligned}$$

and thus $X \vdash_k \pi\sigma(x) = \sigma(\pi x)$ is derivable. By applying $(\mathrm{ax}_{t=u})$ with $\tau = id$, we then have $X \vdash_{m+k} t\sigma = u\sigma$.

Since we have

$$\begin{aligned}
\iota(x) &= [u_k(\iota(x))]_k && \text{because } [-]_k \circ u_k = id \\
&= [\sigma(x)]_k && \text{by definition of } \sigma \\
&= [\![\sigma(x)]\!]_k^{\eta X} && \text{by Lemma A.4,}
\end{aligned}$$

we can conclude that

$$\begin{aligned}
[\![t]\!]^{\iota}_m &= [\![t\sigma]\!]^{\eta X}_{m+k} && \text{by Lemma A.2}\\
&= [t\sigma]_{m+k} && \text{by Lemma A.4}\\
&= [u\sigma]_{m+k} && \text{because } X \vdash_{m+k} t\sigma = u\sigma\\
&= [\![u\sigma]\!]^{\eta X}_{m+k} && \text{by Lemma A.4}\\
&= [\![u]\!]^{\iota}_m && \text{by Lemma A.2.}
\end{aligned}$$

□

In the proof of freeness of FX (Theorem 5.8), we will in fact establish a full-blown adjoint situation with free functor, unit, and counit. We begin by explicitly defining the forgetful functor featuring in the statement of Theorem 5.8:

Definition A.6. We define the forgetful functor $G : \mathsf{Alg}(T, n) \to \mathbf{Nom}$ by

$$\begin{aligned}
GA &= A_0\\
Gh &= h_0.
\end{aligned}$$

We next extend F to a functor:

Definition A.7. We define the free functor $F\colon \mathbf{Nom} \to \mathsf{Alg}(T, n)$ with FX as given above and for every $f \in \mathbf{Nom}(X, Y)$,

$$(Ff)_i([t]_i) = [t\sigma_f]_i,$$

with the substitution $\sigma_f = (X \xrightarrow{f} Y \hookrightarrow \mathsf{Term}_{\Sigma,0}(Y))$.

Lemma A.8. *The above description of F is indeed a well-defined functor, in that*

(1) Ff is a well-defined morphism between (T, n)-algebras for every $f \in \mathbf{Nom}(X, Y)$,
(2) $Fid_X = id_{FX}$ for every nominal set X,
(3) $F(g \circ f) = Fg \circ Ff$ for every $f, g \in \mathbf{Nom}(X, Y)$.

Proof. We will prove the statements individually:

(1) Let $f : X \to Y$ be an equivariant function.

We define a valuation $\kappa : X \to (FY)_0$ with $\kappa(x) = [\![\sigma_f(x)]\!]^{\eta X}_0$. Note that κ is indeed equivariant: For every $\pi \in \mathsf{Perm}(\mathbb{A})$, we have $\kappa(\pi x) = [\![f(\pi x)]\!]^{\eta X}_0 = [\![\pi f(x)]\!]^{\eta X}_0 = \pi[\![f(x)]\!]^{\eta X}_0 = \pi\kappa(x)$ by equivariance of term evaluation and equivariance of f. It then follows that, for every $[t]_m \in (FX)_m$, we have

$$\begin{aligned}
(Ff)_m([t]_m) &= [t\sigma_f]_m && \text{definition}\\
&= [\![t\sigma_f]\!]^{\eta X}_m && \text{Lemma A.4}\\
&= [\![t]\!]^{\kappa}_m && \text{Lemma A.2.}
\end{aligned}$$

Thus, we can conclude well-definedness and equivariance from Theorem 5.7 and equivariance of term evaluation.

To show homomorphy w.r.t. $g/p \in \Sigma_0$, let $[t_1]_m, \dots, [t_p]_m \in (FX)_m$. Then

$$\begin{aligned}
&(Ff)_{m+d(g)}(g_{FX,m}([t_1]_m, \dots, [t_p]_m)) \\
&= (Ff)_{m+d(g)}([g(t_1, \dots, t_p)]_{m+d(g)}) && \text{definition} \\
&= [(g(t_1, \dots, t_p))\sigma_f] && \text{definition} \\
&= [g(t_1\sigma_f, \dots, t_p\sigma_f)] && \text{definition} \\
&= g_{FX,m}([t_1\sigma_f]_m, \dots, [t_p\sigma_f]_m) && \text{definition} \\
&= g_{FX,m}((Ff)_m([t_1]_m), \dots, (Ff)_m([t_p]_m)) && \text{definition.}
\end{aligned}$$

Homomorphy w.r.t. operations in Σ_{f} and Σ_{b} is shown similarly.

(2) Let $[t] \in (FX)_m$.

As seen in (1), we have $(Fid_X)_i([t]_m) = [\![t]\!]^{\kappa}_m$ for $\kappa : X \to (FX)$ with $\kappa(x) = [\![id_X(x)]\!]^{\eta_X}_0 = [\![x]\!]^{\eta_X}_0 = \eta_X(x)$. Thus, $\kappa = \eta_X$ and it follows from Lemma A.4 that $(Fid_X)_i([t]_m) = [t]_m = id([t]_m)$.

(3) Let $t \in \mathsf{Term}_{\Sigma,m}(X)$. We will show that $t\sigma_{g\circ f} = (t\sigma_f)\sigma_g$ by induction on t.

- *For $t = x$ with $x \in X$:* By definition, we have

$$\begin{aligned}
t\sigma_{g\circ f} &= \sigma_{g\circ f}(x) \\
&= g(f(x)) \\
&= \sigma_g(f(x)) \\
&= (f(x))\sigma_g \\
&= (\sigma_f(x))\sigma_g \\
&= (t\sigma_f)\sigma_g.
\end{aligned}$$

- *For $t = f(t_1, \dots, t_p)$ with $f/p \in \Sigma_0$, $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,m'}(X)$, and $m = m' + d(f)$:*

$$\begin{aligned}
t\sigma_{g\circ f} &= f(t_1\sigma_{g\circ f}, \dots, t_p\sigma_{g\circ f}) && \text{assumption} \\
&= f((t_1\sigma_f)\sigma_g, \dots, (t_p\sigma_f)\sigma_g) && \text{induction} \\
&= (f(t_1\sigma_f, \dots, t_p\sigma_f))\sigma_g && \text{definition} \\
&= ((f(t_1, \dots, t_p))\sigma_f)\sigma_g && \text{definition} \\
&= (t\sigma_f)\sigma_g && \text{assumption.}
\end{aligned}$$

- *For $t = a.f(t_1, \dots, t_p)$ and $t = \nu a.f(t_1, \dots, t_p)$:* Analogous to the previous case.

With this, we can conclude that

$$\begin{aligned}
(F(g \circ f))_m([t]_m) &= [t\sigma_{g\circ f}]_m \\
&= [(t\sigma_f)\sigma_g]_m \\
&= (Fg)_m([t\sigma_f]_m) \\
&= (Fg)_m((Ff)_m([t]_m) \\
&= (Fg \circ Ff)_m([t]_m).
\end{aligned}$$

□

With these data, we show, as announced, the following equivalent version of Theorem 5.8:

Theorem A.9. *The functor F is a left adjoint to the forgetful functor G with the unit η and the counit ε given by*

$$\varepsilon_A \colon FA_0 \to A,$$
$$(\varepsilon_A)_i([t]_i) = [\![t]\!]^{id}_i.$$

Proof. We will first show that ε_A is indeed a well-defined morphism between algebras for every (T, n)-model A:

- Well-definedness and equivariance follow directly from Theorem 5.7 and equivariance of term evaluation.
- To show homomorphy w.r.t. $f/p \in \Sigma_0$, let $[t_1]_m, \dots, [t_p]_m \in (FA_0)_m$. By definition, we have

$$\begin{aligned}
&(\varepsilon_A)_{m+d(f)}(f_{FA_0,m}([t_1]_m, \dots, [t_p]_m)) \\
&= (\varepsilon_A)_{m+d(f)}([f(t_1, \dots, t_p)]_{m+d(f)}) \\
&= [\![f(t_1, \dots, t_p)]\!]^{id}_{m+d(f)} \\
&= f_{A,m}([\![t_1]\!]^{id}_m, \dots, [\![t_p]\!]^{id}_m) \\
&= f_{A,m}((\varepsilon_A)_m([t_1]_m), \dots, (\varepsilon_A)_m([t_p]_m)).
\end{aligned}$$

  Homomorphy w.r.t. operations in Σ_{f} and Σ_{b} is shown similarly.

Next, we will show that η is indeed natural. So let X, Y be nominal sets, let $f \colon X \to Y$ be equivariant, and let $x \in X$. Let σ_f be defined as in Definition A.7. By definition, we have

$$\begin{aligned}
\eta_Y(f(x)) &= [f(x)]_m \\
&= [\sigma_f(x)]_m \\
&= [x\sigma_f]_m \\
&= (Ff)_m([x]) \\
&= (Ff)_m(\eta_X(x)).
\end{aligned}$$

To show that ε is also natural, let A, B be (T, n)-models, and let $h \colon A \to B$ be a homomorphism. We will show $(\varepsilon_B)_m((Fh_0)_m([t]_m)) = h_m((\varepsilon_A)_m([t]_m))$ for every $t \in \mathsf{Term}_{\Sigma,m}(A_0)$ by induction on t.

- *For $t = x$ with $x \in A_0$:* By definition, we have

$$\begin{aligned}
(\varepsilon_B)_0((Fh_0)_0([x]_0)) &= (\varepsilon_B)_0([x\sigma_{h_0}]_0) \\
&= (\varepsilon_B)_0([h_0(x)]_0) \\
&= [\![h_0(x)]\!]^{id}_0 \\
&= h_0(x) \\
&= h_0([\![x]\!]^{id}_0) \\
&= h_0((\varepsilon_A)_0([x])).
\end{aligned}$$

- *For $t = f(t_1, \dots, t_p)$ with $f/p \in \Sigma_0$, $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,m'}(X)$, and $m = m' + d(f)$:*

$$
\begin{aligned}
& (\varepsilon_B)_m((Fh_0)_m([f(t_1,\dots,t_p)]_m)) \\
= \quad & \text{(definition)} \\
& (\varepsilon_B)_m((Fh_0)_m(f_{FA_0,m'}([t_1]_{m'},\dots,[t_p]_{m'}))) \\
= \quad & \text{(homomorphy)} \\
& (\varepsilon_B)_m(f_{FB_0,m'}((Fh_0)_{m'}([t_1]_{m'}),\dots,(Fh_0)_{m'}([t_p]_{m'}))) \\
= \quad & \text{(homomorphy)} \\
& f_{B,m'}((\varepsilon_B)_{m'}((Fh_0)_{m'}([t_1]_{m'})),\dots,(\varepsilon_B)_{m'}((Fh_0)_{m'}([t_p]_{m'}))) \\
= \quad & \text{(inductive hypothesis)} \\
& f_{B,m'}(h_{m'}((\varepsilon_A)_{m'}([t_1]_{m'})),\dots,h_{m'}((\varepsilon_A)_{m'}([t_p]_{m'}))) \\
= \quad & \text{(homomorphy)} \\
& h_m(f_{A,m'}((\varepsilon_A)_{m'}([t_1]_{m'}),\dots,(\varepsilon_A)_{m'}([t_p]_{m'}))) \\
= \quad & \text{(homomorphy)} \\
& h_m((\varepsilon_A)_m(f_{FA_0,m'}([t_1]_{m'},\dots,[t_p]_{m'}))) \\
= \quad & \text{(definition)} \\
& h_m((\varepsilon_A)_m([f(t_1,\dots,t_p)]_m)).
\end{aligned}
$$

- *For $t = a.f(t_1, \dots, t_p)$ and $t = \nu a.f(t_1, \dots, t_p)$:* Analogous to the previous case.

Finally, we will prove the adjunction using the counit-unit equations

$$\mathsf{Id}_F = \varepsilon F \circ F\eta, \tag{A.2}$$

$$\mathsf{Id}_G = G\varepsilon \circ \eta G. \tag{A.3}$$

Let A be a (T, n)-model, and let X be a nominal set.

For (A.2), we will show $(\varepsilon_{FX})_m((F\eta_X)_m([t]_m)) = [t]_m$ for every $t \in \mathsf{Term}_{\Sigma,m}(X)$ by induction on t.

- *For $t = x$ with $x \in X$:*

$$
\begin{aligned}
(\varepsilon_{FX})_0((F\eta_X)_0([x]_0)) &= (\varepsilon_{FX})_0([x\sigma_{\eta_X}]_0) && \text{definition} \\
&= (\varepsilon_{FX})_0([[x]_0]_0) && \text{definition} \\
&= [\![[x]_0]\!]_0^{id} && \text{definition of } \varepsilon \\
&= [x]_0 && \text{definition.}
\end{aligned}
$$

- *For* $t = f(t_1, \dots, t_p)$ *with* $f/p \in \Sigma_0$, $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,m'}(X)$, *and* $m = m' + d(f)$*:*

$$\begin{aligned}
&(\varepsilon_{FX})_m((F\eta_X)_m([f(t_1,\dots,t_p)]_m))\\
=\quad &\text{(definition)}\\
&(\varepsilon_{FX})_m((F\eta_X)_m(f_{FX,m'}([t_1]_{m'},\dots,[t_p]_{m'})))\\
=\quad &\text{(homomorphy)}\\
&(\varepsilon_{FX})_m(f_{F((FX)_0),m'}((F\eta_X)_{m'}([t_1]_{m'}),\dots,(F\eta_X)_{m'}([t_p]_{m'})))\\
=\quad &\text{(homomorphy)}\\
&f_{FX,m'}((\varepsilon_{FX}\cdot F\eta_X)_{m'}([t_1]_{m'}),\dots,(\varepsilon_{FX}\cdot F\eta_X)_{m'}([t_p]_{m'}))\\
=\quad &\text{(inductive hypothesis)}\\
&f_{FX,m'}([t_1]_{m'},\dots[t_p]_{m'})\\
=\quad &\text{(definition)}\\
&[f(t_1,\dots,t_p)]_m.
\end{aligned}$$

- *For* $t = a.f(t_1, \dots, t_p)$ *and* $t = \nu a.f(t_1, \dots, t_p)$*:* Analogous to the previous case.

For (A.3), let $x \in A_0$. Then we have

$$\begin{aligned}
(\varepsilon_A)_0(\eta_{A_0}(x)) &= (\varepsilon_A)_0([x]_0) && \text{definition}\\
&= [\![x]\!]_0^{id} && \text{definition of } \varepsilon\\
&= x && \text{definition.}
\end{aligned}$$

With this, it follows that $(\varepsilon, \eta)\colon F \dashv G$. □

A.1. **Details for the Construction of $\mathbb{M}^T$.** We use the adjunction of Theorem A.9 for (T,ω)-models and the strict action $\star\colon \mathbb{N}_0 \times \mathsf{Alg}(T,\omega) \to \mathsf{Alg}(T,\omega)$ of the discrete monoidal category $(\mathbb{N}_0, +, 0)$ on $\mathsf{Alg}(T,\omega)$ defined in full as

$$\begin{gathered}
n \star ((A_i), (f_{A,i})) = ((A_{i+n}), (f_{A,i+n})),\\
n \star (f_i) = (f_{i+n}).
\end{gathered}$$

Using the sandwiching construction [16, Section 3], we obtain a graded monad $((M_n^T), \eta, (\mu^{nk}))$ where

$$\begin{aligned}
M_n^T X &= G(n \star FX) = (FX)_n = \mathsf{Term}_{\Sigma,n}(X)/{\sim},\\
M_n^T f([t]_n) &= G(n \star Ff)([t]_n) = (Ff)_n([t]_n) = [t\sigma_f]_n,
\end{aligned}$$

η being the unit of the adjunction and

$$\begin{aligned}
\mu_X^{nk}\colon M_n^T M_k^T X = G(n \star FG(k \star FX)) &\to G(n \star (k \star FX)) = M_{n+k}^T X,\\
\mu_X^{nk}([t]_n) = G(n \star \varepsilon_{k\star FX})([t]_n) &= (\varepsilon_{k\star FX})_n([t]_n) = [\![t]\!]_n^{id}.
\end{aligned}$$

Details for Theorem 5.13. We start with two lemmas clarifying in general that substitution and the monad multiplications relate in the expected manner:

Lemma A.10. *Let $T = (\Sigma, E)$ be a graded theory and $((M_n), \eta, (\mu^{nk}))$ be the graded monad induced by it. If $t \in \mathsf{Term}_{\Sigma,n}(Y)$ and $\sigma : Y \to \mathsf{Term}_{\Sigma,k}(X)$, then $[t\sigma]_{n+k} = \mu_X^{n,k}([t\bar{\sigma}]_n)$, where $\bar{\sigma} : Y \to \mathsf{Term}_{\Sigma,0}(M_kX)$ is defined as $\bar{\sigma}(x) = [\sigma(x)]_k$.*

Proof. By induction on t.

- *For $t = x$ with $x \in Y$:*

$$[\sigma(x)]_k = \bar{\sigma}(x) = [\![\bar{\sigma}(x)]\!]_0^{id} = \mu_X^{0,k}([\bar{\sigma}(x)]_0).$$

- *For $t = f(t_1, \dots, t_p)$ with $f/p \in \Sigma_0$, $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,m}(Y)$, and $1 = m + d(f)$:* By inductive hypothesis, we know that $[t_i\sigma]_{m+k} = \mu_X^{m,k}([t_i\bar{\sigma}]_m)$ for every i. It then follows that

$$\begin{aligned}
&[t\sigma]_{n+k} \\
&= [f(t_1\sigma, \dots, t_p\sigma)]_{n+k} && \text{definition} \\
&= f_{F(X),m+k}([t_1\sigma]_{m+k}, \dots, [t_p\sigma]_{m+k}) && \text{definition} \\
&= f_{F(X),m+k}(\mu_X^{m,k}([t_1\bar{\sigma}]_m), \dots, \mu_X^{m,k}([t_p\bar{\sigma}]_m)) && \text{induction} \\
&= f_{F(X),m+k}([\![t_1\bar{\sigma}]\!]_m^{id}, \dots, [\![t_p\bar{\sigma}]\!]_m^{id}) && \text{Section A.1} \\
&= [\![f(t_1\bar{\sigma}, \dots, t_p\bar{\sigma})]\!]_n^{id} && \text{definition} \\
&= \mu_X^{n,k}([t\bar{\sigma}]_n) && \text{Section A.1.}
\end{aligned}$$

- *For $t = a.f(t_1, \dots, t_p)$ and $t = \nu a.f(t_1, \dots, t_p)$:* Analogous to the previous case.

□

Lemma A.11. *Let $T = (\Sigma, E)$ be a graded theory and $((M_n), \eta, (\mu^{nk}))$ be the graded monad induced by it. If $\sigma : M_kX \to \mathsf{Term}_{\Sigma,k}(X)$ is a splitting (in that $[-]_k \circ \sigma = id$) and $t \in \mathsf{Term}_{\Sigma,n}(M_kX)$, then $\mu_X^{nk}([t]_n) = [t\sigma]_{n+k}$.*

Proof. First note that, for $[u]_k \in M_kX$, we have

$$\begin{aligned}
[\![\sigma([u]_k)]\!]_k^{\eta_X} &= [\sigma([u]_k)]_k && \text{Lemma A.4} \\
&= id([u]_k) && \text{because } [-]_k \circ \sigma = id.
\end{aligned}$$

It then follows that, for $t \in \mathsf{Term}_{\Sigma,n}(M_kX)$,

$$\begin{aligned}
\mu_X^{nk}([t]_n) &= [\![t]\!]_n^{id} && \text{Section A.1} \\
&= [\![t\sigma]\!]_{n+k}^{\eta_X} && \text{Lemma A.2 with } \kappa = id \\
&= [t\sigma]_{n+k} && \text{Lemma A.4.}
\end{aligned}$$

□

Moreover, we need the following fact on the interpretation of operations in induced graded monads:

Lemma A.12. *Let $T = (\Sigma, E)$ be a graded theory and $((M_n), \eta, (\mu^{nk}))$ be the graded monad induced by it. For $f/p \in \Sigma$ and $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,n}(X)$:*

$$[\eta.f(t_1, \dots, t_p)]_{n+d(f)} = \mu_X^{d(f),n}([\eta.f([t_1]_n, \dots, [t_p]_n)]_{d(f)}).$$

Proof. We will show the statement for $f/p \in \Sigma_0$ by simple computation:

$$\begin{aligned}
&\mu_X^{d(f),n}([f([t_1]_n,\dots,[t_p]_n)]_{d(f)} && \\
&= [\![f([t_1]_n,\dots,[t_p]_n)]\!]^{id}_{d(f)} && \text{Section A.1} \\
&= f_{F(X),n}([\![[t_1]_n]\!]^{id}_0,\dots,[\![[t_p]_n]\!]^{id}_0) && \text{definition} \\
&= f_{F(X),n}([t_1]_n,\dots,[t_p]_n) && \text{definition} \\
&= [f(t_1,\dots,t_p)]_{n+d(f)} && \text{definition.}
\end{aligned}$$

□

Fix a depth-1 graded theory $T = (\Sigma, E)$ and let $((M_n), \eta, (\mu^{nk}))$ be the graded monad induced by it as described in Section A.1.

The monad multiplication $\mu_X^{1,n}$ collapses depth-1 terms with embedded depth-n terms into terms of depth $1+n$. Conversely, we can split a depth-$1+n$ term into such a layered term, given that all operations are at most depth-1.

To do this, let $s^n_X : \mathsf{Term}_{\Sigma,1+n}(X) \to \mathsf{Term}_{\Sigma,1}(M_n X)$ be defined as

$$\begin{aligned}
s^n_X(f(t_1,\dots,t_p)) &= f(s^n_X(t_1),\dots,s^n_X(t_p)) && \text{if } f/p \in \Sigma_0 \text{ and } d(f) = 0, \\
s^n_X(f(t_1,\dots,t_p)) &= f([t_1]_n,\dots,[t_p]_n) && \text{if } f/p \in \Sigma_0 \text{ and } d(f) = 1,
\end{aligned}$$

and similarly for Σ_{f} and Σ_{b}.

Lemma A.13. *The function s^n_X is equivariant.*

Proof. Let $\pi \in \mathsf{Perm}(\mathbb{A})$. We will show $s^n_X(\pi t) = \pi s^n_X(t)$ for all $t \in \mathsf{Term}_{\Sigma,1+n}$ by induction on t, only considering cases where t has at least uniform depth 1.

- *For $t = f(t_1,\dots,t_p)$ with $f/p \in \Sigma_0$, $t_1,\dots,t_p \in \mathsf{Term}_{\Sigma,m}(X)$, and $1+n = m + d(f)$:* Since all operations are at most depth-1, we only have to consider two cases:

  If $d(f) = 0$, then

$$\begin{aligned}
s^n_X(\pi t) &= s^n_X(f(\pi t_1,\dots,\pi t_p)) && \text{definition} \\
&= f(s^n_X(\pi t_1),\dots,s^n_X(\pi t_p)) && \text{definition of } s^n_X \\
&= f(\pi s^n_X(t_1),\dots,\pi s^n_X(t_p)) && \text{induction} \\
&= \pi f(s^n_X(t_1),\dots,s^n_X(t_p)) && \text{definition} \\
&= \pi s^n_X(t) && \text{definition of } s^n_X.
\end{aligned}$$

  If $d(f) = 1$, then

$$\begin{aligned}
s^n_X(\pi t) &= s^n_X(f(\pi t_1,\dots,\pi t_p)) && \text{definition} \\
&= f([\pi t_1]_n,\dots,[\pi t_p]_n) && \text{definition of } s^n_X \\
&= f(\pi[t_1]_n,\dots,\pi[t_p]_n) && \text{equivariance} \\
&= \pi f([t_1]_n,\dots,[t_p]_n) && \text{definition} \\
&= \pi s^n_X(t) && \text{definition of } s^n_X.
\end{aligned}$$

- *For $t = a.f(t_1,\dots,t_p)$ and $t = \nu a.f(t_1,\dots,t_p)$:* Analogous to the previous case.

□

Of course, where we split exactly is chosen arbitrarily: Here, we use the outermost term inside a depth-1 operation, however this term could have more depth-0 operations we could include. We will see that this doesn't matter in the context of morphisms satisfying the universal property of a coequalizer. To do this, we will first show some general properties of such morphisms:

Lemma A.14. *Let (Q,h) be an M_0-algebra and $q : M_1M_nX \to Q$ a morphism between M_0-algebras such that $q \circ M_1\mu_X^{0,n} = q \circ \mu_{M_nX}^{1,0}$.*

(1) If $f/p \in \Sigma$ with $d(f) = 0$ and $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,1}(M_nX)$, then

$$q([\eta.f(t_1,\dots,t_p)]_1) = h([\eta.f(q([t_1]_1),\dots,q([t_p]_1))]_0).$$

(2) If $f/p \in \Sigma$ with $d(f) = 1$ and $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,0}(M_nX)$, then

$$q([\eta.f(t_1,\dots,t_p)]_1) = q([\eta.f(\mu_X^{0,n}([t_1]_0),\dots,\mu_X^{0,n}([t_p]_0))]_1).$$

(3) If $t \in \mathsf{Term}_{\Sigma,0}(Y)$ and $\sigma : Y \to \mathsf{Term}_{\Sigma,1+n}(X)$, then $q([s_X^n(t\sigma)]_1) = h([t\bar{\sigma}]_0)$ for $\bar{\sigma} : Y \to \mathsf{Term}_{\Sigma,0}(Q)$ with $\bar{\sigma}(x) = q([s_X^n(\sigma(x))]_1)$.

(4) If $t \in \mathsf{Term}_{\Sigma,1}(Y)$ and $\sigma : Y \to \mathsf{Term}_{\Sigma,n}(X)$, then $q([s_X^n(t\sigma)]_1) = q([t\bar{\sigma}]_1)$ for $\bar{\sigma} : Y \to \mathsf{Term}_{\Sigma,0}(M_nX)$ with $\bar{\sigma}(x) = [\sigma(x)]_n$.

Proof. Since q is a morphism between the M_0-algebras $(M_1M_nX, \mu_{M_nX}^{0,1})$ and (Q,h), we know that

$$h \circ M_0q = q \circ \mu_{M_nX}^{0,1}. \tag{A.4}$$

We will proceed to show each statement individually:

(1) We will only show the statement for Σ_0, the arguments for the others are completely analogous. By computation, we have

$$\begin{aligned}
& q([f(t_1,\dots,t_p)]_1) && \\
&= q(\mu_{M_nX}^{0,1}([f([t_1]_1,\dots,[t_p]_1)]_0)) && \text{Lemma A.12} \\
&= h(M_0q([f([t_1]_1,\dots,[t_p]_1)]_0)) && \text{Equation (A.4)} \\
&= h([f([t_1]_1,\dots,[t_p]_1)\sigma_q]_0) && \text{Section A.1} \\
&= h([f(q([t_1]_1),\dots,q([t_p]_1))]_0) && \text{definition.}
\end{aligned}$$

(2) Once again, we will only show the statement for Σ_0. By computation, we have

$$\begin{aligned}
& q([f(t_1,\dots,t_p)]_1) && \\
&= q(\mu_{M_nX}^{1,0}([f([t_1]_0,\dots,[t_p]_0)]_1)) && \text{Lemma A.12} \\
&= q(M_1\mu_X^{0,n}([f([t_1]_0,\dots,[t_p]_0)]_1)) && \text{assumption} \\
&= q([f([t_1]_0,\dots,[t_p]_0)\sigma_{\mu_X^{0,n}}]_1) && \text{Section A.1} \\
&= q([f(\mu_X^{0,n}([t_1]_0),\dots,\mu_X^{0,n}([t_p]_0))]_1) && \text{definition.}
\end{aligned}$$

(3) By induction on t.

- *For $t = x$ with $x \in Y$:*

$$\begin{aligned}
& q([s^n_X(t\sigma)]_1) \\
&= q([\![[s^n_X(t\sigma)]_1]\!]^{id}_0) && \text{definition} \\
&= q(\mu^{0,1}_{M_n X}([[s^n_X(t\sigma)]_1]_0)) && \text{Section A.1} \\
&= h(M_0 q([[s^n_X(t\sigma)]_1]_0)) && \text{Equation (A.4)} \\
&= h([q([s^n_X(\sigma(x))]_1)]_0) && \text{Section A.1} \\
&= h([\bar{\sigma}(x)]_0) && \text{definition of } \bar{\sigma} \\
&= h([t\bar{\sigma}]_0) && \text{definition.}
\end{aligned}$$

- *For $t = f(t_1, \dots, t_p)$ with $f/p \in \Sigma_0$:* We know that $d(f) = 0$ (because t is depth-0). It then follows that

$$\begin{aligned}
& q([s^n_X(t\sigma)]_1) \\
&= q([s^n_X(f(t_1\sigma, \dots, t_p\sigma))]_1) && \text{definition} \\
&= q([f(s^n_X(t_1\sigma), \dots, s^n_X(t_p\sigma))]_1) && \text{definition of } s^n_X \\
&= h([f(q([s^n_X(t_1\sigma)]_1), \dots, q([s^n_X(t_p\sigma)]_1))]_0) && (1) \\
&= h([f(h([t_1\bar{\sigma}]_0), \dots, h([t_p\bar{\sigma}]_0))]_0) && \text{induction} \\
&= h(M_0 h([f([t_1\bar{\sigma}]_0, \dots, [t_p\bar{\sigma}]_0)]_0)) && \text{Section A.1} \\
&= h(\mu^{0,0}_Q([f([t_1\bar{\sigma}]_0, \dots, [t_p\bar{\sigma}]_0)]_0)) && (\star) \\
&= h([f(t_1\bar{\sigma}, \dots, t_p\bar{\sigma})]_0) && \text{Lemma A.12} \\
&= h([t\bar{\sigma}]_0) && \text{definition,}
\end{aligned}$$

  where $(\star)$ follows from (Q, h) being an M_0-algebra.
- *For $t = a.f(t_1, \dots, t_p)$ and $t = \nu a.f(t_1, \dots, t_p)$:* Analogous to the previous case.

(4) By induction on t.

- *For $t = x$ with $x \in Y$:* This would imply that t is depth-0, but we know that t has uniform depth 1.
- *For $t = f(t_1, \dots, t_p)$ with $f/p \in \Sigma_0$:* Since t has uniform depth 1, we have to consider two cases:
  For $d(f) = 0$, we have

$$\begin{aligned}
& q([s^n_X(t\sigma)]_1) \\
&= q([s^n_X(f(t_1\sigma, \dots, t_p\sigma))]_1) && \text{definition} \\
&= q([f(s^n_X(t_1\sigma), \dots, s^n_X(t_p\sigma))]_1) && \text{definition of } s^n_X \\
&= h([f(q([s^n_X(t_1\sigma)]_1), \dots, q([s^n_X(t_p\sigma)]_1)]_0) && (1) \\
&= h([f(q([t_1\bar{\sigma}]_1), \dots, q([t_p\bar{\sigma}]_1)]_0) && \text{induction} \\
&= q([f(t_1\bar{\sigma}, \dots, t_p\bar{\sigma})]_1) && (1) \\
&= q([t\bar{\sigma}]_1) && \text{definition.}
\end{aligned}$$

For $d(f) = 1$, we have

$$\begin{aligned}
& q([s^n_X(t\sigma)]_1) \\
&= q([s^n_X(f(t_1\sigma, \dots, t_p\sigma))]_1) && \text{definition} \\
&= q([f([t_1\sigma]_n, \dots, [t_p\sigma]_n)]_1) && \text{definition of } s^n_X \\
&= q([f(\mu^{0,n}_X([t_1\bar{\sigma}]_0), \dots, \mu^{0,n}_X([t_p\bar{\sigma}]_0))]_1) && \text{Lemma A.10} \\
&= q([f(t_1\bar{\sigma}, \dots, t_p\bar{\sigma})]_1) && (2) \\
&= q([t\bar{\sigma}]_1) && \text{definition.}
\end{aligned}$$

- *For $t = a.f(t_1, \dots, t_p)$ and $t = \nu a.f(t_1, \dots, t_p)$:* Analogous to the previous case.

□

With this, we can finally show the coequalizer property for $\mu^{1,n}_X$ and thus the desired result.

Theorem A.15. *The graded monad induced by a depth-1 graded theory $T = (\Sigma, E)$ is depth-1.*

Proof. We show that (4.1) is a coequalizer. So let (Q, h) be an M_0-algebra and $q : M_1M_nX \to Q$ a morphism between M_0-algebras such that $q \circ M_1\mu^{0,n}_X = q \circ \mu^{1,0}_{M_nX}$. We now have to show that there exists a unique morphism $\bar{q} : M_{1+n}X \to Q$ with $\bar{q} \circ \mu^{1,n}_X = q$.

So let $\bar{q}$ be defined as $\bar{q}([t]_{1+n}) = q([s^n_X(t)]_1)$. We will show that this definition satisfies our requirements:

(1) If $\pi \in \mathsf{Perm}(\mathbb{A})$ and $t \in \mathsf{Term}_{\Sigma,1+n}(X)$, then

$$\begin{aligned}
\bar{q}([\pi t]_{1+n}) &= q([s^n_X(\pi t)]_1) && \text{definition of } \bar{q} \\
&= q([\pi s^n_X(t)]_1) && \text{Lemma A.13} \\
&= q(\pi[s^n_X(t)]_1) && \text{Lemma A.1} \\
&= \pi q([s^n_X(t)]_1) && \text{equivariance of } q \\
&= \pi\bar{q}([t]_{1+n}) && \text{definition of } \bar{q}.
\end{aligned}$$

Furthermore, it follows by equivariance of s^n_X, the canonical projection, and q, that

$$\mathsf{supp}(\bar{q}([t]_{1+n})) = \mathsf{supp}(q([s^n_X(t)]_1)) \subseteq \mathsf{supp}(t).$$

(2) $\bar{q}$ is well-defined; i.e., if $X \vdash_{1+n} t = u$ is derivable, then $\bar{q}([t]_{1+n}) = \bar{q}([u]_{1+n})$. We will show this by induction on the derivation.

- *For (refl):* This would imply that $t = x$ for some $x \in X$, however t has uniform depth $1 + n$, whereas x has uniform depth 0.
- *For (symm):* We know that $X \vdash_{1+n} u = t$ is derivable and, by inductive hypothesis, we have $\bar{q}([u]_{1+n}) = \bar{q}([t]_{1+n})$.
- *For (trans):* We know that $X \vdash_{1+n} t = v$ and $X \vdash_{1+n} v = u$ are derivable for some v. Thus, by inductive hypothesis, we have $\bar{q}([t]_{1+n}) = \bar{q}([v]_{1+n}) = \bar{q}([u]_{1+n})$.
- *For (cong):* First, let $f \in \Sigma_0$. We know that $t = f(t_1, \dots, t_p)$, $u = f(u_1, \dots, u_p)$, and $X \vdash_m t_i = u_i$ is derivable for all $i \in \{1, \dots, p\}$ with

$1 + n = m + d(f)$. Since the operations are at most depth-1, we only have to consider two cases:
If $d(f) = 0$, we have

$$\begin{aligned}
&\bar{q}([t]_{1+n}) \\
&= q([s^n_X(f(t_1, \dots, t_p))]_1) && \text{definition of } \bar{q} \\
&= q([f(s^n_X(t_1), \dots, s^n_X(t_p))]_1) && \text{definition of } s^n_X \\
&= h([f(q([s^n_X(t_1)]_1), \dots, q([s^n_X(t_p)]_1))]_0) && \text{Lemma A.14 (1)} \\
&= h([f(\bar{q}([t_1]_{1+n}), \dots, \bar{q}([t_p]_{1+n}))]_0) && \text{definition of } \bar{q},
\end{aligned}$$

and similarly

$$\bar{q}([u]_{1+n}) = h([f(\bar{q}([u_1]_{1+n}), \dots, \bar{q}([u_p]_{1+n})]_0).$$

The equality then follows by inductive hypothesis.
If $d(f) = 1$, then we have

$$\begin{aligned}
&\bar{q}([t]_{1+n}) \\
&= q([s^n_X(f(t_1, \dots, t_p))]_1) && \text{definition of } \bar{q} \\
&= q([f([t_1]_n, \dots, [t_p]_n)]_1) && \text{definition of } s^n_X \\
&= q([f([u_1]_n, \dots, [u_p]_n)]_1) && \text{assumption} \\
&= q([s^n_X(f(u_1, \dots, u_p))]_1) && \text{definition of } s^n_X \\
&= \bar{q}([u]_{1+n}) && \text{definition of } \bar{q}.
\end{aligned}$$

The cases for $f \in \Sigma_{\mathsf{f}}$ and $f \in \Sigma_{\mathsf{b}}$ are analogous.

- *For (ax):* We know that $Y \vdash_m r = s \in E$ with $t = (\tau r)\sigma$ and $u = (\tau s)\sigma$ for a derivably equivariant substitution $\sigma : Y \to \mathsf{Term}_{\Sigma,l}(X)$ and a permutation $\tau \in \mathsf{Perm}(\mathbb{A})$. Since we have assumed T to be depth-1, we have to consider two cases:
If r and s are depth-0, then $l = 1 + n$. In this case, let $\bar{\sigma} : Y \to \mathsf{Term}_{\Sigma,0}(Q)$ be defined as

$$\bar{\sigma}(x) = \bar{q}([\sigma(x)]_{1+n}) = q([s^n_X(\sigma(x))]_1).$$

It then follows by Lemma A.14 (4) that

$$\begin{aligned}
\bar{q}([t]_{1+n}) &= q([s^n_X((\tau r)\sigma)]_1) = h([(\tau r)\bar{\sigma}]_0) \quad \text{and} \\
\bar{q}([u]_{1+n}) &= q([s^n_X((\tau s)\sigma)]_1) = h([(\tau s)\bar{\sigma}]_0).
\end{aligned}$$

Finally, we can show $Q \vdash_0 (\tau r)\bar{\sigma} = (\tau s)\bar{\sigma}$ using $(\mathrm{ax}_{r=s})$ by showing that $\bar{\sigma}$ is derivably equivariant: Let $\pi \in \mathsf{Perm}(\mathbb{A})$ and $x \in Y$. It then follows by inductive hypothesis that

$$\begin{aligned}
\pi\bar{\sigma}(x) &= \pi\bar{q}([\sigma(x)]_{1+n}) && \text{definition of } \bar{\sigma} \\
&= \bar{q}([\pi\sigma(x)]_{1+n}) && (1) \\
&= \bar{q}([\sigma(\pi x)]_{1+n}) && \text{inductive hypothesis} \\
&= \bar{\sigma}(\pi x) && \text{definition of } \bar{\sigma}.
\end{aligned}$$

If r and s are depth-1, then $l = n$. In this case, let $\bar{\sigma} : Y \to \mathsf{Term}_{\Sigma,0}(M_n X)$ be defined as $\bar{\sigma}(x) = [\sigma(x)]_n$. It then follows by

Lemma A.14 (4) that

$$\bar{q}([t]_{1+n}) = q([s^n_X((\tau r)\sigma)]_1) = q([(\tau r)\bar{\sigma}]_1) \quad \text{and}$$
$$\bar{q}([u]_{1+n}) = q([s^n_X((\tau s)\sigma)]_1) = q([(\tau s)\bar{\sigma}]_1).$$

We can then derive $Q \vdash_0 (\tau r)\bar{\sigma} = (\tau s)\bar{\sigma}$ by once again showing that $\bar{\sigma}$ is derivably equivariant: Let $\pi \in \mathsf{Perm}(\mathbb{A})$ and $x \in Y$. Then $\pi\bar{\sigma}(x) = \pi[\sigma(x)]_n = [\pi\sigma(x)]_n = [\sigma(\pi x)]_n$ by assumption.

- *For (perm):* We know that $t = \nu a.f(t_1, \dots, t_p)$ and $u = \nu b.f(u_1, \dots, u_p)$ where $a \neq b$, $a \# u_i$, and $X \vdash_m t_i = (a\ b)u_i$ is derivable for all $i \in \{1, \dots, p\}$. Since the operations are at most depth-1, we only have to consider two cases:
  If $d(f) = 0$, we have

$$\bar{q}([t]_{1+n}) = h([\nu a.f(\bar{q}([t_1]_{1+n}), \dots, \bar{q}([t_p]_{1+n}))]_0) \quad \text{and}$$
$$\bar{q}([u]_{1+n}) = h([\nu b.f(\bar{q}([u_1]_{1+n}), \dots, \bar{q}([u_p]_{1+n}))]_0)$$

  analogously to (cong_f). Furthermore, for $i \in \{1, \dots, p\}$, we have

$$\begin{aligned} \bar{q}([t_i]_{1+n}) &= \bar{q}([(a\ b)u_i]_{1+n} && \text{inductive hypothesis} \\ &= (a\ b)\bar{q}([u_i]_{1+n}) && (1), \end{aligned}$$

  and also $a \notin \mathsf{supp}(\bar{q}([u_i]_{1+n})) \subseteq \mathsf{supp}(u_i)$ by (1). Thus, the equality follows by (perm).
  If $d(f) = 1$, we have

$$\bar{q}([t]_{1+n}) = \bar{q}([\nu a.f([t_1]_n, \dots, [t_p]_n)]_1) \quad \text{and}$$
$$\bar{q}([u]_{1+n}) = \bar{q}([\nu b.f([u_1]_n, \dots, [u_p]_n)]_1)$$

  analogously to (cong_f). Furthermore, for $i \in \{1, \dots, p\}$, we have $[t_i]_n = [(a\ b)u_i]_n = (a\ b)[u_i]_n$ and $a \notin \mathsf{supp}([u_i]_n) \subseteq \mathsf{supp}(u_i)$ by assumption. The result then follows by (perm).

(3) $\bar{q} : (M_{1+n}X, \mu^{0,1+n}_X) \to (Q, h)$ is indeed a morphism between M_0-algebras: First, note that equivariance follows from (1). We will also have to show that $h \circ M_0\bar{q} = \bar{q} \circ \mu^{0,1+n}_X$. So let $[t]_0 \in M_0M_{1+n}X$. Fix a splitting $\sigma : M_{1+n}X \to \mathsf{Term}_{\Sigma,1+n}(X)$, in that $[-]_{1+n} \circ \sigma = id$. Let $\bar{\sigma} : M_{1+n}X \to \mathsf{Term}_{\Sigma,0}(Q)$ be defined as $\bar{\sigma}(x) = \bar{q}(x) = \bar{q}([\sigma(x)]_{1+n}) = q([s^n_X(\sigma(x))]_1)$. It then follows that

$$\begin{aligned} h(M_0\bar{q}([t]_0)) &= h([t\bar{\sigma}]_0) && \text{Section A.1 because } \sigma_{\bar{q}} = \bar{\sigma} \\ &= q([s^n_X(t\sigma)]_1) && \text{Lemma A.14 (3)} \\ &= \bar{q}([t\sigma]_{1+n}) && \text{definition of } \bar{q} \\ &= \bar{q}(\mu^{0,1+n}_X([t]_0)) && \text{Lemma A.11.} \end{aligned}$$

(4) $\bar{q}$ satisfies the given property; i.e., $\bar{q} \circ \mu^{1,n}_X = q$. So let $[t]_1 \in M_1M_nX$. We fix a splitting $\sigma : M_nX \to \mathsf{Term}_{\Sigma,n}(X)$, in that $[-]_n \circ \sigma = id$. Let $\bar{\sigma} : M_nX \to \mathsf{Term}_{\Sigma,0}(M_nX)$ be defined as $\bar{\sigma}(x) = [\sigma(x)]_1 = x$.

It then follows that

$$\begin{aligned}
\bar{q}(\mu_X^{1,n}([t]_1)) &= \bar{q}([t\sigma]_{1+n}) && \text{Lemma A.11}\\
&= q([s_X^n(t\sigma)]_1) && \text{definition of } \bar{q}\\
&= q([t\bar{\sigma}]_1) && \text{Lemma A.14 (4)}\\
&= q(M_1 id([t]_1)) && \text{Section A.1 because } \sigma_{id} = \bar{\sigma}\\
&= q([t]_1) && \text{functoriality of } M_1.
\end{aligned}$$

(5) If there is another such morphism q', then $q'([t]_1) = \bar{q}([t]_1)$ for every $t \in \mathsf{Term}_{\Sigma,1}(M_n X)$.

So assume that $q' : M_{1+n}X \to Q$ is a morphism between M_0-algebras such that

$$q' \circ \mu_X^{1,n} = q. \tag{A.5}$$

We will proceed by induction on t, only considering cases where t has at least uniform depth 1.

- *For $t = f(t_1, \dots, t_p)$ with $f/p \in \Sigma_0$, $t_1, \dots, t_p \in \mathsf{Term}_{\Sigma,m}(X)$, and $1 + n = m + d(f)$:* Since all operations are at most depth-1, we only have to consider two cases:
  If $d(f) = 0$, then

$$\begin{aligned}
&q'([f(t_1, \dots, t_p)]_{1+n})\\
&= q'(\mu^{0,1+n}([f([t_1]_1, \dots, [t_p]_1)]_0)) && \text{Lemma A.12}\\
&= h(M_0 q'([f([t_1]_1, \dots, [t_p]_1)]_0)) && \text{homomorphy}\\
&= h([f(q'([t_1]_1), \dots, q'([t_p]_1)]_0) && \text{Section A.1.}
\end{aligned}$$

  With a similar argument, we have

$$\bar{q}([f(t_1, \dots, t_p)]_{1+n}) = h([f(\bar{q}([t_1]_1), \dots, \bar{q}([t_p]_1))]_0)$$

  The equality then follows by inductive hypothesis.
  If $d(f) = 1$, then

$$\begin{aligned}
&q'([f(t_1, \dots, t_p)]_{1+n})\\
&= q'(\mu_X^{1,n}([f([t_1]_n, \dots, [t_p]_n)]_1)) && \text{Lemma A.12}\\
&= q([f([t_1]_n, \dots, [t_p]_n)]_1) && \text{Equation (A.5)}\\
&= q([s_X^n(f(t_1, \dots, t_p)]_1) && \text{definition of } s_X^n\\
&= \bar{q}([f(t_1, \dots, t_p)]_{1+n}) && \text{definition of } \bar{q}.
\end{aligned}$$

- *For $t = a.f(t_1, \dots, t_p)$ and $t = \nu a.f(t_1, \dots, t_p)$:* Analogous to the previous case.

□

Details for Section 6. Write Σ^{tr} for the signature obtained from Σ^{bar} by removing 0 and $+$, and T^{tr} for the empty theory over Σ^{tr}. Then, pretraces correspond to (uniform-depth) terms in Σ^{tr}. In this view, we have the following characterization of α-equivalence of pretraces:

Lemma A.16. *Pretraces $w, v \in \overline{\mathbb{A}}^n \times X$ are α-equivalent iff $X \vdash_n w = v$ is derivable in T^{tr}.*

Proof. 'If': We will proceed by induction on the derivation of $X \vdash_n w = v$ to show a stronger statement: For all $u \in \overline{\mathbb{A}}^*$, $uw \equiv_\alpha uv$, especially for $u = \epsilon$.

- *For (refl):* We know that $w = v = x \in X$. By reflexivity, $uw = ux \equiv_\alpha ux = uv$.
- *For (symm):* We know that $X \vdash_n v = w$ is derivable and, by inductive hypothesis, $uv \equiv_\alpha uw$. The statement then follows by symmetry.
- *For (trans):* We know that $X \vdash_n w = x$ and $X \vdash_n x = v$ are derivable for some x. By inductive hypothesis and transitivity, we have $uw \equiv_\alpha ux \equiv_\alpha uv$.
- *For (cong):* We know that $w = aw'$ and $v = av'$ for some $a \in \overline{\mathbb{A}}$. By applying the inductive hypothesis with $\tilde{u} = ua$, it follows that $uw = \tilde{u}w' \equiv_\alpha \tilde{u}v' = uv$.
- *For (perm):* We know that $w = |aw'$ and $v = |bv'$, where $a \# v'$ and $X \vdash_m w' = (a\ b)v'$ is derivable. By applying the inductive hypothesis with $\tilde{u} = u|a$, we have $uw = u|aw' \equiv_\alpha u|a((a\ b)v')$. Furthermore, we have $\langle a\rangle((a\ b)v') = \langle b\rangle v'$ and, by definition, $u|a((a\ b)v') \equiv_\alpha u|bv'$. The statement then follows by transitivity of $\equiv_\alpha$.

'Only if': Assume that $w \equiv_\alpha w'$. Since derivable equality is an equivalence, we only have to consider the base case $w = u|av$ and $w' = u|a'v'$ with $\langle a\rangle v = \langle a'\rangle v'$ in $[\mathbb{A}]\,\overline{\mathbb{A}}^*$, where $u \in \overline{\mathbb{A}}^m$ and $v, v' \in \overline{\mathbb{A}}^k$.

If $a = a'$, then $v = v'$ and we are done because derivable equality is reflexive. Consider the case $a \neq a'$ with $a \# v'$ and $v = (a\ a')v'$. By reflexivity, we have $X \vdash_k v = (a\ a')v'$. By applying (perm), we also obtain $X \vdash_{k+1} |av = |a'v'$. Finally, by applying (cong) m times, we have $X \vdash_{m+1+k} u|av = u|a'v'$. □

We will also sometimes use the following standard characterization of $\equiv_\alpha$:

Lemma A.17. *Let $w, v \in \overline{\mathbb{A}}^* \times X$ be traces. Then $t \equiv_\alpha v$ is derivable iff either*

(1) $w = v = x$ for some $x \in X$,
(2) $w = aw'$ and $v = av'$ with $[w']_\alpha = [v']_\alpha$,
(3) $w = |aw'$ and $v = |bv'$ with $\langle a\rangle[w']_\alpha = \langle b\rangle[v']_\alpha$.

Proof. By Lemma A.16, we can use derivability in T^{tr} instead of the definition of $\equiv_\alpha$.
'If': We consider all of the given cases.

(1) We can derive $x \equiv_\alpha x$ by (refl).
(2) We know that $w' \equiv_\alpha v'$ and can thus derive $aw' \equiv_\alpha av'$ by (cong).
(3) If $a = b$, then we have $w' \equiv_\alpha v'$ and $aw' \equiv_\alpha av'$ follows by (cong).
If $a \neq b$, we have $a \# [v']_\alpha$ and $w \equiv_\alpha (a\ b) \cdot v'$. Since $a \# [v']_\alpha$, we can pick some u' with $u' \equiv_\alpha v'$ and $a \# u'$ (and, since $\equiv_\alpha$ is equivariant, also $(a\ b) \cdot v' \equiv_\alpha (a\ b) \cdot u'$). It then follows by (perm) and (cong) that $|aw' \equiv_\alpha |bu' \equiv_\alpha |bv'$.

'Only if': By induction on the derivation of $X \vdash_n w = v$.

- *For (refl):* We know that $w = v = x \in X$. Thus, (1) follows immediately.

- *For (symm):* We know that $X \vdash_n v = w$ is derivable. Since the conditions for all cases are symmetric, we know that one of them must hold by inductive hypothesis.
- *For (trans):* We know that, for some $u \in \overline{\mathbb{A}}^n \times X$, $X \vdash_n w = u$ and $X \vdash_n u = v$ are derivable. By inductive hypothesis for $w \equiv_\alpha u$, we only need to consider the following cases:
  (1) *If $w = u = x$ with $x \in X$:* By inversion on the inductive hypothesis for $u \equiv_\alpha v$, we know $u = v = x$. Thus, (1) follows immediately.
  (2) *If $w = aw'$ and $u = au'$ with $[w']_\alpha = [u']_\alpha$:* By inversion, we know that $v = av'$ with $[u']_\alpha = [v']_\alpha$. Thus, (2) holds.
  (3) *If $w = |aw'$ and $u = |bu'$ with $\langle a\rangle[w']_\alpha = \langle b\rangle[u']_\alpha$.* By inversion, we know that $v = |cv'$ with $\langle b\rangle[u']_\alpha = \langle c\rangle[v']_\alpha$. Thus, we have $\langle a\rangle[w']_\alpha = \langle c\rangle[v']_\alpha$ and (3) holds.
- *For (cong) with $w = aw'$ and $v = av'$:* We know that $w' \equiv_\alpha v'$ and thus (2) holds.
- *For (cong) with $w = |aw'$ and $v = |av'$:* We know that $w' \equiv_\alpha v'$. It follows that $\langle a\rangle[w']_\alpha = \langle a\rangle[v']_\alpha$ and thus (3) holds.
- *For (perm):* We know that $w = |aw'$ and $v = |bv'$, where $a \# v'$ and $w' \equiv_\alpha (a\ b)v'$. It then follows that $a \# [v']_\alpha$ and thus $\langle a\rangle[w']_\alpha = \langle b\rangle[v']_\alpha$. Thus, (3) holds.

□

Proof of Theorem 6.1. We define the following $(\Sigma^{\mathsf{bar}}, \omega)$-model A.

$$\begin{aligned} A_n = M_n^{\mathsf{bar}} X, \qquad & 0_{A,n} = \emptyset, \\ & +_{A,n}(S, V) = S \cup V, \\ & \mathsf{pre}_{A,n}(a, S) = \{[aw]_\alpha \mid [w]_\alpha \in S\}, \\ & \mathsf{abs}_{A,n}(\langle a\rangle S) = \{[|aw]_\alpha \mid [w]_\alpha \in S\} \end{aligned}$$

Note that abs is indeed well-defined: Assume $\langle a\rangle S = \langle a'\rangle S'$ with $a \neq a'$ and thus $a \# S'$ and $S' = (a\ a')S$. We will only show $\mathsf{abs}_{A,n}(\langle a\rangle S) \subseteq \mathsf{abs}_{A,n}(\langle a'\rangle S')$ as the other inclusion is completely analogous. So let $[|aw]_\alpha \in \mathsf{abs}_{A,n}(\langle a\rangle S)$ with $[w]_\alpha \in S$. By equivariance of $\equiv_\alpha$, we have $[(a\ a')w]_\alpha \in S'$ and, by definition, $[|a'((a\ a')w)]_\alpha \in \mathsf{abs}_{A,n}(\langle a'\rangle S')$. Furthermore, since S' is finite (and thus ufs), we have $a \notin \mathsf{supp}([w]_\alpha) \subseteq \mathsf{supp}(S')$. It then follows from Lemma A.17 that $|a'((a\ a')w) \equiv_\alpha |aw$.

Equivariance of $0_{A,n}$ and $+_{A,n}$ are trivial; for $\mathsf{pre}_{A,n}$ (and analogously for $\mathsf{abs}_{A,n}$) observe that

$$\begin{aligned} \pi\mathsf{pre}_{A,n}(a, S) &= \pi\{[aw]_\alpha \mid [w]_\alpha \in S\} \\ &= \{[\pi(a)\pi w]_\alpha \mid [w]_\alpha \in S\} \\ &= \{[\pi(a)w]_\alpha \mid [w]_\alpha \in \pi S\} \\ &= \mathsf{pre}_{A,n}(\pi a, \pi S) \end{aligned}$$

It is straightforward to see that A is a $(T^{\mathsf{bar}}, \omega)$-model. The join-semilattice axioms already hold by the structure of $\mathcal{P}_\omega$. Let $\kappa : X \to A_k$ be an equivariant valuation and $x, y \in X$. For (5.1) (and analogously for (6.1))

we have

$$\begin{aligned}\llbracket a(x+y)\rrbracket_1^\kappa &= \{[aw]_\alpha \mid [w]_\alpha \in \kappa(x) \cup \kappa(y)\}\\ &= \{[aw]_\alpha \mid [w]_\alpha \in \kappa(x)\} \cup \{[aw]_\alpha \mid [w]_\alpha \in \kappa(y)\}\\ &= \llbracket ax + ay\rrbracket_1^\kappa,\end{aligned}$$

and

$$\llbracket a0\rrbracket_n^\kappa = \{[aw]_\alpha \mid [w]_\alpha \in \emptyset\} = \emptyset = \llbracket 0\rrbracket_n^\kappa.$$

We define a map $\mathsf{nf}_n \colon \mathsf{Term}_{\Sigma,n}(X) \to \mathsf{Term}_{\Sigma,n}(X)$ to normalize terms by pushing the join-semilattice operations up (and disposing of 0) via the distributive equations in a derivable way, e.g. $\mathsf{nf}_n(a(x+y)) = \mathsf{nf}_n(ax) + \mathsf{nf}_n(ay)$. Normalized terms have the form

$$\sum\nolimits_{i\in I} v_i, \qquad v_i \in T^{\mathsf{tr}}$$

(where the empty sum is equal to 0). We also define the map $\llbracket\cdot\rrbracket_n^\iota \colon \mathsf{Term}_{\Sigma,n}(X) \to M_n^{\mathsf{bar}}X$ via the valuation $\iota(x) = \{[x]_\alpha\}$.

Combining $\llbracket\cdot\rrbracket_n^\iota$ and nf_n for $n \in \mathbb{N}$ induces a homomorphism $\iota^\sharp \colon FX \to A$, where $\iota_n^\sharp([t]_n) = \llbracket\cdot\rrbracket_n^\iota(\mathsf{nf}_n(t))$. Well-definedness is given by a canonical choice of input for $\llbracket\cdot\rrbracket_n^\iota$. Homomorphy is also preserved, as the equations to check are analogous to those of $(\Sigma^{\mathsf{bar}}, \omega)$-model A. The map $\iota^\sharp$ is surjective as for any set of (depth-n) pretraces there is a straightforward representation as a normalized term. It is also injective, as due to Lemma A.16 we have a correspondence between normalized terms modulo derivable equivalence and sets of α-equivalent pretraces. As such, $\iota^\sharp([t]_n) = \iota^\sharp([s]_n)$ results in the same set of pretraces, which implies they share the same derivable normalized term, so $[t]_n = [s]_n$. □

Proof of Theorem 6.2. First we prove the following statement

$$\gamma^{(n)}(x) = \{[w]_\alpha \mid [wy]_\alpha \in (\overline{\mathbb{A}}^n \times X)/{\equiv_\alpha},\ x \xrightarrow{w} y\} \qquad (n > 0)$$

Through induction on n, for $a \in \overline{\mathbb{A}}$ in the base case

$$\begin{aligned}&[a]_\alpha \in \gamma^{(1)}(x)\\ \Leftrightarrow\ &[a]_\alpha \in (\mu_X^{1,0} \cdot M_1\gamma^{(0)} \cdot \gamma)(x) && \text{definition}\\ \Leftrightarrow\ &[a]_\alpha \in (\mu_X^{1,0} \cdot M_1M_0! \cdot M_1\eta_X \cdot \gamma)(x) && \text{definition}\\ \Leftrightarrow\ &[a]_\alpha \in (M_1! \cdot \gamma)(x) && \text{naturality, monad laws}\\ \Leftrightarrow\ &\exists y.\, [ay]_\alpha \in \gamma(x) && \text{definition}\\ \Leftrightarrow\ &\exists y, a', y'.\, ay \equiv_\alpha a'y' \text{ and } x \xrightarrow{a'} y' && \text{definition}\end{aligned}$$

and in the induction step

$$\begin{aligned}
&[aw]_\alpha \in \gamma^{(n)}(x) \\
\Leftrightarrow\ &[aw]_\alpha \in (\mu_X^{1,n-1} \cdot M_1\gamma^{(n-1)} \cdot \gamma)(x) && \text{definition} \\
\Leftrightarrow\ &[aw]_\alpha \in \{[bv]_\alpha \mid [by]_\alpha \in \gamma(x),\ [v]_\alpha \in \gamma^{(n-1)}(y)\} && \text{definition} \\
\Leftrightarrow\ &\exists y.\, [ay]_\alpha \in \gamma(x),\ [w]_\alpha \in \gamma^{(n-1)}(y) && \text{definition} \\
\Leftrightarrow\ &\exists y, z.\, x \xrightarrow{a} y,\ y \xrightarrow{w} z && \text{induction} \\
\Leftrightarrow\ &\exists z.\, x \xrightarrow{aw} z && \text{definition}
\end{aligned}$$

where the last two equivalences hold up to α-equivalence (c.f. the last line of the base case, omitted in favour of readability).

To prove the theorem, it suffices to show the following as a consequence of injectivity of the global freshness operator N.

$$L(A, x) = L(A, y) \Leftrightarrow \forall n.\, \gamma^{(n)}(x) = \gamma^{(n)}(y)$$

This follows from straightforward induction on n via the previously shown equality. □

Proof of Theorem 6.3. We give the outline of the proof, relying on several key lemmas (Lemmas A.18 to A.20) for which we give proofs, in turn involving dedicated lemmas, in the subsequent sections.

The local freshness semantics is computed from the bar language semantics by applying the D operator, which takes all α-equivalent representatives of bar strings and then drops the bars (Section 3). We can extend D to pretraces and sets of pretraces in the evident manner. In order to show coincidence of the graded semantics $(\mathbb{M}^{\mathsf{loc}}, \theta_1 \cdot \beta)$ with local freshness semantics, it then suffices to show that D and θ make the same identifications on $M_n^{\mathsf{bar}}X$.

To show this, we define a $(T^{\mathsf{drop}}, \omega)$-model $F'X$ (despite the notation, F' will not be functorial) into which the free model FX embeds along a subalgebra inclusion e. We will then show that, for the canonical transformation

$$\kappa_n \colon M_n^{\mathsf{loc}} \to M_n^{\mathsf{drop}},$$

$e_n \cdot \kappa_n \cdot \theta_n$ makes the same identifications as D. Since we will show that $e_n \cdot \kappa_n$ is already injective, this also means that θ makes the same identifications as D.

As the first step in this programme, we take $F'X$ to have carriers

$$(F'X)_0 = \mathcal{P}_\omega(X), \qquad (F'X)_{n+1} = \{f \colon \mathbb{A} \to_{\mathsf{fs}} (F'X)_n \mid (\star)\},$$

(recall that $\mathbb{A} \to_{\mathsf{fs}} (F'X)_n$ is the set of finitely supported maps from $\mathbb{A}$ to $(F'X)_n$) where $(\star)$ requires that, for all $a \in \mathbb{A}$ and $c \# (f, a)$, we have

$$f(a) = f(a) + (a\ c) \cdot \mathsf{res}_{F'X,n}(\langle a \rangle f(c)). \tag{A.6}$$

This ensures a form of closure under restriction that allows unblocking blocked α-renamings in the same spirit as the method of name dropping [37]. We use the usual (nominal) join-semilattice structure at depth 0 (e.g. $+$ is union) and let $0_{F'X,m+1}$, $+_{F'X,m+1}$ be given by

$$0_{F'X,m+1}(b) = 0 \quad \text{and} \quad +_{F'X,m+1}(f_1, f_2)(b) = f_1(b) + f_2(b)$$

Furthermore, we add name restriction by

$$\mathsf{res}_{F'X,0}(\langle a\rangle S) = \{x \in S \mid a \# x\},$$
$$\mathsf{res}_{F'X,m+1}(\langle a\rangle f)(b) = \mathsf{res}_{F'X,m}(\langle c\rangle((a\ c)\cdot f)(b)) \text{ for } c \# (f,a,b),$$

and interpret the depth-1 operations by

$$\mathsf{pre}_{F'X,m}(a,x)(b) = \begin{cases} x & a = b \\ 0 & \text{otherwise, and} \end{cases}$$
$$\mathsf{abs}_{F'X,m}(\langle a\rangle x)(b) = \mathsf{res}_{F'X,m}(\langle c\rangle(b\ c)\cdot(a\ c)\cdot x) \text{ for } c \# (x,a,b).$$

The above definition of the restriction operation matches that of the name restriction operation defined on finitely supported functions given in Pitts [34], while the abstraction operation resembles the definition of α-equivalence. The requirement $(\star)$ is necessary to ensure that the traces for an atom in the support include all of the relevant traces for fresh atoms, which is used to prove axiom (6.4) (2).

Lemma A.18. *The $(\Sigma^{\mathsf{drop}},\omega)$-algebra $F'X$ is a $(T^{\mathsf{drop}},\omega)$-model.*

By freeness, we thus obtain a morphism $e\colon FX \to F'X$. Furthermore, by the obvious isomorphism $(\mathcal{P}_{\mathsf{fs}}(Y)^X)_{\mathsf{fs}} \cong \mathcal{P}_{\mathsf{fs}}(X\times Y)$ for nominal sets X, Y, we can define an injective operator $\overline{(-)}\colon (F'X)_n \to \mathcal{P}_{\mathsf{fs}}(\mathbb{A}^n \times X)$ that "flattens" our carrier set into a set of pretraces.

Lemma A.19. *For $S \in M_n^{\mathsf{bar}}X$, we have $\overline{e_n(\kappa_n(\theta_n(S)))} = D(S)$.*

The remaining last step in our programme is injectivity of $e_n \cdot \kappa_n$:

Lemma A.20. *For all components $e_n : (FX)_n \to (F'X)_n$ of the $(\Sigma^{\mathsf{drop}},\omega)$-homomorphism $e\colon FX \to F'X$, $e_n \cdot \kappa_n$ is injective.*

Details for Lemma A.18. First, note that all operations preserve the condition stated in (A.6) by the observation that the condition does not depend on the choice of c:

Lemma A.21. *Let X be a nominal set with a name restriction operation $[\mathbb{A}]X \to X, \langle a\rangle x \mapsto a \setminus x$, $f\colon \mathbb{A} \to_{\mathsf{fs}} X$, and $a, b \in \mathbb{A}$. For $c, c' \# (f,a,b)$, we have $c \setminus ((a\ c)\cdot f)(b) = c' \setminus ((a\ c')\cdot f)(b)$.*

Proof. If $b = a$, it follows by computation that

$$\begin{aligned}
& c \setminus ((a\ c)\cdot f)(a) && \\
&= c \setminus ((a\ c)\cdot(c\ c')\cdot f)(a) && \text{because } c, c' \# f \\
&= c \setminus (a\ c)\cdot(c\ c')\cdot f(c') && \text{definition} \\
&= (a\ c)\cdot(c\ c')\cdot a \setminus f(c') && \text{equivariance} \\
&= (a\ c')\cdot a \setminus f(c') && \text{by } (\star) \\
&= c' \setminus (a\ c')\cdot f(c') && \text{equivariance} \\
&= c' \setminus ((a\ c')\cdot f)(a) && \text{definition,}
\end{aligned}$$

where $(\star)$ follows by the fact that $(a\ c)\cdot(c\ c')\cdot d = (a\ c')\cdot d$ for all $d \in \mathsf{supp}(a \setminus f(c'))$.

If $b \neq a$, we have

$$\begin{aligned}
& c \setminus ((a\ c) \cdot f)(b) \\
&= c \setminus (a\ c) \cdot f(b) && \text{definition} \\
&= (a\ c) \cdot a \setminus f(b) && \text{equivariance} \\
&= a \setminus f(b) && \text{because } a, c \# (a \setminus f(b)),
\end{aligned}$$

and similarly, $d \setminus ((a\ d) \cdot f)(b) = a \setminus f(b)$. □

One can then verify that the definitions of abs and res also do not depend on the choice c.

Most of the depth-0 axioms are easy to verify, we will restrict ourselves to the most intersting cases for $(F'X)_{n+1}$:

- For (6.4): First, we have to show that $\mathsf{res}_{F'X,n+1}(\langle a\rangle f + g) = \mathsf{res}_{F'X,n+1}(\langle a\rangle f) + \mathsf{res}_{F'X,n+1}(\langle a\rangle g)$ for all a, f, g.
  Let $b \in \mathbb{A}$ and pick a $c \in \mathbb{A} \setminus \mathsf{supp}(f, g, a, b)$. We then have

$$\begin{aligned}
& \mathsf{res}_{F'X,n+1}(\langle a\rangle f + g)(b) \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle((a\ c) \cdot (f + g))(b)) && \text{definition} \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle(a\ c) \cdot (f + g)((a\ c) \cdot b)) && \text{definition} \\
&= (a\ c) \cdot \mathsf{res}_{F'X,n}(\langle c\rangle f((a\ c) \cdot b) + g((a\ c) \cdot b)) && \text{equivariance} \\
&= (a\ c) \cdot \mathsf{res}_{F'X,n}(\langle c\rangle f((a\ c) \cdot b)) + \\
&\quad (a\ c) \cdot \mathsf{res}_{F'X,n}(\langle c\rangle g((a\ c) \cdot b)) && (6.4) \text{ for } (F'X)_n \\
&= \mathsf{res}_{F'X,n+1}(\langle a\rangle f)(b) + \mathsf{res}_{F'X,n+1}(\langle a\rangle g)(b) && \text{definition.}
\end{aligned}$$

  We also have to show that $f = f + \mathsf{res}_{F'X,n+1}(\langle a\rangle f)$ for all a, f. So let $b \in \mathbb{A}$.
  If $a = b$, this follows directly from the definition of $(F'X)_{n+1}$.
  If $a \neq b$, pick $c \in \mathbb{A} \setminus \mathsf{supp}(f, a, b)$.

$$\begin{aligned}
f(b) &= f(b) + \mathsf{res}_{F'X,n}(\langle a\rangle f(b)) && (6.4) \text{ for } (F'X)_n \\
&= f(b) + (a\ c) \cdot \mathsf{res}_{F'X,n}(\langle a\rangle f(b)) && \text{because } a, c \# \langle a\rangle f(b) \\
&= f(b) + \mathsf{res}_{F'X,n}(\langle c\rangle(a\ c) \cdot f(b)) && \text{equivariance} \\
&= f(b) + \mathsf{res}_{F'X,n}(\langle c\rangle((a\ c) \cdot f)(b)) && \text{because } (a\ c) \cdot b = b \\
&= f(b) + \mathsf{res}_{F'X,n+1}(\langle a\rangle f)(b) && \text{definition.}
\end{aligned}$$

For the depth-1 axioms, we will use the following lemma:

Lemma A.22. *For $a \in \mathbb{A}$ and $x \in (F'X)_n$, we have $\mathsf{abs}_{F'X,n}(\langle a\rangle x)(a) = x$.*

Proof. For sufficiently fresh c, we have

$$\begin{aligned}
& \mathsf{abs}_{F'X,n}(\langle a\rangle x)(a) \\
&= \mathsf{res}_{F'X,n'}(\langle c\rangle(a\ c) \cdot (a\ c) \cdot x) && \text{definition} \\
&= \mathsf{res}_{F'X,n'}(\langle c\rangle x) && \text{because } (a\ c) \cdot (a\ c) = id \\
&= x && \text{axioms, because } c \# x.
\end{aligned}$$

□

Again, we will restrict ourselves to the most interesting axioms:

- For (6.5) (2): We have to show that $\mathsf{res}_{F'X,n+1}(\langle a\rangle \mathsf{pre}_{F'X,n}(a,x)) = 0_{F'X,n+1}$ for all a, x. So let $b \in \mathbb{A}$.
  Pick $c \in \mathbb{A} \setminus \mathsf{supp}(x,a,b)$. We then have $(a\ c)\cdot b \neq a$ by case distinction on whether $b = a$.
  It then follows immediately that

$$
\begin{aligned}
&\mathsf{res}_{F'X,n+1}(\langle a\rangle \mathsf{pre}_{F'X,n}(a,x))(b) \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle (a\ c)\cdot \mathsf{pre}_{F'X,n}(a,x))((a\ c)\cdot b) && \text{definition} \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle (a\ c)\cdot 0_{F'X,n}) && \text{because } (a\ c)\cdot b \neq a \\
&= 0_{F'X,n} = 0_{F'X,n+1}(b) && \text{definition.}
\end{aligned}
$$

- For (6.6): We have to show that $\mathsf{abs}_{F'X,n}(\langle a\rangle \mathsf{res}_{F'X,n}(\langle b\rangle x)) = \mathsf{res}_{F'X,n+1}(\langle b\rangle \mathsf{abs}_{F'X,n}(\langle a\rangle x))$ for all a, b, x with $a \neq b$. So let $e \in \mathbb{A}$.
  If $b = e$, pick $c \in \mathbb{A} \setminus \mathsf{supp}(x,a,b)$ and $d \in \mathbb{A} \setminus \mathsf{supp}(x,a,b,c)$. It then follows that

$$
\begin{aligned}
&\mathsf{abs}_{F'X,n}(\langle a\rangle \mathsf{res}_{F'X,n}(\langle b\rangle x))(b) \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle (b\ c)\cdot (a\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle x)) && \text{definition} \\
&= (b\ c)\cdot (a\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle \mathsf{res}_{F'X,n}(\langle b\rangle x)) && \text{equivariance} \\
&= (b\ c)\cdot (a\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle x) && \text{axioms} \\
&= (b\ c)\cdot (a\ c)\cdot (c\ d)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle x) && \text{because } c, d \# x \\
&= (b\ c)\cdot (c\ d)\cdot (a\ d)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle x) && \text{computation} \\
&= (b\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle (c\ d)\cdot (a\ d)\cdot x) && \text{equivariance} \\
&= (b\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle (c\ d)\cdot \mathsf{res}_{F'X,n}(\langle c\rangle (a\ d)\cdot x)) && \text{axioms} \\
&= (b\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle (c\ d)\cdot \mathsf{abs}_{F'X,n}(\langle a\rangle x)(c)) && \text{definition} \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle (b\ c)\cdot \mathsf{abs}_{F'X,n}(\langle a\rangle x)(c)) && \text{equivariance} \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle ((b\ c)\cdot \mathsf{abs}_{F'X,n}(\langle a\rangle x))(b)) && \text{definition} \\
&= \mathsf{res}_{F'X,n+1}(\mathsf{abs}_{F'X,n}(\langle a\rangle x))(b) && \text{definition.}
\end{aligned}
$$

  If $b \neq e$, pick $c \in \mathbb{A} \setminus \mathsf{supp}(x,a,b,e)$. We then have

$$
\begin{aligned}
&\mathsf{abs}_{F'X,n}(\langle a\rangle \mathsf{res}_{F'X,n}(\langle b\rangle x))(e) \\
&= \mathsf{res}_{F'X,n}(\langle c\rangle (e\ c)\cdot (a\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle x)) && \text{definition} \\
&= (e\ c)\cdot \mathsf{res}_{F'X,n}(\langle e\rangle \mathsf{res}_{F'X,n}(\langle b\rangle (a\ c)\cdot x)) && \text{equivariance} \\
&= (e\ c)\cdot \mathsf{res}_{F'X,n}(\langle b\rangle \mathsf{res}_{F'X,n}(\langle e\rangle (a\ c)\cdot x)) && \text{axioms} \\
&= \mathsf{res}_{F'X,n}(\langle b\rangle (e\ c)\cdot \mathsf{res}_{F'X,n}(\langle e\rangle (a\ c)\cdot x)) && \text{equivariance} \\
&= \mathsf{res}_{F'X,n}(\langle b\rangle \mathsf{abs}_{F'X,n}(\langle a\rangle x)(e)) && \text{definition} \\
&= \mathsf{res}_{F'X,n+1}(\langle b\rangle \mathsf{abs}_{F'X,n}(\langle a\rangle x))(e) && \text{because } b \neq e.
\end{aligned}
$$

- For (6.2): We have to show that $\mathsf{abs}_{F'X,n}(\langle a\rangle x) = \mathsf{abs}_{F'X,n}(\langle a\rangle x) + \mathsf{pre}_{F'X,n}(a,x)$.
  This follows directly from the definition of pre and Lemma A.22.

Proof of Lemma A.19. Using the characterization of $\equiv_\alpha$ given in Lemma A.17, we first obtain the following properties of D:

Lemma A.23. *Let $t' \in \overline{\mathbb{A}}^n \times X$, $w' \in \mathbb{A}^n \times X$, and $a, b \in \mathbb{A}$.*

(1) $bw' \in D([at']_\alpha)$ *iff* $w' \in D([t']_\alpha)$ *and* $a = b$.
(2) $aw' \in D([\mathsf{l}at']_\alpha)$ *iff* $w' \in D([t']_\alpha)$.
(3) If $a \neq b$, *then* $bw' \in D([\mathsf{l}at']_\alpha)$ *iff* $b \# [t']_\alpha$ *and* $w' \in D([(a\ b)t']_\alpha)$.

Proof. We will show each statement individually.

(1) *'If':* Assume $w' \in D([t']_\alpha)$ and $a = b$.
Then there exists a $\tilde{w}' \in [t']_\alpha$ with $\mathsf{ub}(\tilde{w}') = w'$. Since $\tilde{w}' \equiv_\alpha t'$, we have $a\tilde{w}' \equiv_\alpha at'$ by A.17 (2). Thus, it follows that $aw' = (a, \mathsf{ub}(\tilde{w}')) = \mathsf{ub}(a\tilde{w}') \in D([at']_\alpha)$.
'Only if': Assume $bw' \in D([at']_\alpha)$.
Then there exists a $\tilde{w} \in [at']_\alpha$ with $\mathsf{ub}(\tilde{w}) = bw'$. Since $\tilde{w} \equiv_\alpha at'$, we know from A.17 that $\tilde{w} = a\tilde{w}'$ for some $\tilde{w}' \in \overline{\mathbb{A}}^n \times X$ with $\tilde{w}' \equiv_\alpha t'$. Thus, we have $\tilde{w}' \in [t']_\alpha$. Furthermore, we have $(a, \mathsf{ub}(\tilde{w}')) = \mathsf{ub}(a\tilde{w}') = \mathsf{ub}(\tilde{w}) = bw'$, concluding that $a = b$ and $w' = \mathsf{ub}(\tilde{w}') \in D([t']_\alpha)$.

(2) Analogous to the above proof.

(3) *'If':* Assume $w' \in D([(a\ b)t']_\alpha$ and $b \# [t']_\alpha$.
Then there exists a $\tilde{w}' \in [(a\ b)t']_\alpha$ with $\mathsf{ub}(\tilde{w}') = w'$. Since $\tilde{w}' \equiv_\alpha (a\ b)t'$ and $b \# [t']_\alpha$ with $a \neq b$, it follows from A.17 (3) that $\mathsf{l}b\tilde{w}' \equiv_\alpha \mathsf{l}at'$. Thus, we have

$$bw' = (b, \mathsf{ub}(\tilde{w}')) = \mathsf{ub}(\mathsf{l}b\tilde{w}') \in D([\mathsf{l}at']_\alpha).$$

'Only if': Assume $bw' \in D([\mathsf{l}at']_\alpha)$.
Then there exists some $\tilde{w} \in [\mathsf{l}at']_\alpha$ with $\mathsf{ub}(\tilde{w}) = bw'$. Since $\tilde{w} \equiv_\alpha \mathsf{l}at'$, we know from A.17 that $\tilde{w} = \mathsf{l}c\tilde{w}'$ for some $\tilde{w}' \in \overline{\mathbb{A}}^n \times X$ and $c \in \mathbb{A}$. Since $(c, \mathsf{ub}(\tilde{w}')) = \mathsf{ub}(\mathsf{l}c\tilde{w}') = \mathsf{ub}(\tilde{w}) = bw'$, we know that $c = b$ and $\mathsf{ub}(\tilde{w}') = w'$. Since $b \neq a$ and $\mathsf{l}b\tilde{w}' \equiv_\alpha \mathsf{l}at'$, it follows from A.17 that $b \# [t']_\alpha$ and $\tilde{w}' \equiv_\alpha (a\ b)t'$. Finally, it follows that $w' = \mathsf{ub}(\tilde{w}') \in D([(a\ b)t']_\alpha)$.

□

Lemma A.24. *Let* $t, u \in \overline{\mathbb{A}}^{n+1} \times X$. *If* $D([t]_\alpha) = D([u]_\alpha)$, *then one of the following is true:*

(1) $t = at'$ *and* $u = au'$ *for some* $a \in \mathbb{A}$ *and* $t', u' \in \overline{\mathbb{A}}^n \times X$, *or*
(2) $t = \mathsf{l}at'$ *and* $u = \mathsf{l}bu'$ *for some* $a, b \in \mathbb{A}$ *and* $t', u' \in \overline{\mathbb{A}}^n \times X$.

Proof. We will consider the following exhaustive list of cases for t, u:

- *For* $t = at'$ *and* $u = bu'$ *with* $t', u' \in \overline{\mathbb{A}}^{n'} \times X$*:* By definition, $\mathsf{ub}(at') \in D([t]_\alpha) = D([u]_\alpha)$. It then follows by Lemma A.23 (1) that $a = b$. Thus, (1) holds.
- *For* $t = \mathsf{l}at'$ *and* $u = bu'$ *with* $t', u' \in \overline{\mathbb{A}}^{n'} \times X$*:* Pick a $c \in \mathbb{A} \setminus \mathsf{supp}(t', a, b)$. Then we have $\mathsf{l}at' \equiv_\alpha \mathsf{l}c((a\ c)t')$ and, by definition, $\mathsf{ub}(c((a\ c)t')) \in D([\mathsf{l}at']_\alpha) = D([bu']_\alpha)$. However, by Lemma A.23 (1), this would imply $c = b$, contradicting the assumption.
- *For* $t = at'$ *and* $u = \mathsf{l}bu'$ *with* $t', u' \in \overline{\mathbb{A}}^{n'} \times X$*:* Analogous to the above case.
- *For* $t = \mathsf{l}at'$ *and* $u = \mathsf{l}bu'$ *with* $t', u' \in \overline{\mathbb{A}}^{n'} \times X$*:* Then (2) holds.

□

Lemma A.25. *The operator D is injective for single pretraces: If $D([t]_\alpha) = D([u]_\alpha)$ for $t, u \in \overline{\mathbb{A}}^n \times X$, then $[t]_\alpha = [u]_\alpha$.*

Proof. By induction on n.

- *For $t = x$ with $x \in X$:* Let $u = y \in X$. By Lemma A.17 (1), we have
$$D([t]_\alpha) = \{\mathsf{ub}(x)\} = \{x\} = \{y\} = \{\mathsf{ub}(y)\} = D([u]_\alpha).$$
Thus, we have $x = y$.
- *For $t = at'$ with $t' \in \overline{\mathbb{A}}^{n'} \times X$:* We know from Lemma A.24 that $u = au'$ for some u'. First, observe that $D([t']_\alpha) = D([u']_\alpha)$:
$$\begin{aligned} & w' \in D([t']_\alpha) \\ \Leftrightarrow\ & aw' \in D([at']_\alpha) && \text{Lemma A.23 (1)} \\ \Leftrightarrow\ & aw' \in D([au']_\alpha) && \text{assumption} \\ \Leftrightarrow\ & w' \in D([u']_\alpha) && \text{Lemma A.23 (1).} \end{aligned}$$
It then follows by inductive hypothesis that $t' \equiv_\alpha u'$ and thus $t = at' \equiv_\alpha au' = u$.
- *For $t = \mathsf{l}at'$ with $t' \in \overline{\mathbb{A}}^{n'} \times X$:* We know from Lemma A.24 that $u = \mathsf{l}bu'$ for some $u' \in \overline{\mathbb{A}}^{n'} \times X$ and $b \in \mathbb{A}$.
If $a = b$, one can show that $t = \mathsf{l}at' \equiv_\alpha \mathsf{l}au' = u$ analogously to the above case. So assume $a \neq b$.
First note that $\mathsf{ub}(\mathsf{l}at') \in D([t]_\alpha) = D([\mathsf{l}bu']_\alpha)$. By Lemma A.23 (3), we have $a \# [u']_\alpha$. By Lemma A.17 (3) then suffices to show $[t']_\alpha = (a\ b)[u']_\alpha$ to conclude $\mathsf{l}at' \equiv_\alpha \mathsf{l}bu'$.
First, observe that
$$\begin{aligned} & w' \in D([t']_\alpha) \\ \Rightarrow\ & aw' \in D([\mathsf{l}at']_\alpha) && \text{Lemma A.23 (2)} \\ \Rightarrow\ & aw' \in D([\mathsf{l}bu']_\alpha) && \text{assumption} \\ \Rightarrow\ & w' \in D([(a\ b)u']_\alpha) && \text{Lemma A.23 (3),} \end{aligned}$$
and conversely,
$$\begin{aligned} & w' \in D([(a\ b)u']_\alpha) \\ \Rightarrow\ & (a\ b)w' \in D([u']_\alpha) && \text{equivariance} \\ \Rightarrow\ & b((a\ b)w')) \in D([\mathsf{l}bu']_\alpha) && \text{Lemma A.23 (2)} \\ \Rightarrow\ & b((a\ b)w') \in D([\mathsf{l}at']_\alpha) && \text{assumption} \\ \Rightarrow\ & (a\ b)w' \in D([(a\ b)t']_\alpha) && \text{Lemma A.23 (3)} \\ \Rightarrow\ & w' \in D([t']_\alpha) && \text{equivariance.} \end{aligned}$$
The statement then follows by inductive hypothesis.

□

As seen in Theorem 6.1, a set $S \in M_n^{\mathsf{bar}}$ can be seen in T^{bar} as a big join of all pretraces in S. By the join-semilattice structure of $F'X$ and the interpretation of $+$, we thus have
$$\overline{e_n(\kappa_n(\theta_n(S)))} = \bigcup\nolimits_{[t]_\alpha \in S} \overline{e_n([t]_n)}.$$

As a prerequisite, we will characterize the behavior of the restriction operation for such terms:

Lemma A.26. *For any $t \in \overline{\mathbb{A}}^n \times X$ and $b \in \mathsf{FN}(t)$, we have $\mathsf{res}_{F'X,n}(\langle b\rangle e_n([t]_n)) = 0_{F'X,n}$.*

Proof. By induction on n.

- *For $t = x \in X$:* By definition, we have
$$\begin{aligned}\mathsf{res}_{F'X,0}(\langle b\rangle e_0([x]_0)) &= \mathsf{res}_{F'X,0}(\langle b\rangle\{x\})\\ &= \{x \in \{x\} \mid b \# x\}\\ &= \emptyset = 0_{F'X,0}.\end{aligned}$$
- *For $t = at'$ with $t' \in \overline{\mathbb{A}}^{n'} \times X$:* We know that $b \in \mathsf{FN}(at') = \mathsf{FN}(t') \cup \{a\}$.
  If $b = a$, then
$$\begin{aligned}&\mathsf{res}_{F'X,n}(\langle b\rangle e_n([at]_n))\\ =\ &\mathsf{res}_{F'X,n}(\langle a\rangle \mathsf{pre}_{F'X,n'}(a, e_{n'}([t']_{n'}))) && \text{definition}\\ =\ &0_{F'X,n} && \text{axioms.}\end{aligned}$$
  If $b \neq a$ (and thus $b \in \mathsf{FN}(t')$), we have
$$\begin{aligned}&\mathsf{res}_{F'X,n}(\langle b\rangle e_n([at]_n))\\ =\ &\mathsf{res}_{F'X,n}(\langle b\rangle \mathsf{pre}_{F'X,n'}(a, e_{n'}([t']_{n'}))) && \text{definition}\\ =\ &\mathsf{pre}_{F'X,n'}(a, \mathsf{res}_{F'X,n'}(\langle b\rangle e_{n'}([t']_{n'}))) && \text{axioms}\\ =\ &\mathsf{pre}_{F'X,n'}(a, 0_{F'X,n'}) && \text{induction}\\ =\ &0_{F'X,n} && \text{axioms.}\end{aligned}$$
- *For $t = \mathsf{l}at'$ with $t' \in \overline{\mathbb{A}}^{n'} \times X$:* We know that $b \in \mathsf{FN}(\mathsf{l}at') = \mathsf{FN}(t') \setminus \{a\}$; i.e., $b \in \mathsf{FN}(t')$ and $b \neq a$. The argument is then analogous to the one in the previous case.

□

By definition of D, it now suffices to show that $\overline{e_n([t]_n)} = D([t]_\alpha)$ for $t \in \overline{\mathbb{A}}^n \times X$.

We will do this by induction on n.

- *For $t = x \in X$:* We know by Lemma A.17 (1) that $[x]_\alpha = \{x\}$. It then follows that
$$\overline{e_0([x]_0)} = \overline{\{x\}} = \{x\} = \{\mathsf{ub}(x)\} = D([x]_\alpha).$$
- *For $t = at'$ with $t' \in \overline{\mathbb{A}}^{n'} \times X$:*
$$\begin{aligned}&\overline{e_n([at']_n)}\\ =\ &\{aw \mid w \in \overline{e_{n'}([t']_{n'})}\} && \text{definition of } \mathsf{pre}\\ =\ &\{aw \mid w \in D([t']_\alpha)\} && \text{induction}\\ =\ &D([at']_\alpha) && \text{Lemma A.23 (1).}\end{aligned}$$

- *For $t = |at'$ with $t' \in \overline{\mathbb{A}}^{n'} \times X$:* First, assume $bv \in \overline{e_n([|at']_n)}$; i.e., $v \in \overline{\mathsf{abs}_{F'X,n'}(\langle a\rangle e_{n'}([t']_{n'}))(b)}$.
  If $a = b$, then we have

  $$\begin{aligned} v &\in \overline{\mathsf{abs}_{F'X,n'}(\langle a\rangle e_{n'}([t']_{n'}))(a)} \\ &= \overline{e_{n'}([t']_{n'})} && \text{Lemma A.22} \\ &= D([t']_\alpha) && \text{induction.} \end{aligned}$$

  By Lemma A.23 (2), we thus have $bv \in D([|at']_\alpha)$.
  If $a \neq b$, we will first show that $b \# e_{n'}([t']_{n'})$: Otherwise, we would have $b \in \mathsf{supp}(e_{n'}([t']_{n'})) = \mathsf{supp}(D([t']_\alpha)) = \mathsf{supp}([t']_\alpha) = \mathsf{FN}(t')$ by inductive hypothesis and injectivity. Thus, for sufficiently fresh c, we would obtain

  $$\begin{aligned} v &\in \overline{\mathsf{abs}_{F'X,n'}(\langle a\rangle e_{n'}([t']_{n'}))(b)} \\ &= \overline{\mathsf{res}_{F'X,n'}(\langle c\rangle (b\ c) \cdot (a\ c) \cdot e_{n'}([t']_{n'}))} && \text{definition} \\ &= \overline{(b\ c) \cdot (a\ c) \cdot \mathsf{res}_{F'X,n'}(\langle b\rangle e_{n'}([t']_{n'}))} && \text{equivariance} \\ &= \overline{(b\ c) \cdot (a\ c) \cdot 0_{F'X,n'}} && \text{Lemma A.26} \\ &= \overline{0_{F'X,n'}} = \emptyset. \end{aligned}$$

  It then follows that

  $$\begin{aligned} v &\in \overline{\mathsf{abs}_{F'X,n'}(\langle a\rangle e_{n'}([t']_{n'}))(b)} \\ &= \overline{\mathsf{abs}_{F'X,n'}(\langle b\rangle (a\ b) \cdot e_{n'}([t']_{n'}))(b)} && \text{because } b \# e_{n'}([t']_{n'}) \\ &= (a\ b) \cdot \overline{e_{n'}([t']_{n'})} && \text{Lemma A.22} \\ &= (a\ b) \cdot D([t']_\alpha) && \text{induction} \\ &= D([(a\ b) \cdot t']_\alpha) && \text{equivariance,} \end{aligned}$$

  and, by Lemma A.23 (3), $bv \in D([|at']_\alpha)$.
  Conversely, assume $bv \in D([|at']_\alpha)$. By Lemma A.23, there exists some $u' \in \overline{\mathbb{A}}^{n'} \times X$ with $v \in D([u']_\alpha)$ and $[|at']_\alpha = [|bu']_\alpha$. It then follows that

  $$\begin{aligned} v &\in D([u']_\alpha) \\ &= \overline{e_{n'}([u']_{n'})} && \text{induction} \\ &= \overline{\mathsf{abs}_{F'X,n'}(\langle b\rangle e_{n'}([u']_{n'}))(b)} && \text{Lemma A.22} \\ &= \overline{\mathsf{abs}_{F'X,n'}(\langle a\rangle e_{n'}([t']_{n'}))(b)} && \text{assumption} \\ &= \overline{e_n([|at']_n)(b)} && \text{definition.} \end{aligned}$$

□

Proof of Lemma A.20. All of the following computations are done in T^{loc} unless stated otherwise.

Given $s, t \in \mathsf{Term}_{\Sigma^{\mathsf{bar}},n}(X)$, we write $X \vdash_n s \leq t$ for $X \vdash_n t + s = t$; intuitively, a term is "larger" than another one if adding that other term does not change anything.

Lemma A.27. *$X \vdash_n s \leq t$ and $X \vdash_n t \leq s$ iff $X \vdash_n t = s$.*

Proof. 'If': By definition, we have $X \vdash_n t + s = t$ and $X \vdash_n s + t = s$. It then follows by symmetry of $+$ that $X \vdash_n t = s$.
'Only if': By idempotence of $+$.

□

For sum terms (with 0 being the neutral element), we have the following additional properties:

Lemma A.28. *Let $T \in \mathcal{P}_\omega(\mathsf{Term}_{\Sigma^{\mathsf{bar}},n}(X))$ and $s \in \mathsf{Term}_{\Sigma^{\mathsf{bar}},n}(X)$. Then we have*

(1) If $X \vdash_n t \leq s$ for every $t \in T$, then $X \vdash_n \sum_{t\in T} t \leq s$.
(2) If $X \vdash_n s \leq t$ for some $t \in T$, then $X \vdash_n s \leq \sum_{t\in T} t$.
(3) If $s \in T$, then $X \vdash_n s \leq \sum_{t\in T} t$.

Proof. By unfolding the definitions for $\leq$ and sums. (3) is a special case of (2) using Lemma A.27. □

Furthermore, we have the following properties for (free and bound) prefixing:

Lemma A.29. *Let $t, s \in \mathsf{Term}_{\Sigma^{\mathsf{bar}},n}(X)$ and $u \in \mathsf{Term}_{\Sigma^{\mathsf{bar}},n+1}(X)$. Then we have*

(1) If $X \vdash_n t \leq s$ and $X \vdash_{n+1} as \leq u$, then $X \vdash_{n+1} at \leq u$.
(2) If $X \vdash_n t \leq s$ and $X \vdash_{n+1} \mathsf{l}as \leq u$, then $X \vdash_{n+1} \mathsf{l}at \leq u$.

Proof. We will only show the first statement, the second one is completely analogous. By simple computation, we have

$$\begin{aligned}
[u]_{n+1} &= [u + as]_{n+1} && \text{definition}\\
&= [u + a(s + t)]_{n+1} && \text{definition}\\
&= [(u + as) + at]_{n+1} && \text{axioms}\\
&= [u + at]_{n+1} && \text{definition.}
\end{aligned}$$

□

The most important characterization, relating the generated language to terms, is the following one:

Lemma A.30. *Let $w \in \overline{\mathbb{A}}^n \times X$ and $s \in \mathsf{Term}_{\Sigma^{\mathsf{bar}},n}(X)$. If $D([w]_\alpha) \subseteq \overline{e_n([s]_n)}$, then $X \vdash_n w \leq s$.*

Proof. Assume w.l.o.g. that s is in normal form; i.e., $s = \sum_{v\in S} v$ for a set $S \in \mathcal{P}_\omega(\overline{\mathbb{A}}^n \times X)$ of pretraces.

Note that, by Lemma A.19 and the join-semilattice structure of $F'X$, we have

$$D([w]_\alpha) \subseteq \overline{e_n([s]_n)} = \bigcup_{v\in S} D([v]_\alpha). \tag{A.7}$$

We will proceed by induction on n.

- *For $w = x \in X$:* Since $n = 0$, we know that $S \subseteq X$.
  It follows from Lemma A.17 that $D([w]_\alpha) = \{\mathsf{ub}(x)\} = \{x\}$. By (A.7), it then follows that $x \in D([v]_\alpha = \{v\})$ for some $v \in S$; i.e., $x \in S$. The statement then follows by Lemma A.28 (3).

- *For $w = aw'$ with $w' \in \overline{\mathbb{A}}^{n'} \times X$:* We will define a new sum term

$$s' := \sum\nolimits^{k}_{av' \in S} v' + \sum\nolimits^{k}_{|av' \in S} v' + \sum\nolimits^{k}_{\substack{|bv' \in S \\ a \neq b, a \notin \mathsf{FN}(v')}} (a\ b)v'.$$

  Note that, by (5.1), as' is derivably equal to a form where a is distributed over all terms of the sums.

  Next, we will show $X \vdash_n as' \leq s$ using Lemma A.28 (1) for all three sums in s'.

  (1) *For $av' \in S$:* Then we already have $X \vdash_n av' \leq s$ by Lemma A.28 (3).
  (2) *For $|av' \in S$:* Analogous to the above case.
  (3) *For $|bv' \in S$ with $a \neq b$ and $a \notin \mathsf{FN}(v')$:* Since $a \# [v']_\alpha$, we know that $|bv' \equiv_\alpha |a((a\ b)v')$ by Lemma A.17 (3). By Lemma A.16, this equality also holds in T^{loc}. By (6.2), we then have $X \vdash_n a((a\ b)v') \leq |a((a\ b)v') = |bv'$. Finally, we can conclude $X \vdash_n a((a\ b)v' \leq s$ using Lemma A.28 (2).

  Furthermore, we will show that $D([w']_\alpha) \subseteq \overline{e_{n'}([s']_{n'})}$. So let $x \in D([w']_\alpha)$. By the join-semilattice structure of $F'X$, it is enough to show that $x \in D([u]_\alpha)$ for some summand u in s'. By Lemma A.23 (1), we know that $ax \in D([aw']_\alpha)$ and, by (A.7), $ax \in D([v]_\alpha)$ for some $v \in S$. We consider the cases for v as outlined in Lemma A.23:

  If $v = av'$ or $v = |av'$, then $x \in D([v']_\alpha)$ and, by definition, v' is a summand in s'.

  If $v = |bv'$ with $a \neq b$, then $a \# [v']_\alpha$ and $x \in D([(a\ b)v']_\alpha$. By definition, $(a\ b)v'$ is a summand in s'.

  With this, we have $X \vdash_{n'} w' \leq s'$ by inductive hypothesis. Since $X \vdash_n as' \leq s$, we can then conclude $X \vdash_n aw' \leq s$ by Lemma A.29.

- *For $w = |aw'$ with $w' \in \overline{\mathbb{A}}^{n'} \times X$:* Pick $c \in \mathbb{A} \setminus \mathsf{FN}(w')$ s.t. $c \neq a$ and there is no $v \in S$ with $v = cv'$ or $v = |cv'$ (so there is no term prefixed with c). This is always possible because there are only k terms and thus only finitely many different prefixes used. Note that $|aw' \equiv_\alpha |c((a\ c)w')$.

  We will now define a new sum term

$$s' := \sum\nolimits^{k}_{\substack{|bv' \in S \\ c \notin \mathsf{FN}(v')}} (b\ c)v'.$$

  Note that, by (6.1), $|cs'$ is derivably equal to a form where $|c$ is distributed over all terms of the sum.

  Next, we will once again show $X \vdash_n |cs' \leq s$ using Lemma A.28 (1) on s'. So let $|bv' \in S$ with $c \notin \mathsf{FN}(v')$, implying that $c \# [v']_\alpha$. Since $b \neq c$ by choice of c, we have $|bv' \equiv_\alpha |c((b\ c)v')$. The inequality then follows by Lemma A.27 and Lemma A.16.

  Furthermore, we will show that $D([(a\ c)w']_\alpha) \subseteq \overline{e_{n'}([s']_{n'})}$. So let $x \in D([(a\ c)w']_\alpha)$. By the join-semilattice structure of $F'X$, it is once again enough to show that $x \in D([u]_\alpha)$ for some summand u in s'. By Lemma A.23 (2), we know that $cx \in D([|c((a\ c)w')]_\alpha) = D([|aw']_\alpha)$ and, by (A.7), $cx \in D([v]_\alpha)$ for some $v \in S$. By Lemma A.23 and because there are no pretraces beginning with c or $|c$, we know that

$v = |bv'$ with $c \# [v']_\alpha$ and $x \in D([(b\ c)v']_\alpha)$. By definition, $(b\ c)v'$ is a summand in s'.

With this, we have $X \vdash_{n'} (a\ c)w' \leq s'$ by inductive hypothesis. Since $X \vdash_n |cs' \leq s$, we can then conclude $X \vdash_n |aw' = |c((a\ c)w') \leq s$ by Lemma A.29.

□

We now have to show that, for $t, s \in \mathsf{Term}_{\Sigma^{\mathsf{bar}},n}(X)$ with $e_n([t]_m) = e_n([s]_n)$, we have $[t]_n = [s]_n$. Assume w.l.o.g. that $t = \sum_{w \in T} w$ for a set $T \in \mathcal{P}_\omega(\overline{\mathbb{A}}^n \times X)$ of pretraces. By Lemma A.27, it suffices to show that $X \vdash_n t \leq s$ and $X \vdash_n s \leq t$. We will only show the first inequality as the second one is completely symmetrical. To achieve this, we will use Lemma A.28 (1).

So let $w \in T$. By the join-semilattice structure of $F'X$ and Lemma A.19, we know that

$$D([w]_\alpha) = \overline{e_n([w]_n)} \subseteq \overline{e_n([t]_n)} = \overline{e_n([s]_n)}.$$

It then follows by Lemma A.30 that $X \vdash_n w \leq rs$. □

A.2. **Details for Remark 6.4.** Intuitively, we can think of $x|_N$ as the free restriction of x's support to $N \subseteq \mathsf{supp}(x)$. As such, we have the following properties:

Lemma A.31. *Given a nominal set X, let $x \in X$ and $N \subseteq \mathsf{supp}(x)$. Then we have:*

(1) The least support $\mathsf{supp}(x|_N)$ is exactly N.
(2) For $\pi \in \mathsf{Perm}(\mathbb{A})$, we have $\pi \cdot (x|_N) = (\pi \cdot x)|_{\pi \cdot N}$.
(3) For $x' \in X$ and $N' \subseteq \mathsf{supp}(x')$, $x|_N = x'|_{N'}$ iff $N' = N$ and $x' = \pi \cdot x$ for some $\pi \in \mathsf{Fix}(N)$.
(4) We have $x|_N = \{y\}$ iff $x = y$ and $N = \mathsf{supp}(y)$.

Proof. We show the statements individually:

(1) Shown elsewhere [37, Lemma 5.8].
(2) Let $\pi \cdot \tau \cdot x \in \pi \cdot (x|_N)$ for some $\tau \in \mathsf{Fix}(N)$. Then $\pi \cdot \tau \cdot x = \pi \cdot \tau \cdot \pi^{-1} \cdot \pi \cdot x \in (\pi \cdot x)|_{\pi \cdot N}$, because $\pi \cdot \tau \cdot \pi^{-1} \in \mathsf{Fix}(\pi \cdot N)$.
Conversely, let $\tau \cdot \pi \cdot x \in (\pi \cdot x)|_{\pi \cdot N}$ for some $\tau \in \mathsf{Fix}(\pi \cdot N)$. Then $\tau \cdot \pi \cdot x = \pi \cdot \pi^{-1} \cdot \tau \cdot \pi \cdot x \in \pi \cdot (x|_N)$, because $\pi^{-1} \cdot \tau \cdot \pi \in \mathsf{Fix}(N)$.
(3) *'If':* Obvious by definition of $x|_N$.
'Only If': By (1), we have $N = \mathsf{supp}(x|_N) = \mathsf{supp}(x'|_{N'}) = N'$. Furthermore, since $\pi = 1 \in \mathsf{Fix}(N')$, we have $x' \in x'|_{N'} = x|_N$. Thus, by definition, there exists a $\pi \in \mathsf{Fix}(N)$ with $x' = \pi \cdot x$.
(4) First, note that $y|_{\mathsf{supp}(y)} = \{y\}$, thus the 'if' case is immediately discharged. For 'only if', we also have $N = \mathsf{supp}(y)$ by (3). Furthermore, for $\pi = id \in \mathsf{Fix}(N)$, we have $x = \pi \cdot x \in x|_N = \{y\}$ and thus $x = y$.

□

We can then make our claimed description into a $(T^{\mathsf{drop}}, 0)$-model A with the operations defined as for $(F'X)_0$. We can easily see that all operations preserve the requirement that the sets are closed under further name restriction.

Given a finite set $A = \{a_1, \dots a_n\} \subseteq \mathsf{supp}(x)$ where $a_1, \dots, a_n$ are pairwise distinct, we abbreviate the term

$$A \setminus x = \nu a_1.\mathsf{res}(\dots(\nu a_n.\mathsf{res}(x))),$$

where the order does not matter since we can swap name restrictions by axiom (6.3).

To prove the claimed description, first note that we can normalize terms as follows: First, we can distribute name restrictions over joins using (6.3) and (6.4). We can then reduce every summand to a form $\nu a_1.\mathsf{res}(\dots(\nu a_n.\mathsf{res}(x)))$ where $a_1, \dots, a_n \in \mathsf{supp}(x)$ are pairwise distinct using (6.3). Furthermore, we can add all further possible such restrictions for the summands using (6.4).

The result then follows using the following observation:

Lemma A.32. *Let $x, y \in X$, $A \subseteq \mathsf{supp}(x)$, and $B \subseteq \mathsf{supp}(y)$. If $x|_{\mathsf{supp}(x)\setminus A} = y|_{\mathsf{supp}(y)\setminus B}$, then $X \vdash_0 A \setminus x = B \setminus y$ is derivable in T^{drop}.*

Proof. We proceed by induction on $|A|$.

- For $A = \emptyset$: We have $y|_{\mathsf{supp}(y)\setminus B} = x|_{\mathsf{supp}(x)} = \{x\}$ and thus, by Lemma A.31 (4), $y = x$ and $\mathsf{supp}(y) \setminus B = \mathsf{supp}(x)$ (i.e., $B = \emptyset$). The statement then follows by reflexivity.
- For $A = A' \cup \{a\}$ with $a \notin A'$: By Lemma A.31 (3), we know that $\mathsf{supp}(x) \setminus A = \mathsf{supp}(y) \setminus B$ and $y = \pi \cdot x$ for some $\pi \in \mathsf{Fix}(\mathsf{supp}(x) \setminus A)$.

  If $\pi(a) = a$, we have $a = \pi(a) \in \pi \cdot \mathsf{supp}(x) = \mathsf{supp}(y)$ by equivariance of supp. Since $a \notin \mathsf{supp}(x) \setminus A = \mathsf{supp}(y) \setminus B$, we must have $a \in B$. Set $B' = B \setminus \{a\}$. Since $\mathsf{supp}(x) \setminus A' = \mathsf{supp}(y) \setminus B'$ and $\pi \in \mathsf{Fix}(\mathsf{supp}(x) \setminus A')$, we have $x|_{\mathsf{supp}(x)\setminus A'} = y|_{\mathsf{supp}(y)\setminus B'}$ by Lemma A.31 (3). It then follows by inductive hypothesis and the derivation rules that

  $$X \vdash_0 \nu a.\mathsf{res}(A' \setminus x) = \nu a.\mathsf{res}(B' \setminus y)$$

  is derivable.

  If $\pi(a) \neq a$, then $\pi(a) \notin \mathsf{supp}(x) \setminus A = \mathsf{supp}(y) \setminus B$. Since $a \in A \subseteq \mathsf{supp}(x)$, we have $\pi(a) \in \pi \cdot \mathsf{supp}(x) = \mathsf{supp}(y)$ by equivariance of supp, and thus $\pi(a) \in B$. Set $B' = B \setminus \{\pi(a)\}$.

  We are done once we have shown that

  $$X \vdash_0 \nu a.\mathsf{res}(A' \setminus x) = \nu \pi(a).\mathsf{res}(B' \setminus y).$$

  Since $a \notin \mathsf{supp}(x) \setminus A = \mathsf{supp}(y) \setminus B$, we already have $a \notin \mathsf{supp}(y) \setminus B'$. Using the inductive hypothesis, it then suffices to show that

  $$x|_{\mathsf{supp}(x)\setminus A'} = ((a\ \pi(a)) \cdot y)|_{(a\ \pi(a))\cdot(\mathsf{supp}(y)\setminus B')}$$

First, note that

$$\begin{aligned}
&\mathsf{supp}(x) \setminus A' \\
&= (\mathsf{supp}(x) \setminus A) \cup \{a\} && \text{assumption} \\
&= (a\ \pi(a)) \cdot (\mathsf{supp}(x) \setminus A) \cup \{a\} && \text{by } a, \pi(a) \notin \mathsf{supp}(x) \setminus A \\
&= (a\ \pi(a)) \cdot ((\mathsf{supp}(x) \setminus A) \cup \{\pi(a)\}) && \text{equivariance} \\
&= (a\ \pi(a)) \cdot ((\mathsf{supp}(y) \setminus B) \cup \{\pi(a)\}) && \text{assumption} \\
&= (a\ \pi(a)) \cdot (\mathsf{supp}(y) \setminus B') && \text{assumption.}
\end{aligned}$$

Furthermore, for $\tau = (a\ \pi(a)) \cdot \pi$, we have

$$\begin{aligned}
&(a\ \pi(a)) \cdot y \\
&= (a\ \pi(a)) \cdot \pi \cdot x && \text{assumption} \\
&= \tau \cdot x && \text{definition of } \tau,
\end{aligned}$$

and, since $\tau \in \mathsf{Fix}(\mathsf{supp}(x) \setminus A')$, the statement follows by Lemma A.31 (3).

□

A.3. **Details for Section 7.** The functor $\overline{M}_1$ is described in purely categorical terms by taking $\overline{M}_1(A, a)$, for an M_0-algebra (A, a) with structure map $a\colon M_0A \to A$, to be the coequalizer

$$M_1M_0A \underset{\mu_A^{10}}{\overset{M_1a}{\rightrightarrows}} M_1A \longrightarrow \overline{M}_1(A, a) \qquad \text{(in the category of } M_0\text{-algebras).}$$

Equivalently, $\overline{M}_1(A, a)$ is described as follows. We have forgetful functors $(-)_0$, $(-)_1$ from the category of M_1-algebras to the category of M_0-algebras, sending an M_1-algebra $((B_i), (b_{ij}))$ to (B_0, b_{00}) and (B_1, b_{01}), respectively. Then, the free M_1-algebra $((B_i), (b_{ij}))$ w.r.t. $(-)_0$ over an M_0-algebra (A, a) has $B_0 = A$ and $b_{00} = a$; applying $(-)_1$ to $((B_i), (b_{ij}))$ yields $\overline{M}_1(A, a)$ [11, 12].

A.4. **Details for Example 7.1.** *Global freshness:* Since, at depth 0, we have the join-semilattice axioms, an M_0-algebra A has an equivariant join-semilattice structure. Using the distributivity axioms and (perm), depth-1 terms over A can thus be normalized to the form $a_1x_1 + \cdots + a_nx_n + \lfloor cy$ for some sufficiently fresh $c \in \mathbb{A}$ and $x_1, \ldots, x_n, y \in A$, requiring that $x_i \neq 0$ for all i. It follows that $\overline{M}_1A = A_\omega^{\mathbb{A}} \times [\mathbb{A}]A$.
Local feshness: We prove the claimed description of $\overline{M}_1$ for local freshness semantics with name-dropping. As this description makes $\overline{M}_1$ a subfunctor of the functor $\mathbb{A} \to_{\mathsf{fs}} (-)$, it follows that $\overline{M}_1$ preserves monos.

Let (A, h_A) be an M_0-algebra. First note that we can define a $(\Sigma^{\mathsf{drop}}, 1)$-model with A at depth 0 and $\overline{M}_1(A, h_A)$ at depth 1. The proof that this model actually satisfies the axioms is the same as for Lemma A.18, observing that the proof for $(F'X)_{n+1}$ does not depend on the actual structure of $(F'X)_n$ and only uses the fact that $(F'X)_n$ is an M_0-algebra. Since M_0A and M_1A form the free M_1-algebra over the set A, we obtain in particular an M_0-homomorphism $m\colon M_1A \to \overline{M}_1(A, h_A)$ that acts by term evaluation.

Let (Q, h_Q) be another M_0-algebra, and let $q\colon M_1 A \to Q$ be an M_0-homomorphism such that $q \cdot M_1 h_A = q \cdot \mu_A^{10}$. We define $m'\colon \overline{M}_1(A, h_A) \to Q$ by

$$m'(f) = q\left(\left[\textstyle\sum_{a \in \mathsf{supp}(f)} a f(a) + \mathsf{I}c f(c)\right]\right) \quad \text{for } c \# f.$$

This is indeed an M_0-homomorphism: We will show that $m'(f_1 + f_2) = m'(f_1) + m'(f_2)$, the proof for name restriction is largely analogous. Since q is an M_0-algebra morphism and thus distributes over $+$, it suffices to show that

$$\begin{aligned}&\textstyle\sum_{a \in \mathsf{supp}(f_1+f_2)} a(f_1 + f_2)(a) + \mathsf{I}c(f_1 + f_2)(c)\\ =\ &\textstyle\sum_{a \in \mathsf{supp}(f_1)} a f_1(a) + \mathsf{I}c f_1(c) + \sum_{a \in \mathsf{supp}(f_2)} a f_2(a) + \mathsf{I}c f_2(c)\end{aligned}$$

is derivable from the axioms and the assumption on q that $q \cdot M_1 h_A = q \cdot \mu_A^{10}$ which, algebraically speaking, allows us to collapse terms at depth 0 into a single variable. For example, the term $x + y$ with variables $x, y \in A$ can be collapsed into a single variable $x + y \in A$, utilizing the join-semilattice structure of A. Under these additional equalities, we will once again use Lemma A.28 and show that the individual summands on each side are contained in the other side. For the bound summands, this is clear because $+$ distributes over $\mathsf{I}c$. For free summands, this reduces to the following cases:

- For terms $a f_1(a)$ with $a \in \mathsf{supp}(f_1) \cap \mathsf{supp}(f_1 + f_2)$, we have

  $$\begin{aligned} a f_1(a) &\le a(f_1(a) + f_2(a)) && \text{axioms}\\ &= a(f_1 + f_2)(a) && \text{assumption on } q.\end{aligned}$$

- For terms $a f_1(a)$ with $a \in \mathsf{supp}(f_1) \setminus \mathsf{supp}(f_1 + f_2)$ and sufficiently fresh c, we have

  $$\begin{aligned} & a f_1(a)\\ \le\ & a(f_1 + f_2)(a) && \text{axioms, assumption on } q\\ =\ & a(\nu c.\mathsf{res}((f_1 + f_2)(a))) && \text{axioms, } c \# (f_1 + f_2)(a)\\ =\ & a(\nu c.\mathsf{res}(((a\ c) \cdot (f_1 + f_2))(a))) && \text{because } a, c\#(f_2 + f_2)\\ =\ & a(\nu c.\mathsf{res}((a\ c) \cdot (f_1 + f_2)(c))) && \text{definition}\\ =\ & a((a\ c) \cdot \nu a.\mathsf{res}((f_1 + f_2)(c))) && \text{equivariance}\\ \le\ & \mathsf{I}a((a\ c) \cdot \nu a.\mathsf{res}((f_1 + f_2)(c))) && \text{axioms}\\ =\ & \mathsf{I}c(\nu a.\mathsf{res}((f_1 + f_2)(c))) && \text{derivation rules}\\ \le\ & \mathsf{I}c(f_1 + f_2)(c) && \text{axioms,}\end{aligned}$$

  notably using axiom (6.4) in the last step.

- For terms $a(f_1 + f_2)(a)$ with $a \in \mathsf{supp}(f_1 + f_2) \setminus \mathsf{supp}(f_1)$ (and thus $a \in \mathsf{supp}(f_2)$) and sufficiently fresh c, we have

$$\begin{aligned}
& a(f_1 + f_2)(a) && \\
&= af_1(a) + af_2(a) && \text{axioms, assumption on } q \\
&= a(\nu c.\mathsf{res}(f_1(a))) + af_2(a) && \text{axioms, } c \# f_1(a) \\
&= a(\nu c.\mathsf{res}(((a\ c) \cdot f_1)(a))) + af_2(a) && \text{because } a, c \# f_1 \\
&= a(\nu c.\mathsf{res}((a\ c) \cdot f_1(c))) + af_2(a) && \text{definition} \\
&= a((a\ c) \cdot \nu a.\mathsf{res}(f_1(c))) + af_2(a) && \text{equivariance} \\
&\leq \mathsf{l}a((a\ c) \cdot \nu a.\mathsf{res}(f_1(c))) + af_2(a) && \text{axioms} \\
&= \mathsf{l}c(\nu a.\mathsf{res}(f_1(c))) + af_2(a) && \text{derivation rules} \\
&\leq \mathsf{l}c(f_1(c)) + af_2(a) && \text{axioms.}
\end{aligned}$$

- The cases for f_2 are completely analogous.

To show that m' is unique, it then suffices to show that, for $f \in \overline{M}_1(A, h_A)$ and sufficiently fresh c,

$$m\left(\left[\textstyle\sum_{a \in \mathsf{supp}(f)} af(a) + \mathsf{l}cf(c)\right]\right) = f.$$

This is clear from the definition of pre and abs, as well as the additional requirement for f that traces for all atoms include those from bound atoms.

Categorical Phrasing of the Equivalence Game. The categorical formulation of the game is as follows.

Definition A.33. Let $(\mathbb{M}, \beta)$ be a graded semantics for a functor F on **Nom**, and let $\gamma\colon C \to FC$ be a F-coalgebra. For an equivariant subset $Z \subseteq M_0C \times M_0C$, we define c_Z, C_Z as the coequalizer

$$M_0Z \underset{r_0^*}{\overset{l_0^*}{\rightrightarrows}} M_0C \xrightarrow{c_Z} C_Z$$

in **Nom**, where l, r are the left and right projection maps $Z \to M_0C$ (recall that $(-)_0^*$ is part of the *graded Kleisli star* of $\mathbb{M}$; cf. Definition 4.1). We then have an equivariant map

$$\overline{Z} = (M_0C \xrightarrow{(\beta\cdot\gamma)_0^*} M_1C = \overline{M}_1M_0C \xrightarrow{\overline{M}_1c_Z} \overline{M}_1C_Z). \tag{A.8}$$

The *n-round $(\mathbb{M}, \beta)$-behavioural equivalence game* $\mathsf{G}_n(\gamma)$ on (C, γ), played by Spoiler and Duplicator, has *configurations* $(s, t) \in M_0C \times M_0C$. The game proceeds in n rounds. A round starting in configuration (s, t) is played as follows:

(1) Duplicator picks an equivariant subset $Z \subseteq M_0C \times M_0C$ such that $\overline{Z}(s) = \overline{Z}(t)$ (for $\overline{Z}$ as per (A.8)).
(2) Spoiler picks $(s', t') \in Z$.

The next configuration is then (s', t'). Any player who is unable to move on their turn loses. After n rounds have been played, ending in configuration (s, t), Duplicator wins if $M_0!(s) = M_0!(t)$; otherwise, Spoiler wins. This final check is referred to as *calling the bluff*.

This formulation is analogous to the one in the set-based case [11], from which it is obtained essentially by replacing the base category **Set** with **Nom** and (correspondingly) requiring Z to be an equivariant subset, i.e. a subobject.

When $\mathbb{M}$ is induced by a graded nominal theory T, equivalence of this formulation of the game to the algebraic formulation given in the main body of the paper is immediate from the way $\mathbb{M}$ is constructed from T as per Section 5. In more detail, the correspondence works as follows.

As indicated in the main body, we regard $Z \subseteq M_0C \times M_0C$ as a set of equalities, understanding $(s,t) \in Z$ as a depth-0 equation $C \vdash_0 s = t$. Requiring Z to be equivariant in the categorical game corresponds to allowing equations in Z to be used in renamed form in algebraic derivations. The algebraic game indeed effectively restricts to orbit-finite Z; this is justified by the fact that by Remark 5.4, the system can be made finitary, so that logical consequence is compact, i.e. any equation derivable from a given set Z of equalities follows already from a finite subset. The coequalizer c_Z as per Definition A.33 merges all pairs (s,t) of terms in M_0C such that $Z \Rightarrow C \vdash_0 s = t$. As indicated in the main body, we view the map $\beta \cdot \gamma \colon C \to M_1C$ as a uniform-depth substitution σ. The map $(\beta \cdot \gamma)_0^* \colon M_0C \to M_1C$ applies σ to depth-0 terms over C, and the map $\overline{M}_1 c_z \colon M_1C = \overline{M}_1 M_0 C \to \overline{M}_1 C_Z$ identifies two depth-1 terms u, v over C if $Z \Rightarrow C \vdash_1 u = v$. Thus, Z is a legal move for Duplicator in configuration (s,t) if $Z \Rightarrow C \vdash s\sigma = t\sigma$, so the notion of admissible move in the two games is the same. Finally, it is easy to see that the two notions of calling the bluff agree.

Proof of Theorem 7.3. We conduct the proof using the categorical formulation of the game. It does not actually require the graded monad $\mathbb{M}$ to be presented by a graded nominal theory; it does however need this property for the monad M_0. We introduce corresponding terminology:

Definition A.34. A monad on **Nom** is *regularly algebraic* if it is presented by a depth-0 nominal theory as per Section 5.

The proof then needs the following fact:

Lemma A.35. *Let M be a regularly algebraic monad on* **Nom**. *Then the following hold.*

(1) Coequalizers in the category of M-algebras are surjective.

(2) Let $f \colon A \to B$ be a morphism of M-algebras. Let $l, r \colon K \to A$ denote the kernel of f, and let $c \colon A \to C$ be the coequalizer of l, r. Then the unique factorizing morphism $m \colon C \to B$ such that $m \cdot c = f$ is monic.

Proof. Claim (1): Since operations in our nominal theories are total, the coequalizer of M_0-homomorphisms $l, r \colon A \to B$ just quotients B by the equivariant congruence generated by the relation $\{(l(x), r(x) \mid x \in A\}$.

Claim (2): Let $m(x) = m(y)$. By (1), there are $x', y' \in B$ such that $c(x') = x$, $c(y') = y$. Then $f(x') = m(c(x')) = m(c(y')) = f(y')$, i.e. (x', y') is in the Kernel of f. Thus, $x = c(x') = c(y') = y$. □

Proof (Theorem 7.3). The proof for the corresponding result over **Set** [11, Theorem 6.6] is in fact formulated in categorical generality. Specifically, the

base category is only assumed to be concrete over **Set** and to have certain limits and colimits (kernels and coequalizers), conditions which both hold for **Nom**. Additionally, the proof uses a property that over **Set** follows from regularity of the category of M_0-algebras [11, full version, Remark A.3], which may fail over **Nom**; however, we have established the requisite property for regularly algebraic monads on **Nom** directly as Lemma A.35.2. □

Details for Example 7.4. *Global freshness:* Terms t at depth 1 can be brought into the form

$$t = \sum_{a\in S} at_a + |ct_{|c},$$

where $c \in \mathbb{A}$ is sufficiently fresh and $S \in \mathcal{P}_\omega(\mathbb{A})$ can be extended to any finite superset $S' \supseteq S$ (by setting any new t_a to 0). Given a configuration (s,t) where w.l.o.g. $(\beta\cdot\gamma)_0^*(s)$ and $(\beta\cdot\gamma)_0^*(t)$ are viewed as terms as described above, a move Z is then admissible iff $Z \Rightarrow C \vdash_0 s_\rho = t_\rho$ is derivable for every $\rho \in S \cup \{|c\}$, where the 'only if' direction follows from the above description of $\overline{M}_1$. More explicitly, let A be the M_0-algebra obtained by identifying terms in M_0C that are equal under Z. By the coequalizer property of $\overline{M}_1$ (and the fact that equalities in Z cannot be used in substituted form), we then know that $Z \Rightarrow C \vdash_1 u = v$ iff u and v are equal in $\overline{M}_1A$.

Duplicator may now choose to only play inequalities, using the fact that, over a join-semilattice, $u = v$ iff $u \leq v$ and $v \leq u$. Treating elements of M_0C as finite sets of states, we have $u \leq v$ iff $x \leq v$ for all $x \in u$ and, by the above discussion, Duplicator can thus limit themselves to playing only one inequality $x \leq t'$ with $t' \subseteq t_\rho$ for every $x \in s_\rho$ (and symmetrically). The game can then be rearranged as follows: In configuration (s,t), Spoiler begins by picking a $\rho \in S \cup \{|c\}$ and $x \in s_\rho$ (or symmetrically, $y \in t_\rho$). Duplicator then responds with a finite set $t' \subseteq t_\rho$ to play the inequality $x \leq t'$ (i.e., configuration $(x + t', t')$). This also means that Spoiler cannot switch sides after the initial move.

Finally, the given description of $(\beta\cdot\gamma)_0^*$ stated in the main body follows directly by computation using the description of $\overline{M}_1$.

Local freshness: Terms t at depth 1 can also be brought into the above form, additionally requiring that $C \vdash (a\ c)\cdot\nu a.\mathsf{res}(t_{|c}) \leq t_a$ for all $a \in S$. Once again, S can be extended to finite supersets $S' \supseteq S$, now setting any new $t_a = (a\ c)\cdot\nu a.\mathsf{res}(t_{|c})$. Admissible moves can be described as above, using the additional requirement of the normal form to show the 'only if' direction for $a \in S$. As detailed in Remark 6.4, the elements of M_0C can be thought of as sets of name restrictions closed under further restriction. We can thus rearrange the game so Spoiler can move first, now picking a possibly restricted $x|_N \in s_\rho$ (or symmetrically), to which Duplicator responds with a finite set $t' \subseteq t_\rho$ of possibly restricted names closed under further restrictions, playing the inequality $x|_N \leq t'$.

We will also show the explicit description of $(\beta\cdot\gamma)_0^\star$ by computation using the term representation outlined in Sections A.2 and A.4. Note that, with this description, the case where Spoiler picks $\rho = |c$ is subsumed by simply picking a sufficiently fresh c by Lemma A.22.

We will once again use the notation

$$A \setminus x = \nu a_1.\mathsf{res}(\dots(\nu a_n.\mathsf{res}(x)))$$

for a finite set $A = \{a_1, \dots a_n\} \in \mathcal{P}_\omega(\mathbb{A})$. By the description given in Section A.2, a term $(\mathsf{supp}(x) \setminus N) \setminus x$ then corresponds to a name restriction $x|_{\mathsf{supp}(x)\cap N}$.

We will first prove the following helper lemma:

Lemma A.36. *Let (C, γ) be a H_ω-coalgebra, $x, y \in C$, and $N \subseteq \mathsf{supp}(x)$.*

(1) If $x \xrightarrow{a} y$, then $a \notin \mathsf{supp}(x) \setminus N$ iff $a \in N$.
(2) If $x \xrightarrow{a} y$, then $C \vdash_0 (\mathsf{supp}(x) \setminus N) \setminus y = (\mathsf{supp}(y) \setminus N) \setminus y$ is derivable.
(3) If $x \xrightarrow{|c} y$, then $C \vdash_0 (\mathsf{supp}(x) \setminus (N \cup \{c\})) \setminus y = (\mathsf{supp}(y) \setminus (N \cup \{c\})) \setminus y$ is derivable.

Proof. We will prove the statements individually:

(1) Since $x \xrightarrow{a} y$, we have $\{a\} \subseteq \{a\} \cup \mathsf{supp}(y) \subseteq \mathsf{supp}(x)$. Thus, $a \in \mathsf{supp}(x)$ holds and the statement follows immediately.
(2) Since (6.3) (1) allows us to leave out name restrictions for names that are not in the support, it suffices to show $\mathsf{supp}(y) \cap (\mathsf{supp}(x) \setminus N) = \mathsf{supp}(y) \setminus N$. This follows from the fact that $\mathsf{supp}(y) \subseteq \{a\} \cup \mathsf{supp}(y) \subseteq \mathsf{supp}(x)$ since $x \xrightarrow{a} y$.
(3) Once again, it suffices to show $\mathsf{supp}(y) \cap (\mathsf{supp}(x) \setminus (N \cup \{c\})) = \mathsf{supp}(y) \setminus (N \cup \{c\})$. This follows from the fact that $\mathsf{supp}(y) \subseteq \mathsf{supp}(x) \cup \{c\}$ because $x \xrightarrow{|c} y$.

□

Let $s \in M_0 C$ and fix some $c \# s$. Choosing one representative for every element $x|_N \in s$, we obtain

$$\begin{aligned}
(\beta \cdot \gamma)_0^\star(s) &= (\beta \cdot \gamma)_0^\star \left(\left[\sum\nolimits_{x|_N \in s} (\mathsf{supp}(x) \setminus N) \setminus x \right]_0 \right) \\
&= \left[\sum\nolimits_{x|_N \in s} (\mathsf{supp}(x) \setminus N) \setminus \beta(\gamma(x)) \right]_1 \\
&= \left[\sum\nolimits_{x|_N \in s} (\mathsf{supp}(x) \setminus N) \setminus \left(\sum\nolimits_{x \xrightarrow{a} y} ay + \sum\nolimits_{x \xrightarrow{|c} y} |cy \right) \right]_1 \\
&= \left[\sum\nolimits_{x|_N \in s, x \xrightarrow{a} y} (\mathsf{supp}(x) \setminus N) \setminus (ay) + \sum\nolimits_{x|_N \in s, x \xrightarrow{|c} y} (\mathsf{supp}(x) \setminus N) \setminus (|cy) \right]_1 .
\end{aligned}$$

For the first sum, we have the derivable equality

$$\begin{aligned}
&\sum_{\substack{x|_N \in s,\\ x \xrightarrow{a} y}} (\mathsf{supp}(x) \setminus N) \setminus (ay) \\
&= \sum_{\substack{x|_N \in s,\\ x \xrightarrow{a} y,\\ a \in \mathsf{supp}(x) \setminus N}} (\mathsf{supp}(x) \setminus N) \setminus (ay) + \sum_{\substack{x|_N \in s,\\ x \xrightarrow{a} y,\\ a \notin \mathsf{supp}(x) \setminus N}} (\mathsf{supp}(x) \setminus N) \setminus (ay) && \text{(case distinction)} \\
&= \sum_{\substack{x|_N \in s,\\ x \xrightarrow{a} y,\\ a \notin \mathsf{supp}(x) \setminus N}} a((\mathsf{supp}(x) \setminus N) \setminus y) && \text{(axioms)} \\
&= \sum_{\substack{x|_N \in s,\\ x \xrightarrow{a} y,\\ a \in N}} a((\mathsf{supp}(x) \setminus N) \setminus y) && \text{(Lemma A.36 (1))} \\
&= \sum_{\substack{x|_N \in s,\\ x \xrightarrow{a} y,\\ a \in N}} a((\mathsf{supp}(y) \setminus N) \setminus y) && \text{(Lemma A.36 (2))}
\end{aligned}$$

For the second sum, we have

$$\begin{aligned}
&\sum_{\substack{x|_N \in s,\\ x \xrightarrow{\mathsf{lc}} y}} (\mathsf{supp}(x) \setminus N) \setminus (\mathsf{lc}y) \\
&= \sum_{\substack{x|_N \in s,\\ x \xrightarrow{\mathsf{lc}} y}} \mathsf{lc}((\mathsf{supp}(x) \setminus (N \cup \{c\})) \setminus y) && \text{(axioms)} \\
&= \sum_{\substack{x|_N \in s,\\ x \xrightarrow{\mathsf{lc}} y}} \mathsf{lc}((\mathsf{supp}(y) \setminus (N \cup \{c\})) \setminus y) && \text{(Lemma A.36 (3)).}
\end{aligned}$$

Using the model from Section A.4, we then obtain the claimed description.

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